

BFS Traversal

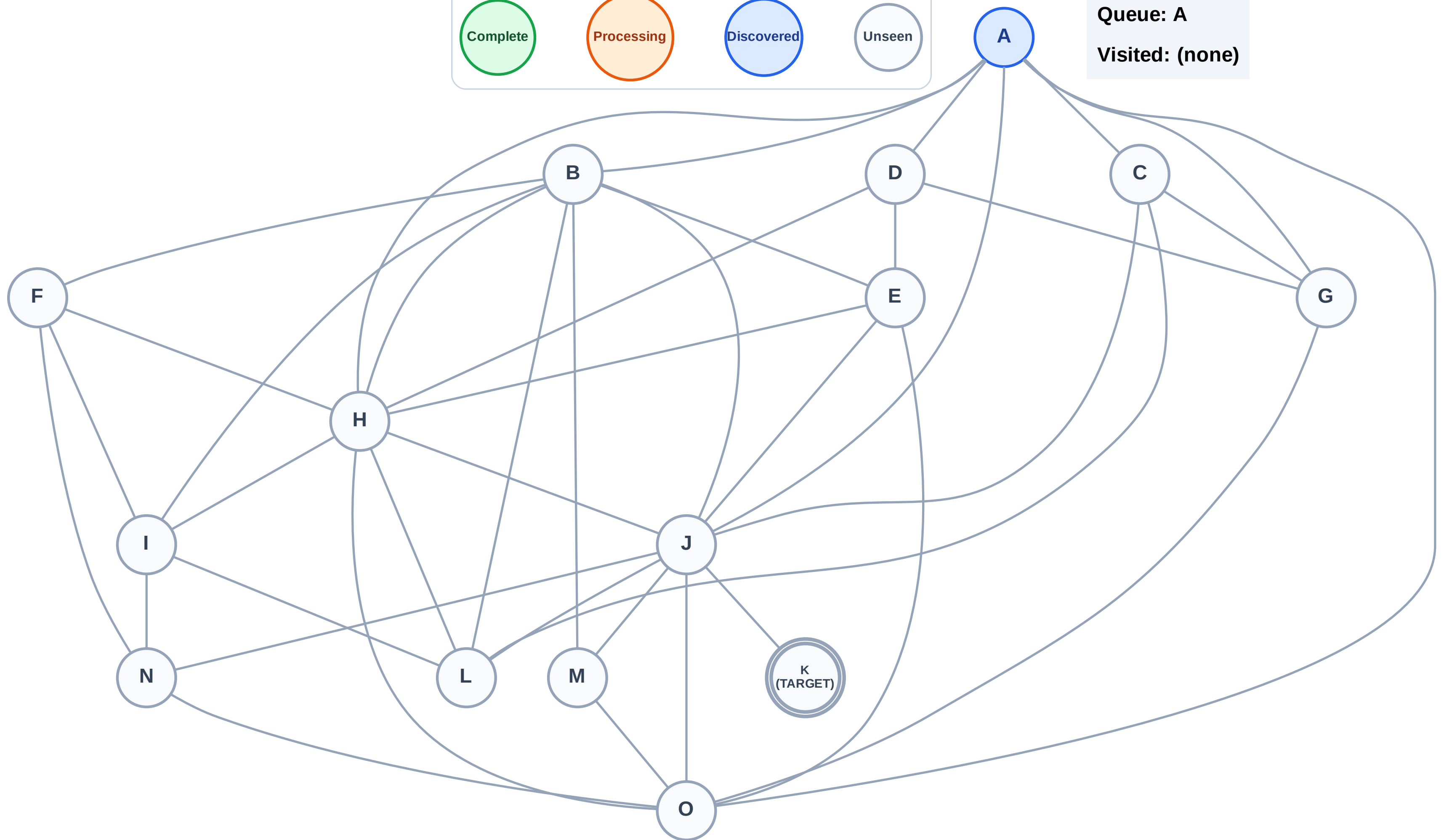
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

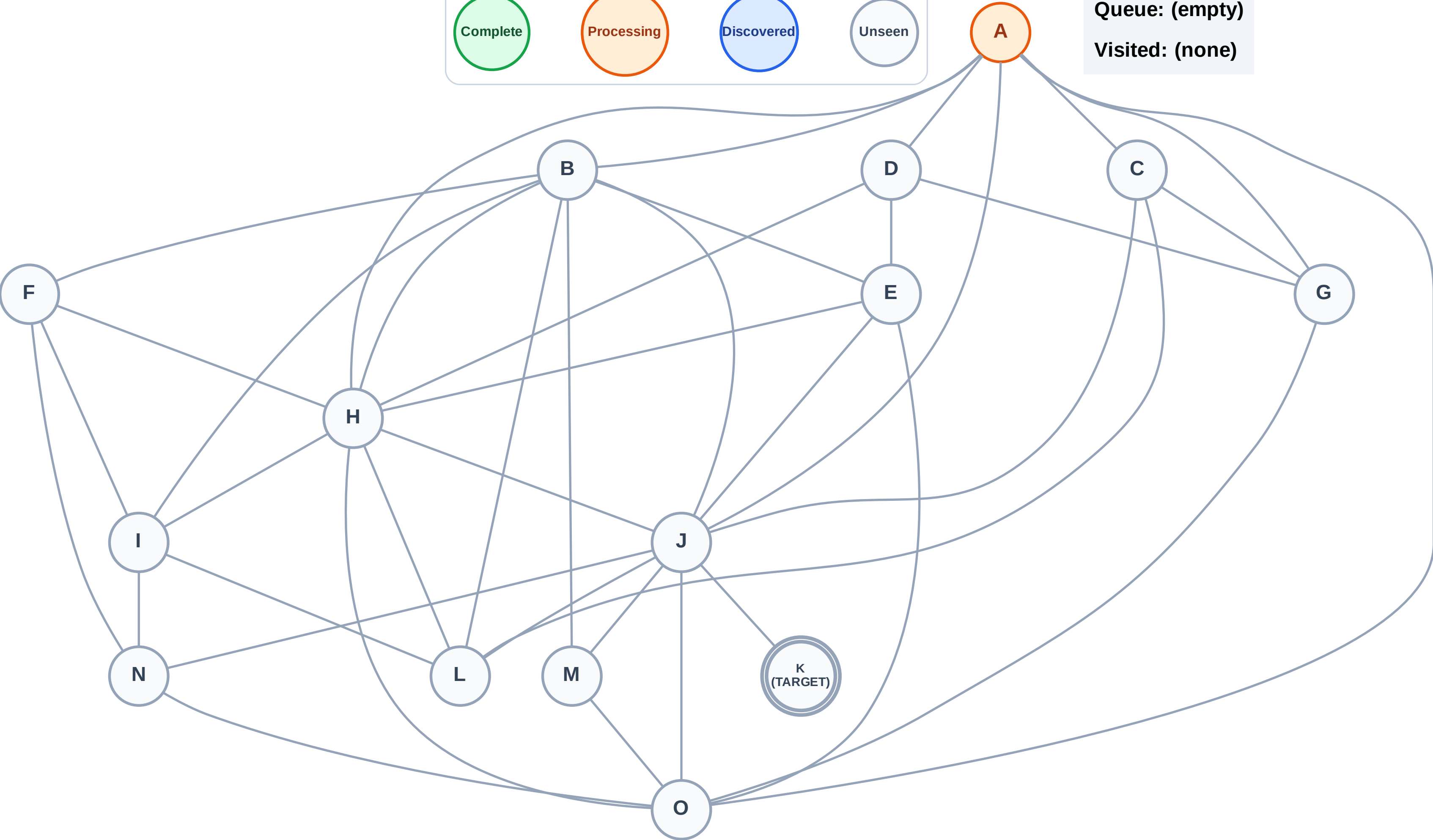
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



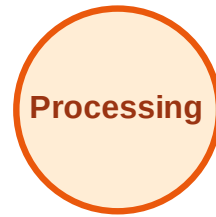
BFS Traversal

Step 3: Discover B, C, D, G, H, J, O from A

Legend



Complete



Processing



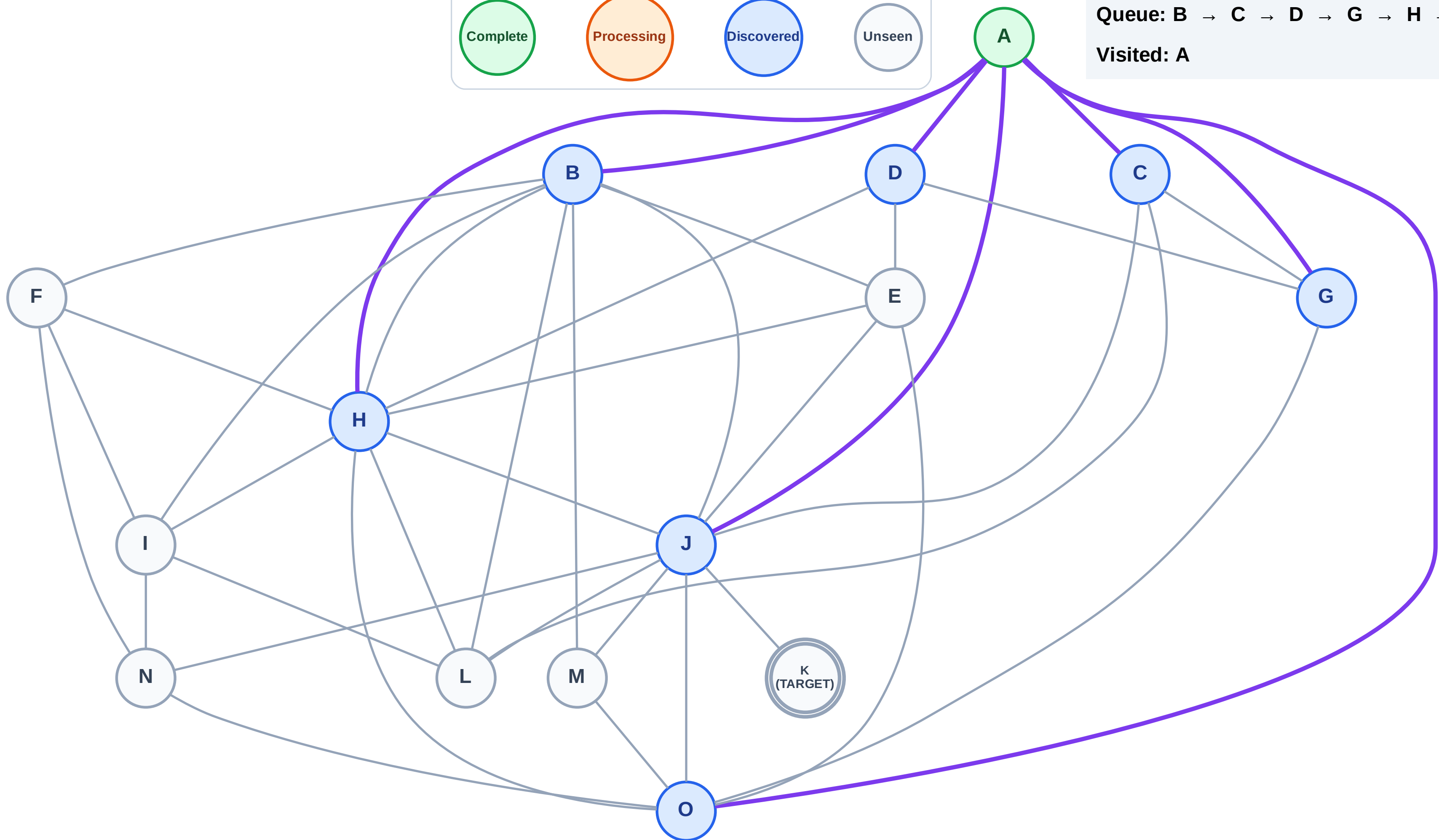
Discovered



Unseen

Queue: B → C → D → G → H → J → O

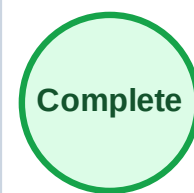
Visited: A



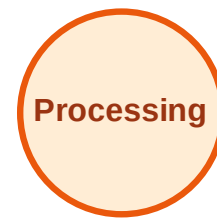
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing

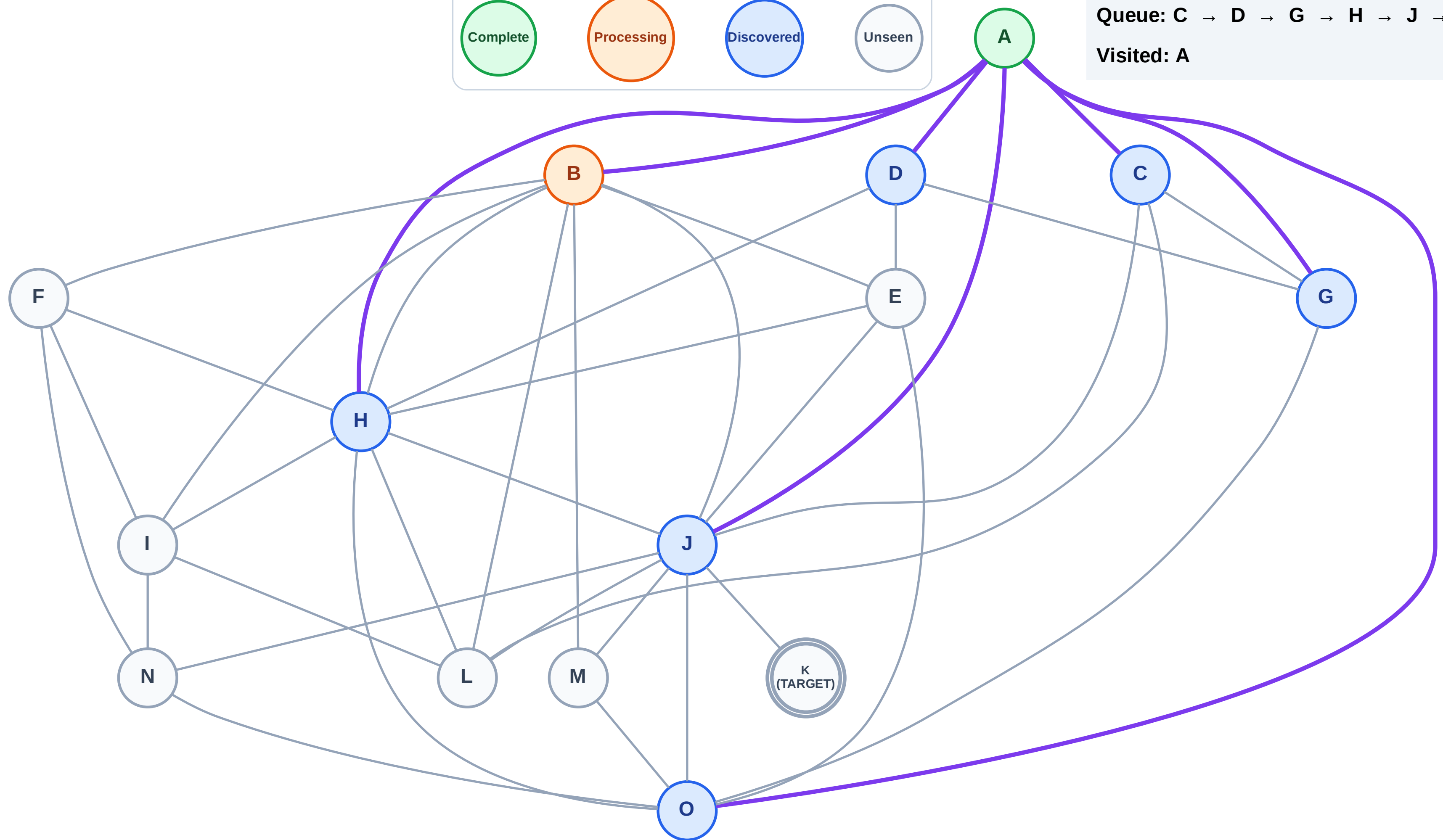


Discovered



Unseen

Queue: C → D → G → H → J → O
Visited: A



BFS Traversal

Step 5: Discover E, F, I, L, M from B

Legend



Complete



Processing



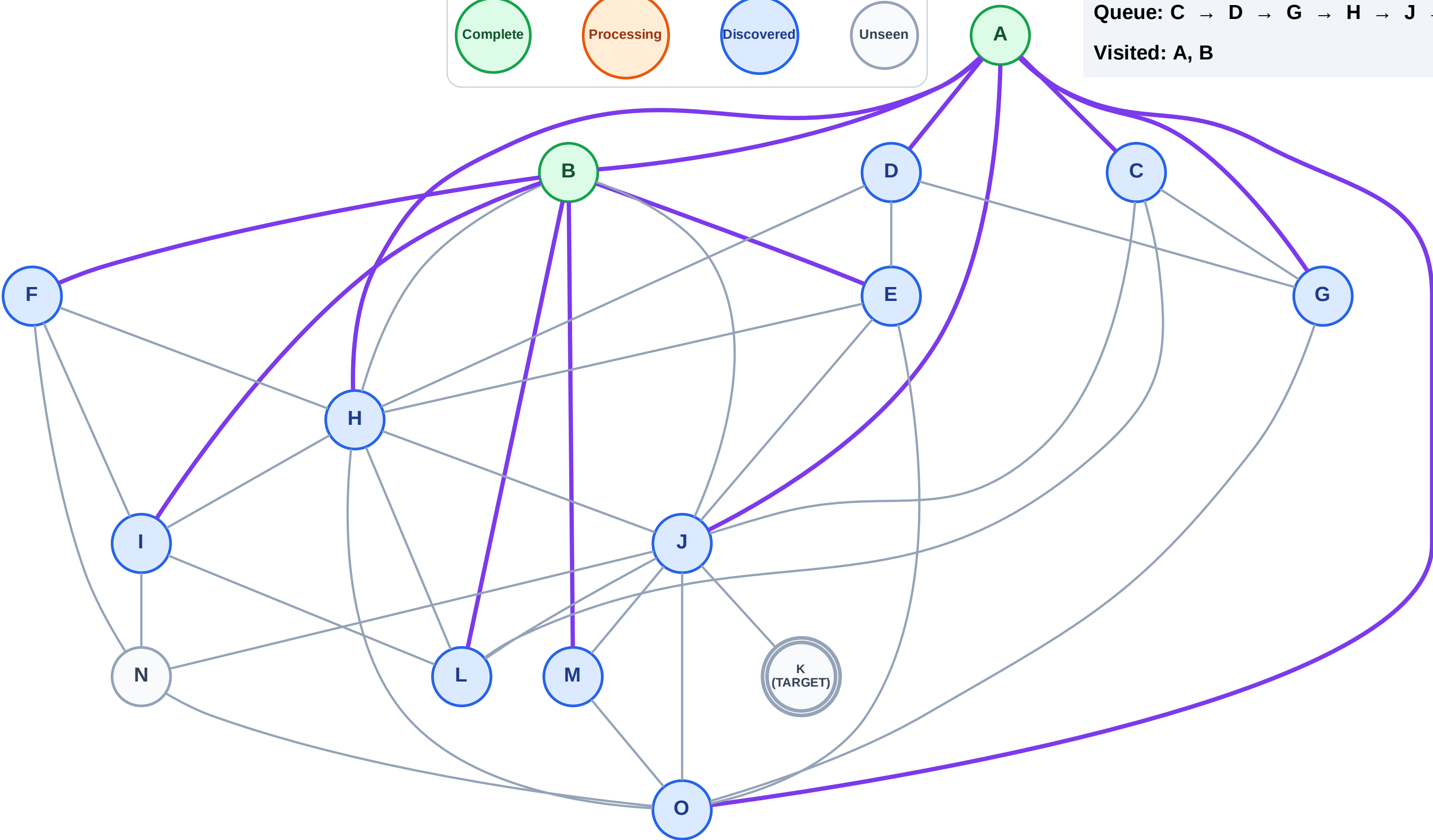
Discovered



Unseen

Queue: C → D → G → H → J → O → E → F → I → L → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



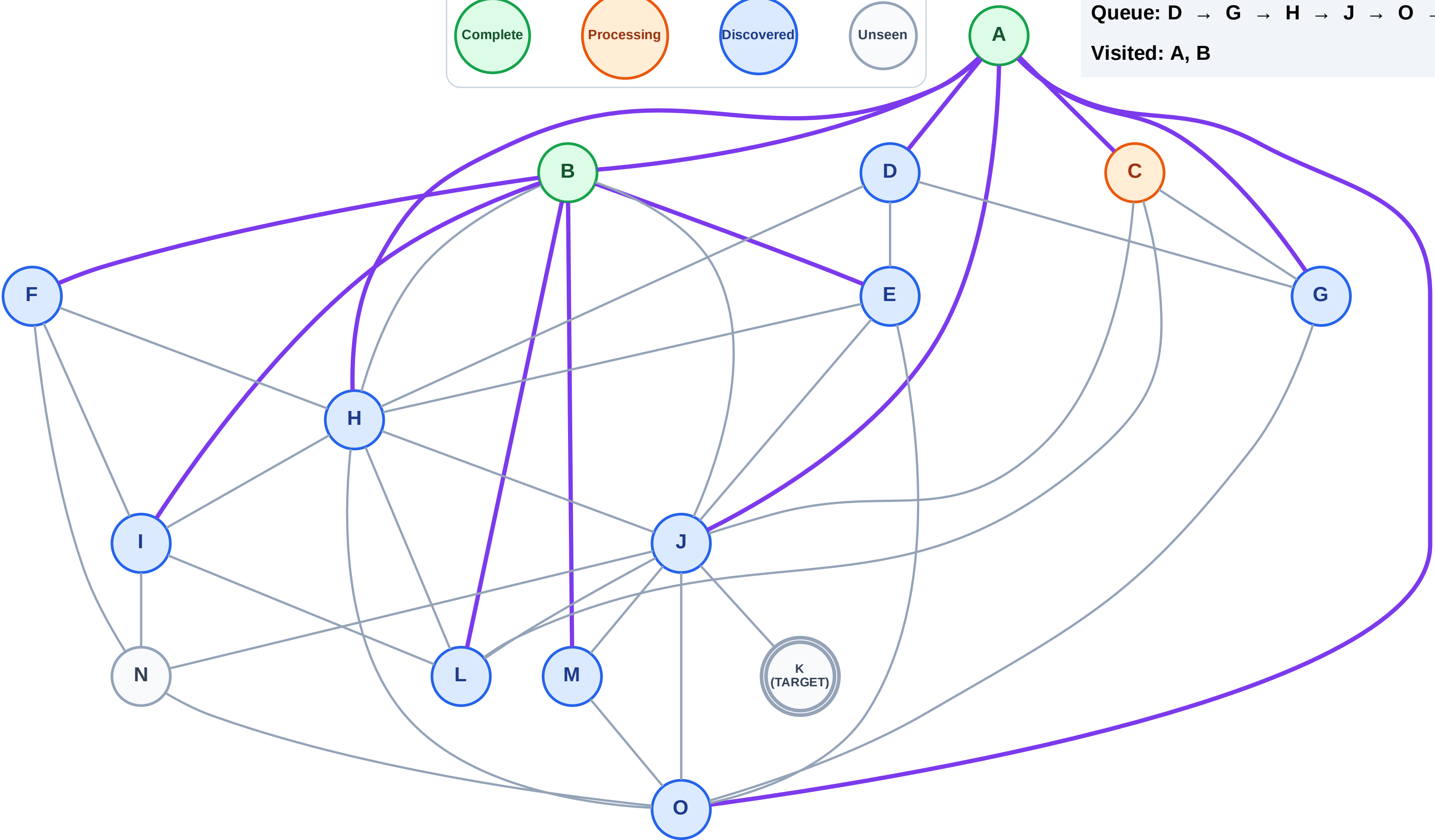
Discovered



Unseen

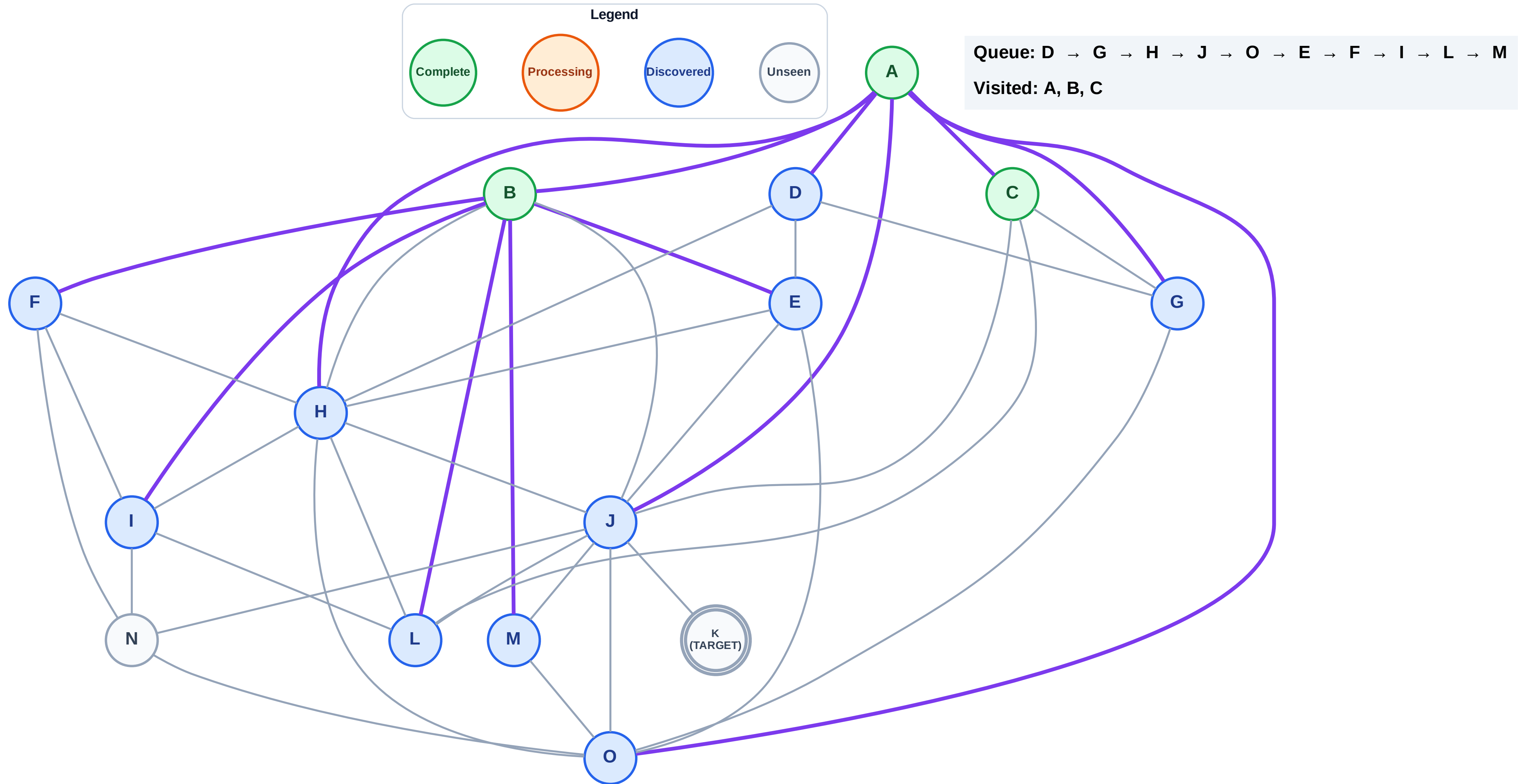
Queue: D → G → H → J → O → E → F → I → L → M

Visited: A, B



Step 7: No new neighbors from C

Step 7: No new neighbors from C



BFS Traversal

Step 8: Process node D

Legend

Complete

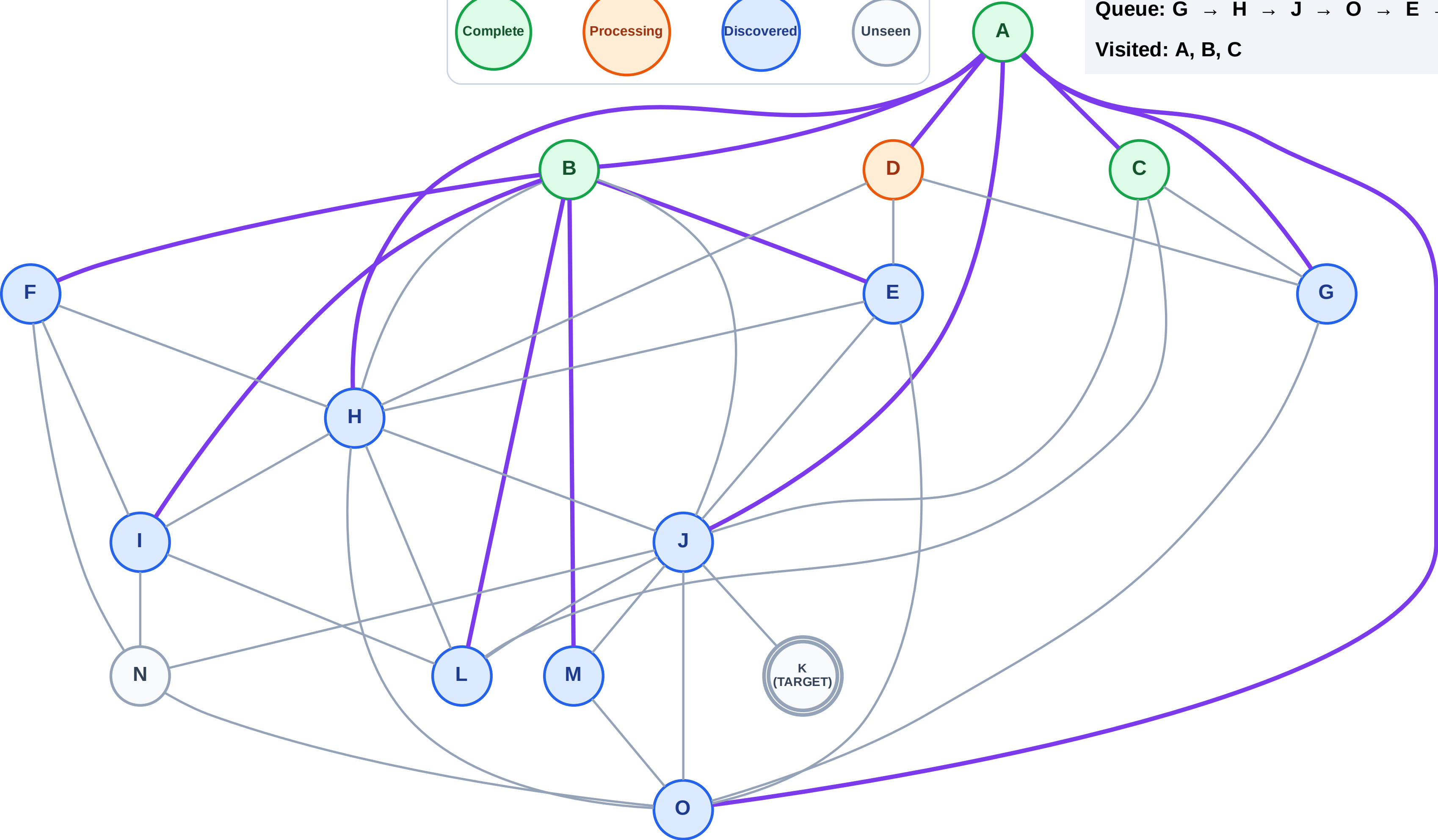
Processing

Discovered

Unseen

Queue: G → H → J → O → E → F → I → L → M

Visited: A, B, C



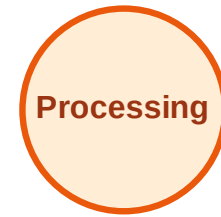
BFS Traversal

Step 9: No new neighbors from D

Legend



Complete



Processing



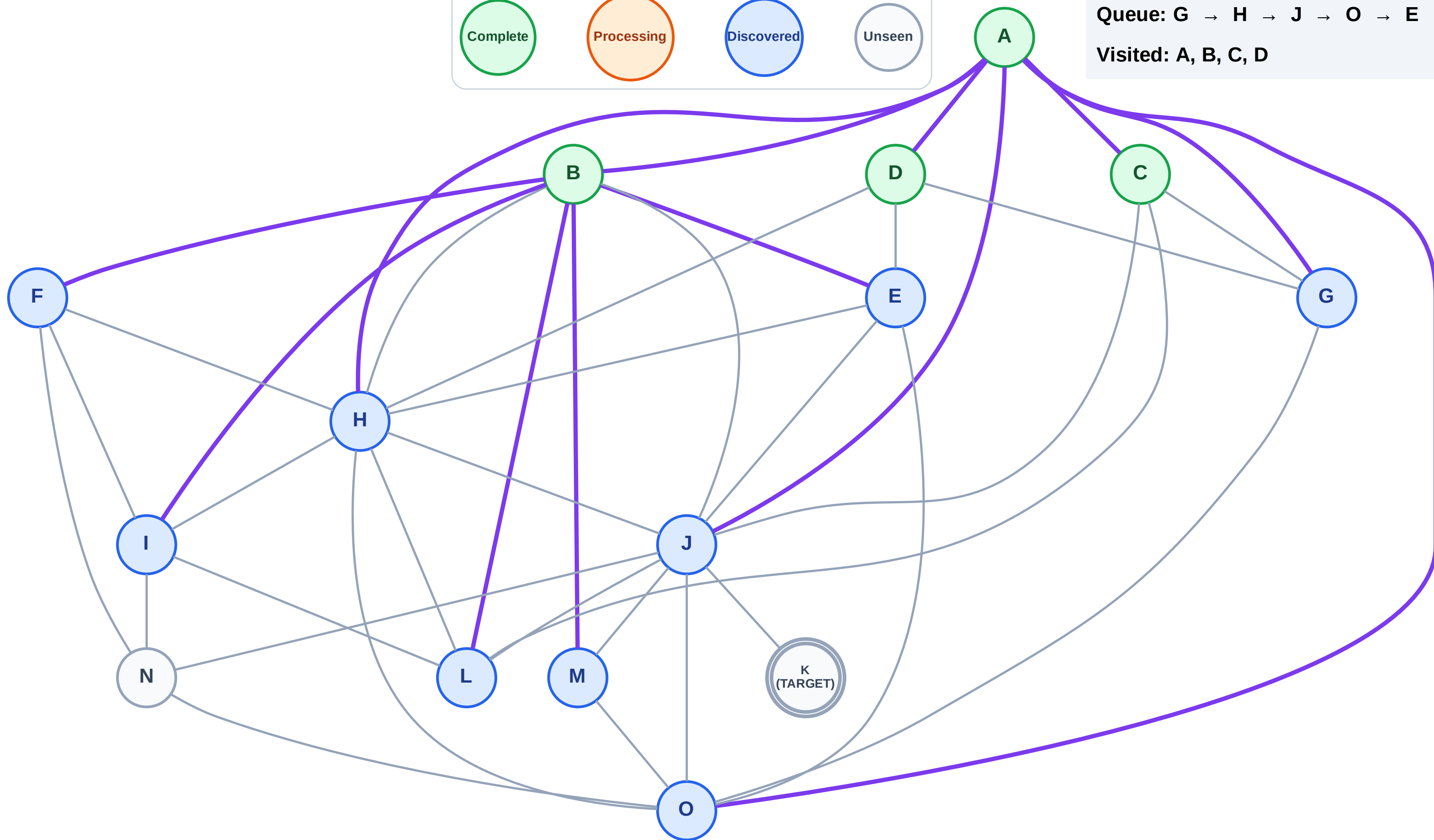
Discovered



Unseen

Queue: G → H → J → O → E → F → I → L → M

Visited: A, B, C, D



BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



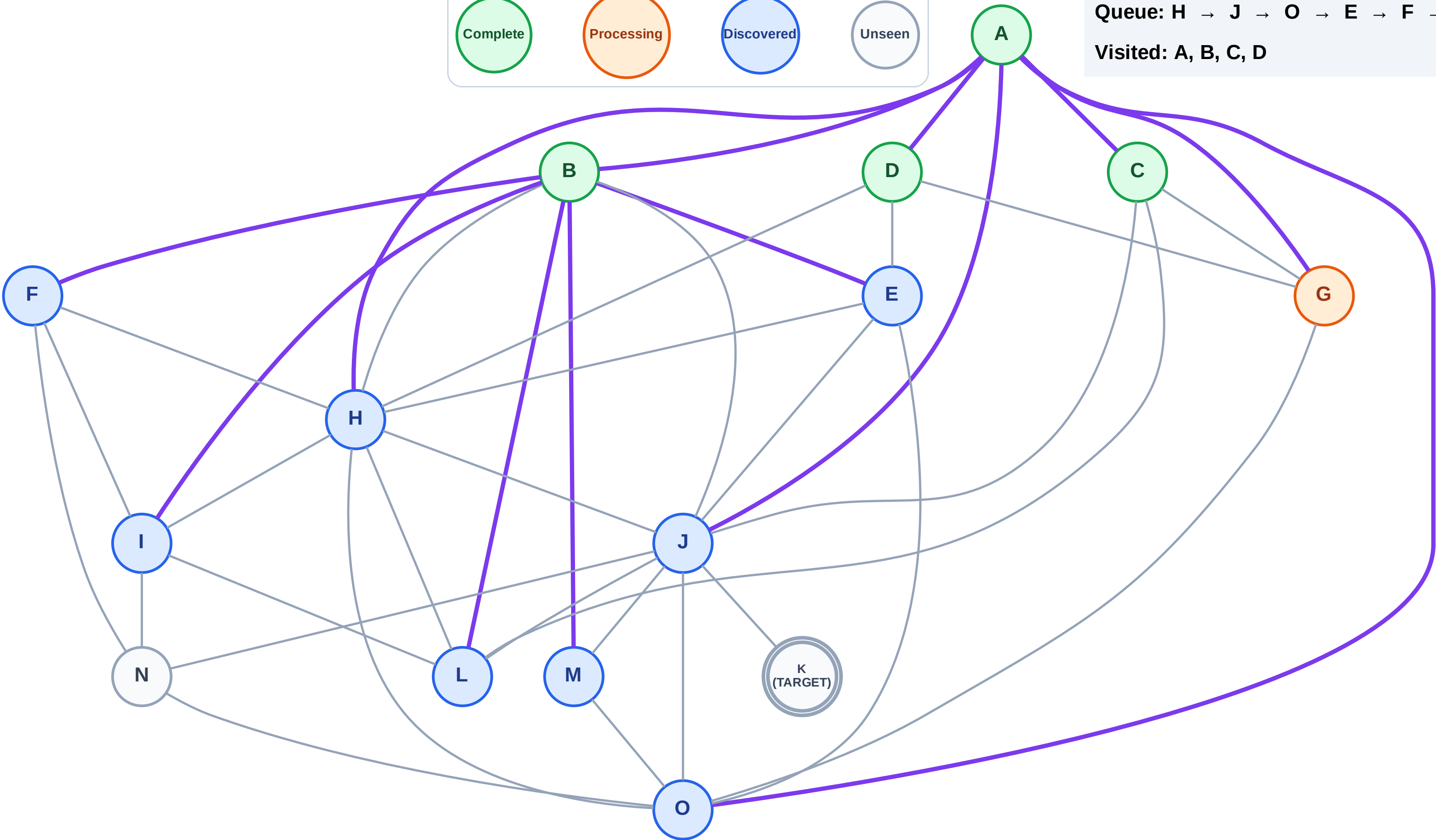
Discovered



Unseen

Queue: H → J → O → E → F → I → L → M

Visited: A, B, C, D



BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



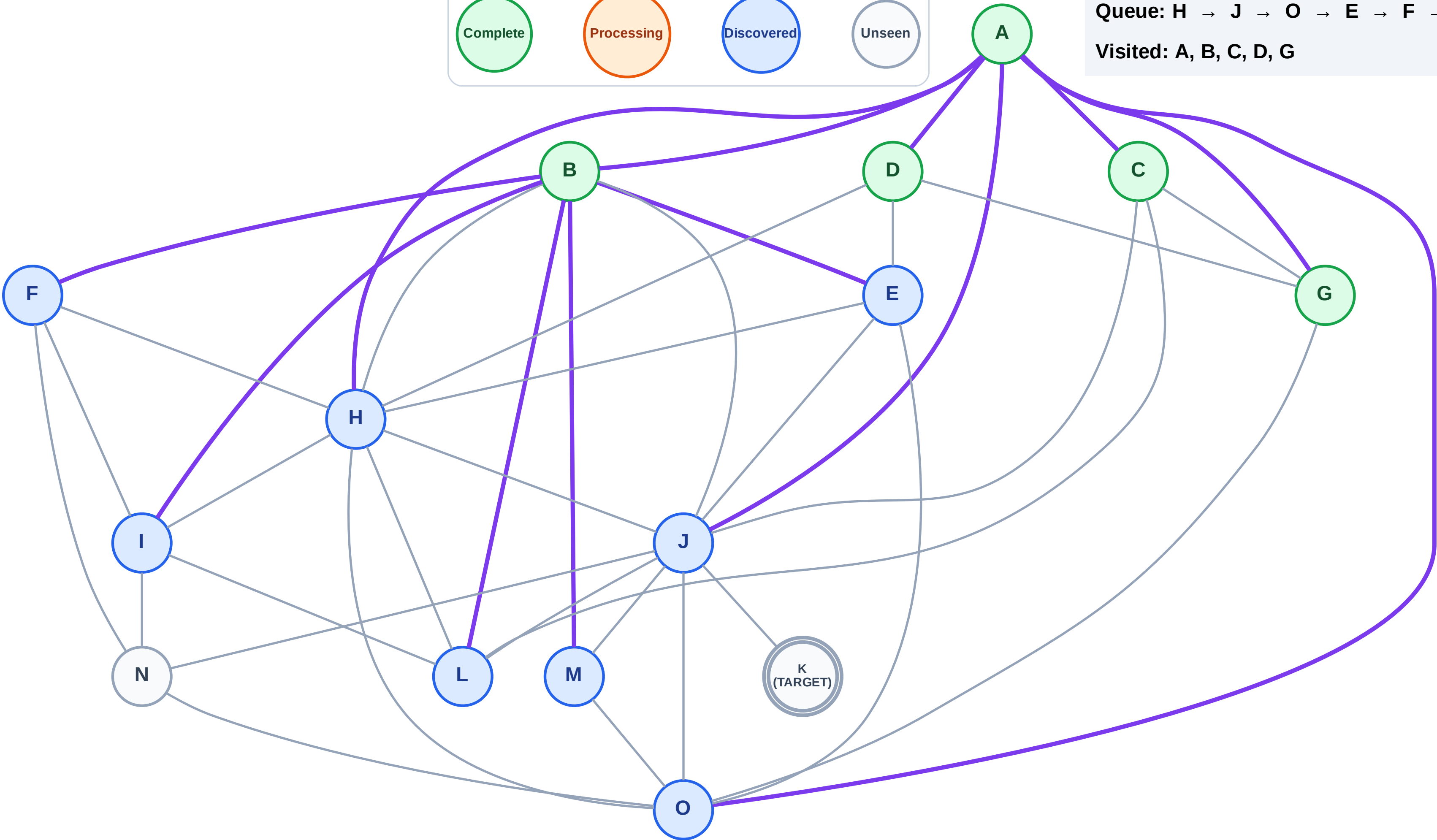
Discovered



Unseen

Queue: H → J → O → E → F → I → L → M

Visited: A, B, C, D, G



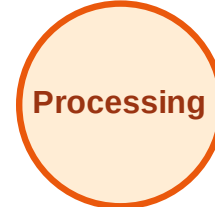
BFS Traversal

Step 12: Process node H

Legend



Complete



Processing



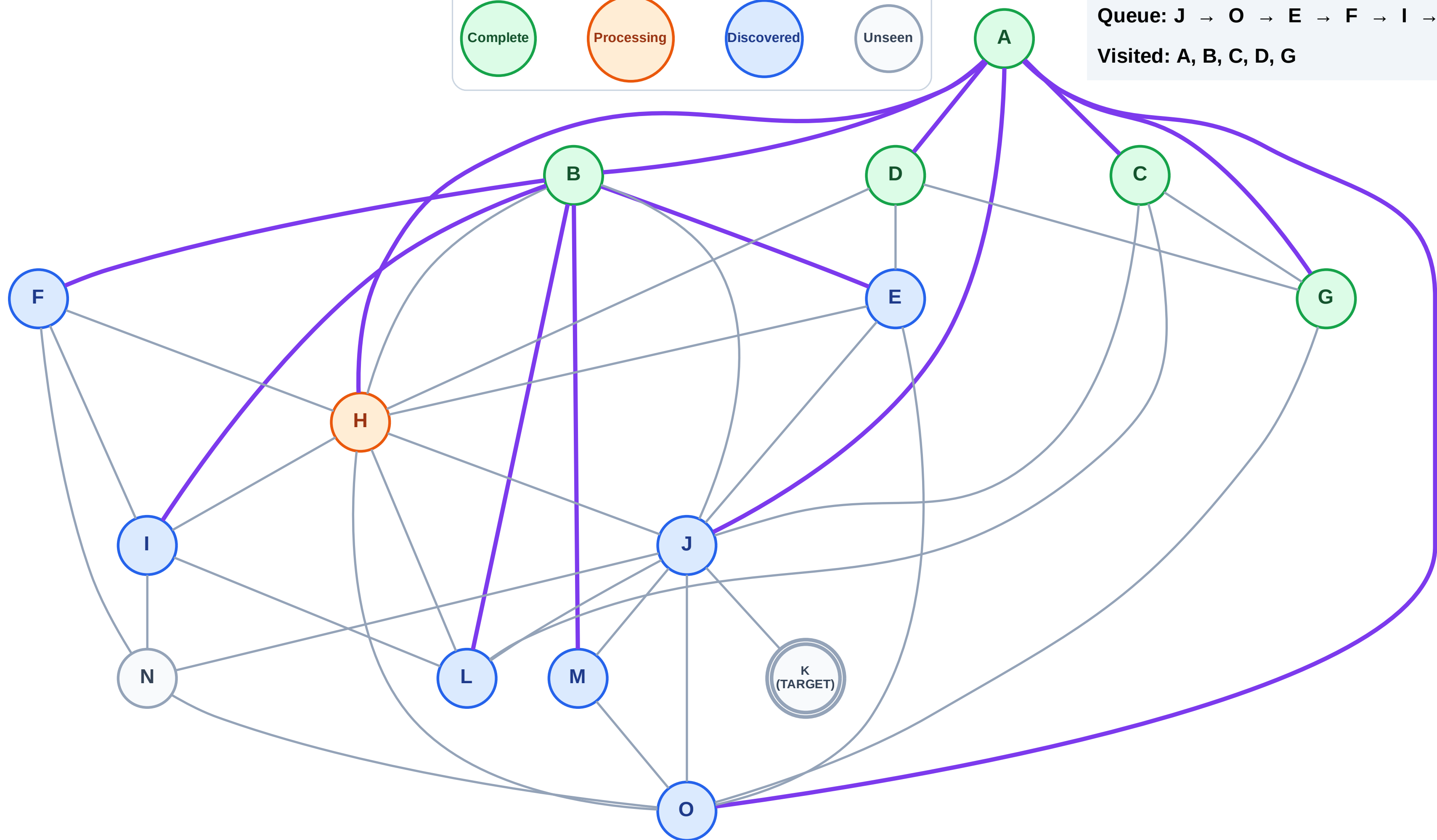
Discovered



Unseen

Queue: J → O → E → F → I → L → M

Visited: A, B, C, D, G



BFS Traversal

Step 13: No new neighbors from H

Legend



Complete



Processing



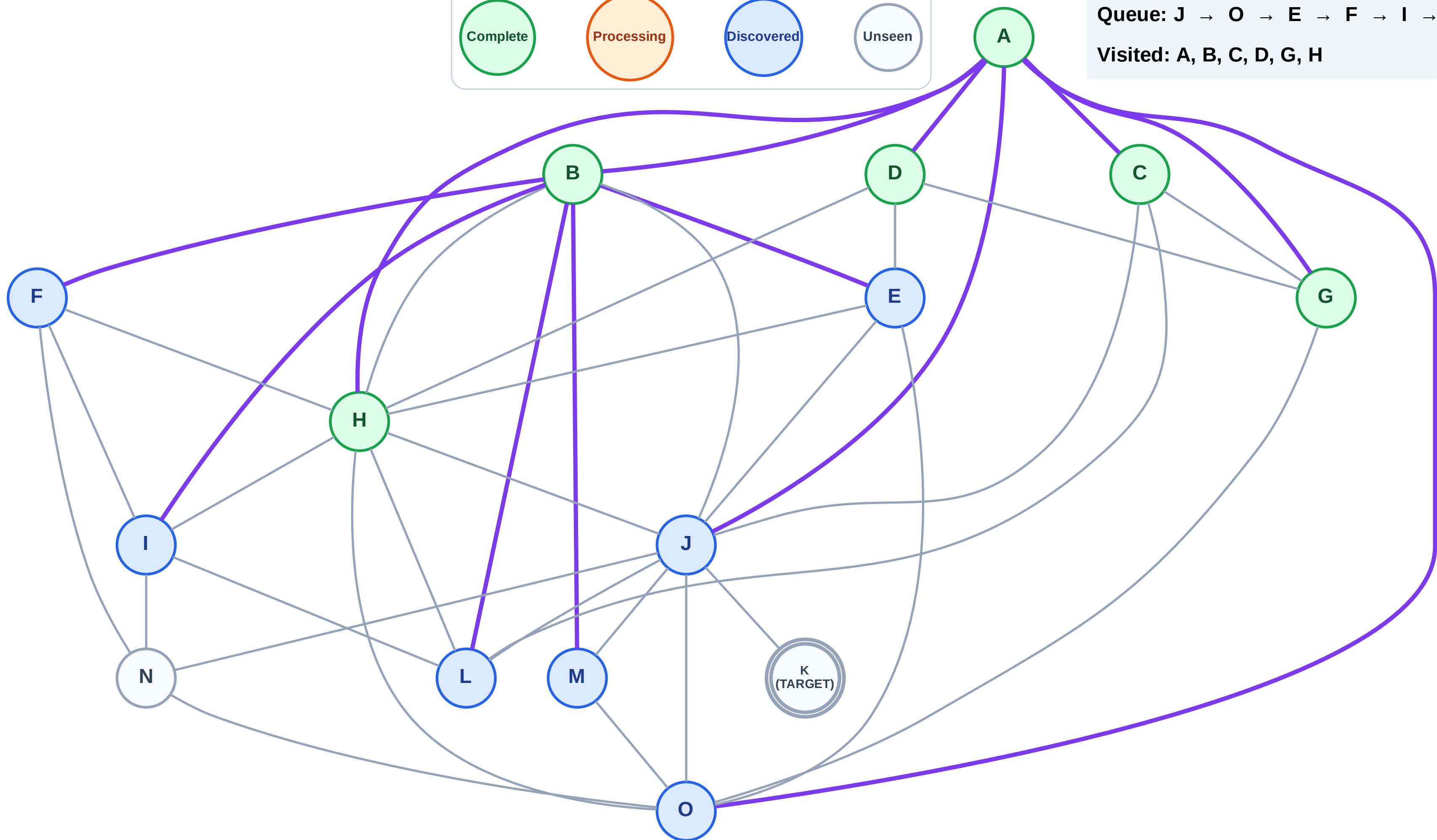
Discovered



Unseen

Queue: J → O → E → F → I → L → M

Visited: A, B, C, D, G, H



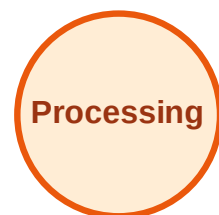
BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



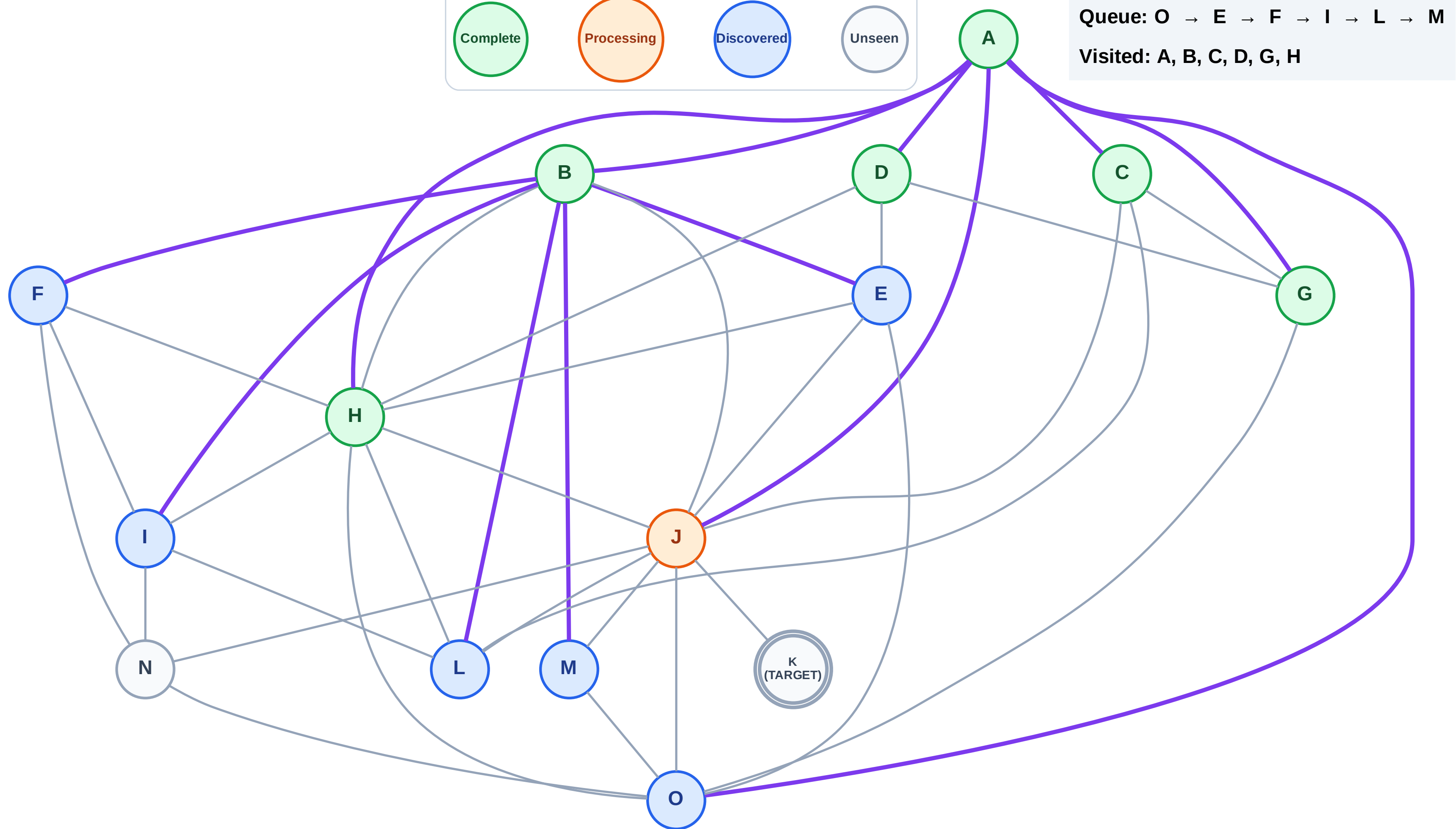
Discovered



Unseen

Queue: O → E → F → I → L → M

Visited: A, B, C, D, G, H



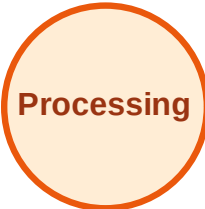
BFS Traversal

Step 15: Discover K, N from J

Legend



Complete



Processing



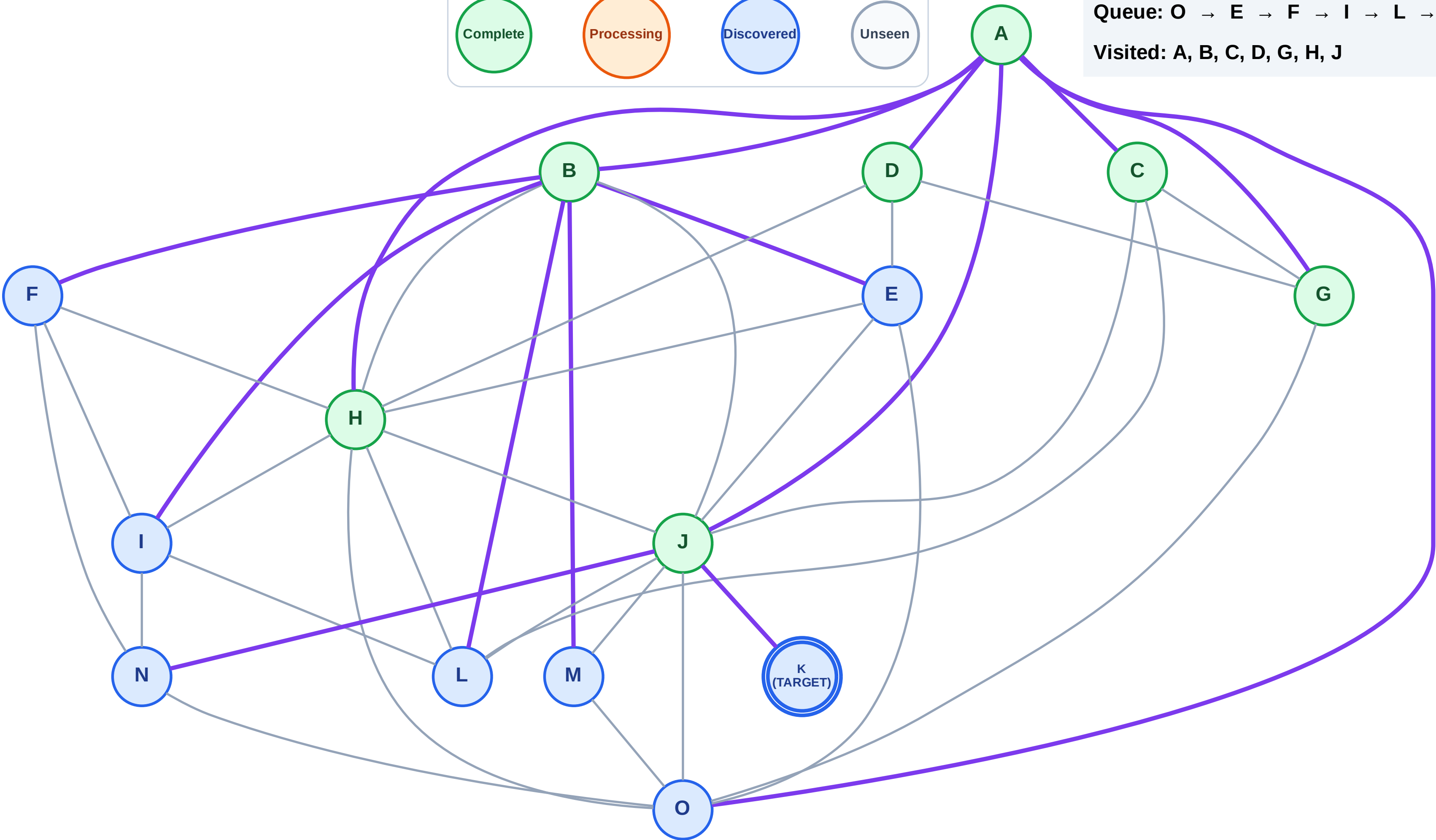
Discovered



Unseen

Queue: O → E → F → I → L → M → K → N

Visited: A, B, C, D, G, H, J



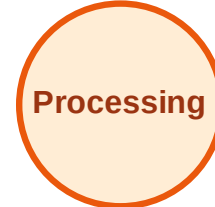
BFS Traversal

Step 16: Process node O

Legend



Complete



Processing



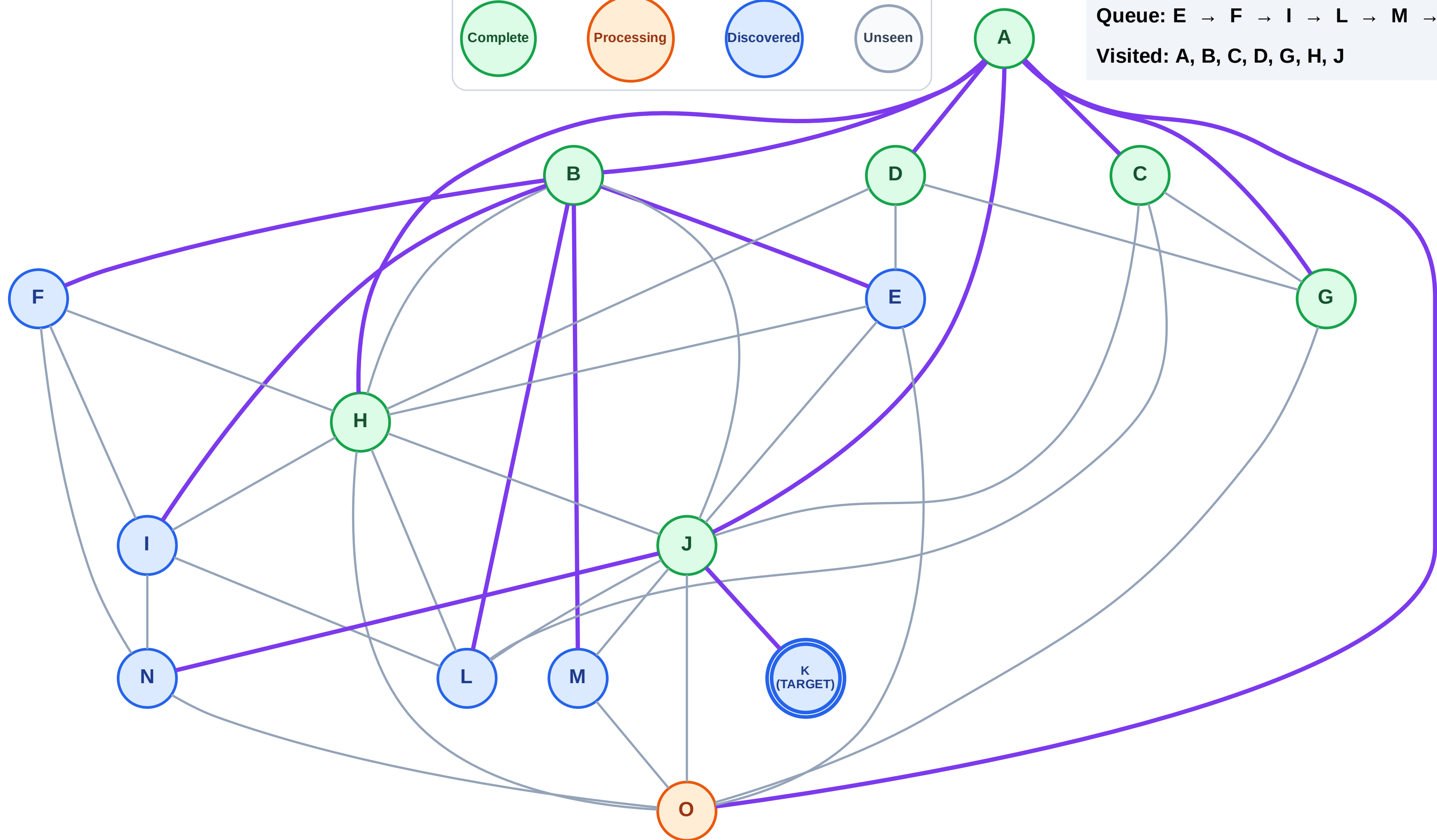
Discovered



Unseen

Queue: E → F → I → L → M → K → N

Visited: A, B, C, D, G, H, J



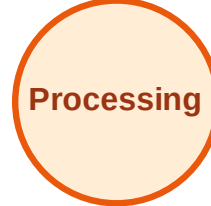
BFS Traversal

Step 17: No new neighbors from O

Legend



Complete



Processing



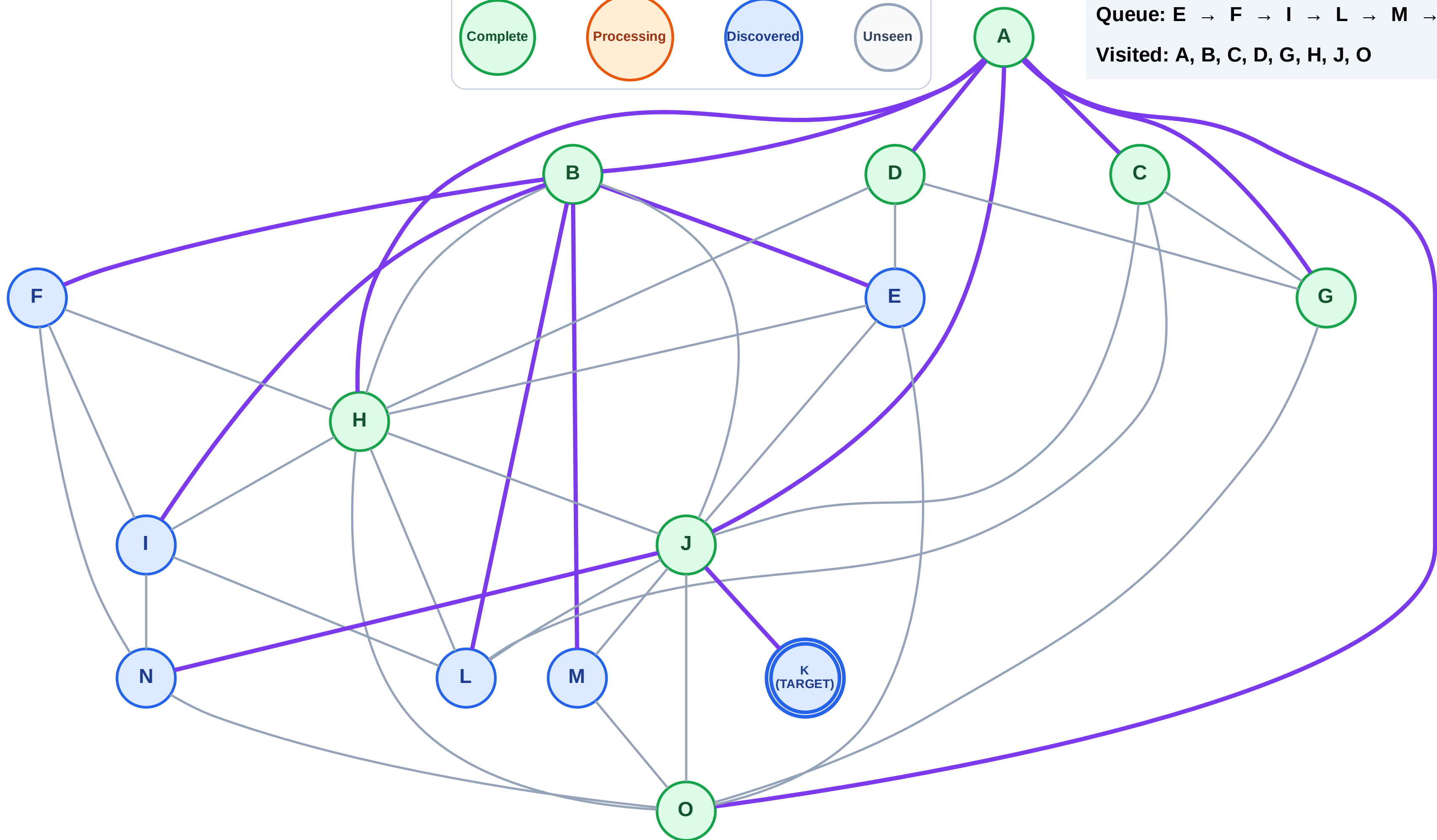
Discovered



Unseen

Queue: E → F → I → L → M → K → N

Visited: A, B, C, D, G, H, J, O



BFS Traversal

Step 18: Process node E

Legend

Complete

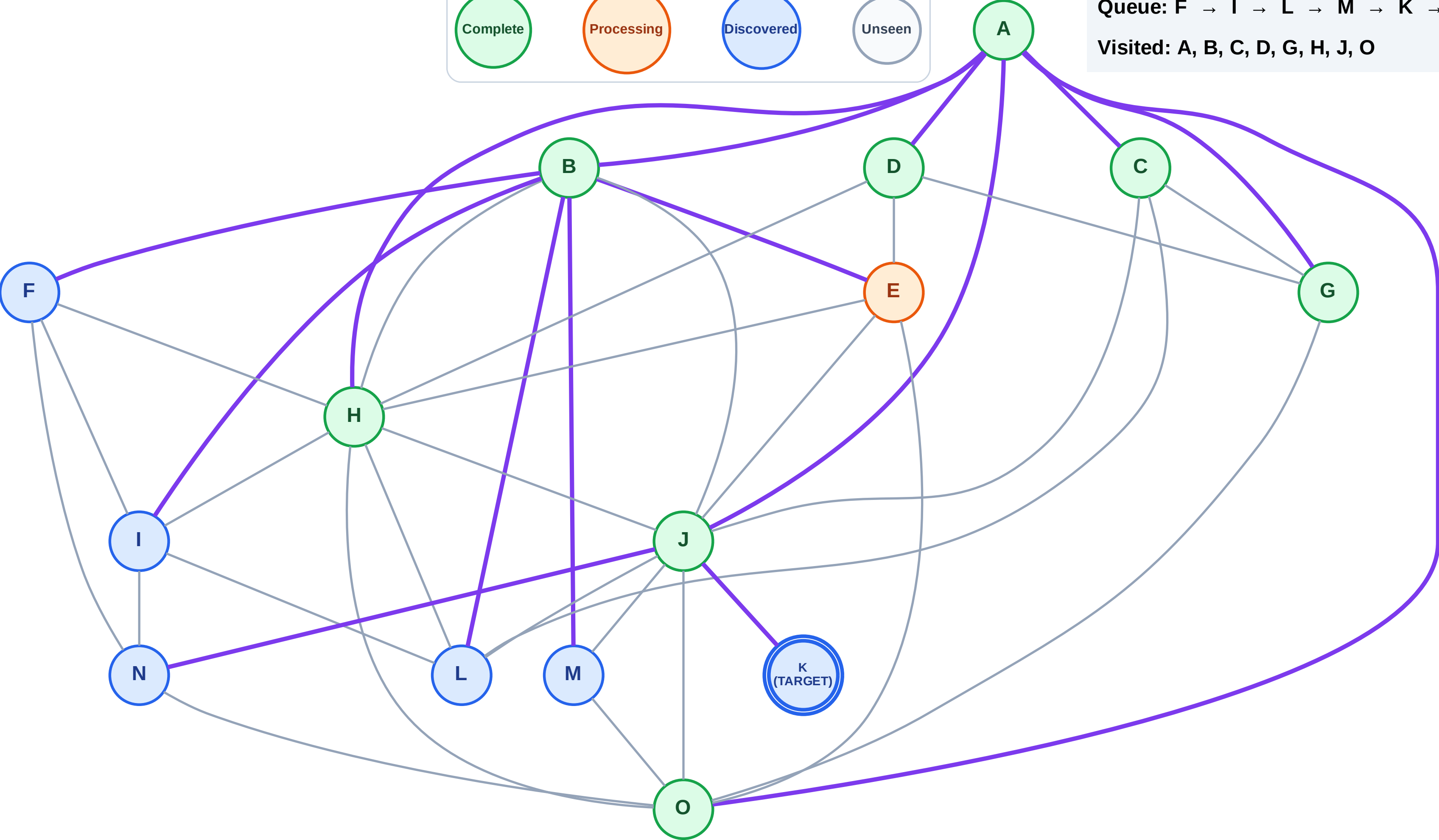
Processing

Discovered

Unseen

Queue: F → I → L → M → K → N

Visited: A, B, C, D, G, H, J, O



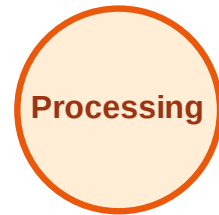
BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing

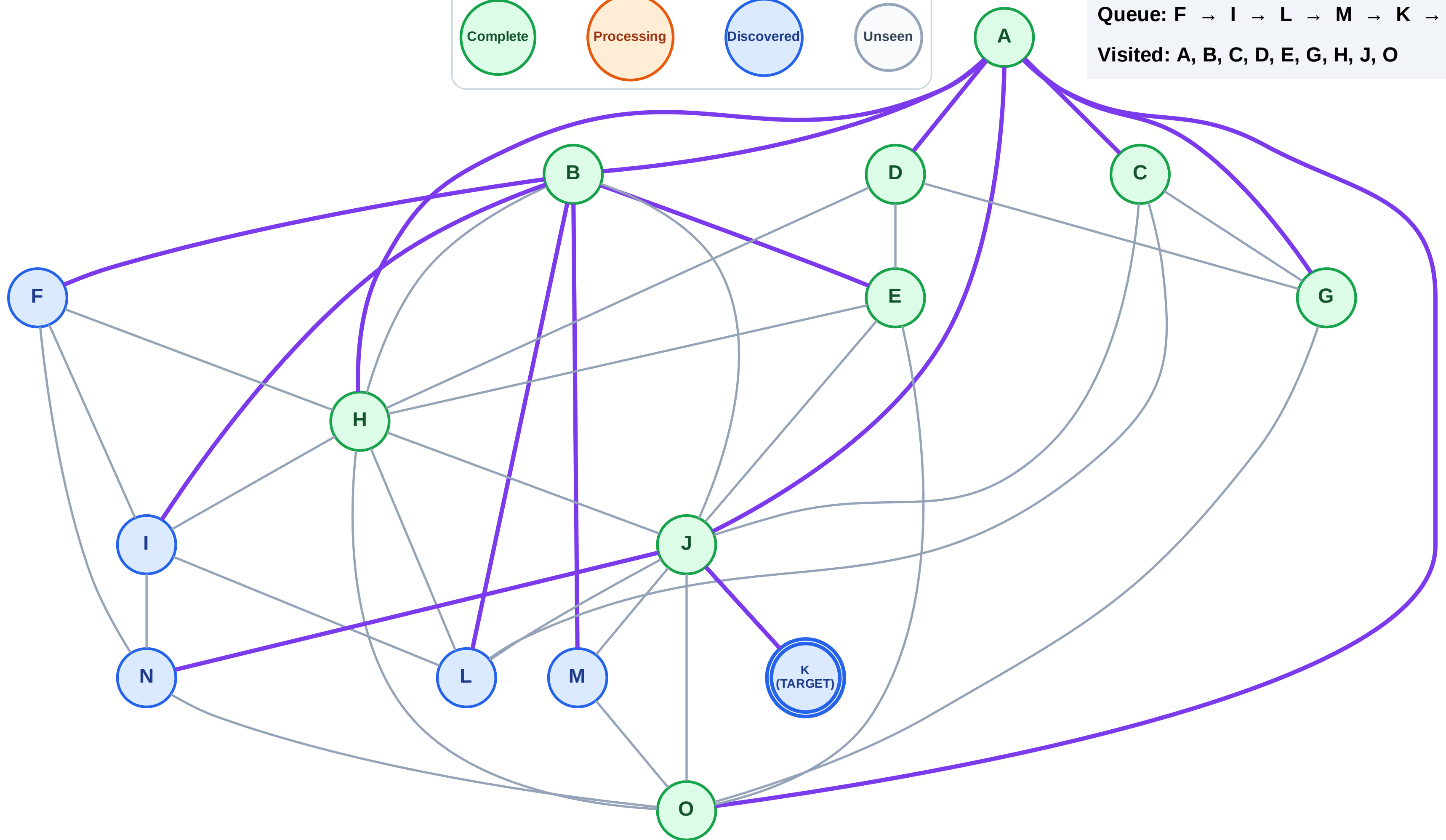


Discovered



Unseen

Queue: F → I → L → M → K → N
Visited: A, B, C, D, E, G, H, J, O



BFS Traversal

Step 20: Process node F

Legend

Complete

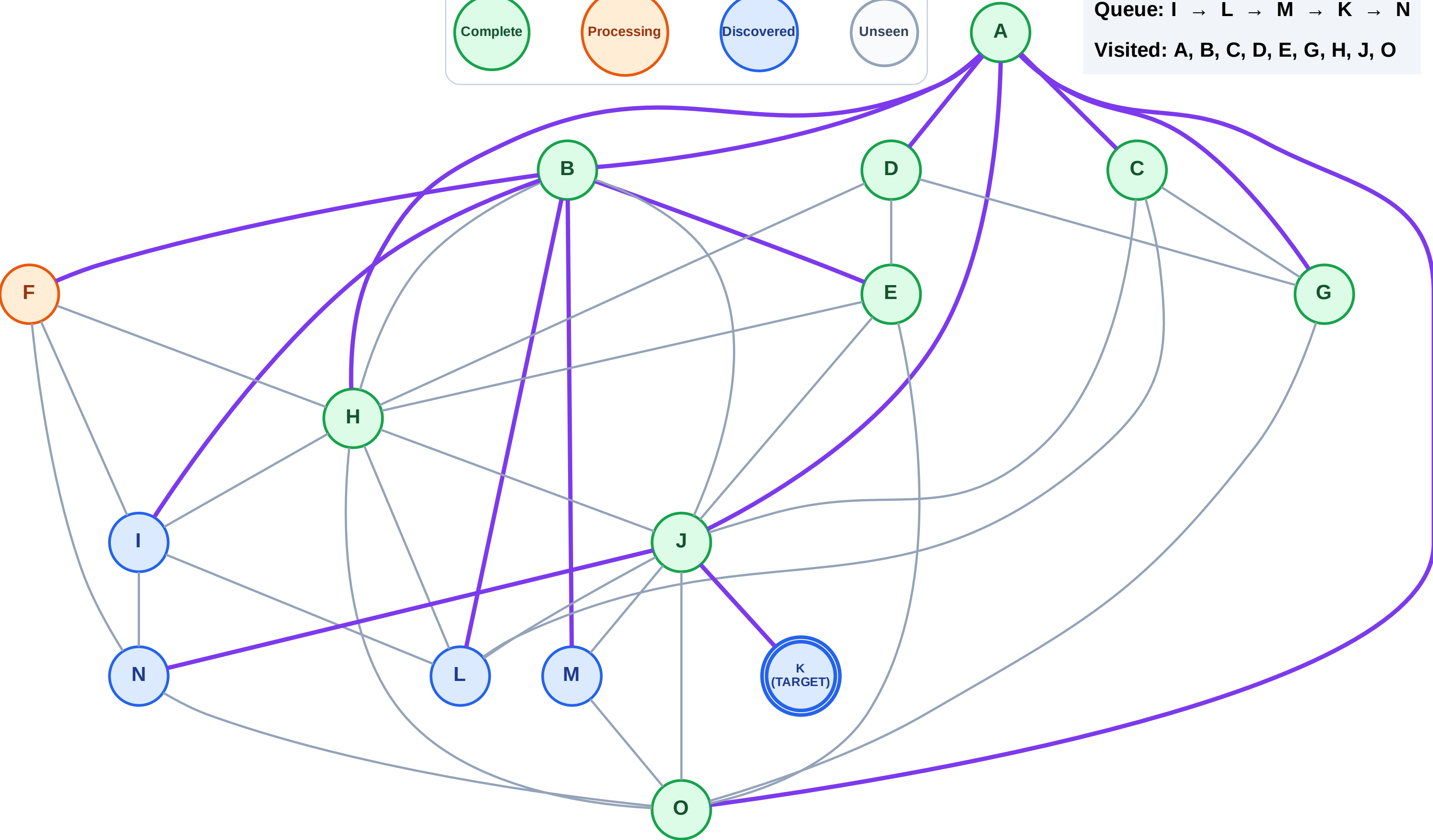
Processing

Discovered

Unseen

Queue: I → L → M → K → N

Visited: A, B, C, D, E, G, H, J, O



BFS Traversal

Step 21: No new neighbors from F

Legend

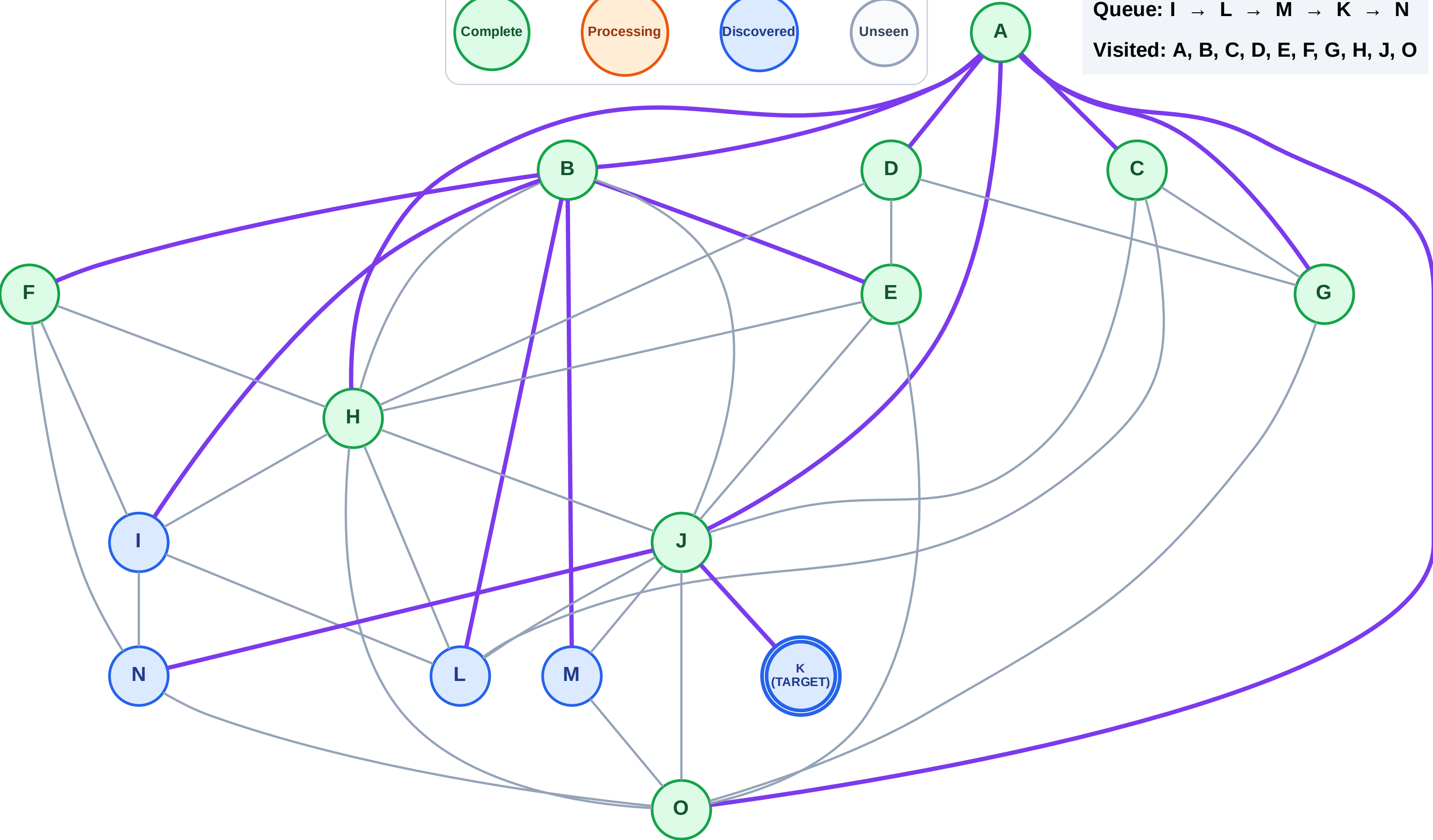
Complete

Processing

Discovered

Unseen

Queue: I → L → M → K → N
Visited: A, B, C, D, E, F, G, H, J, O



BFS Traversal

Step 22: Process node I

Legend

Complete

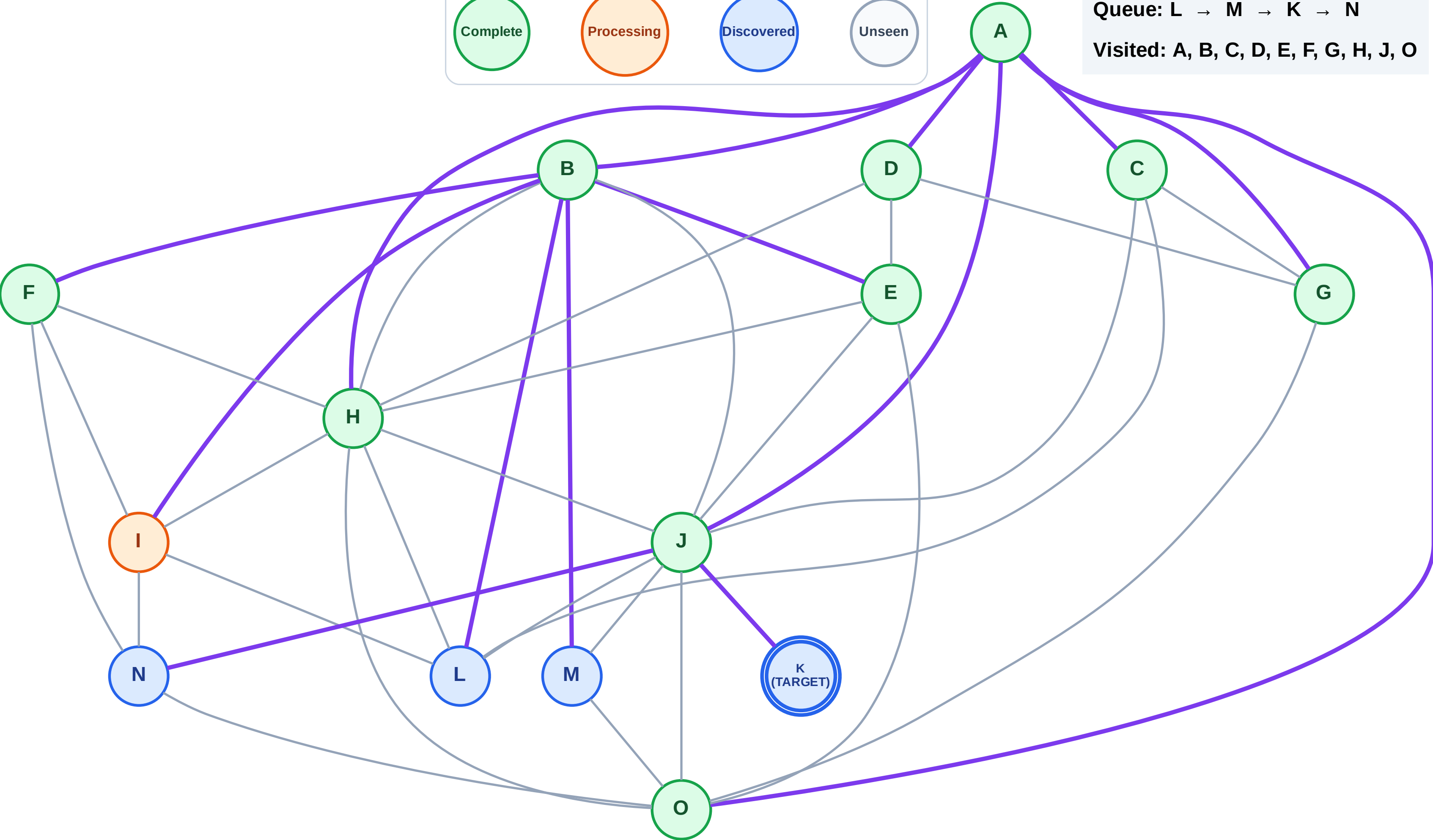
Processing

Discovered

Unseen

Queue: L → M → K → N

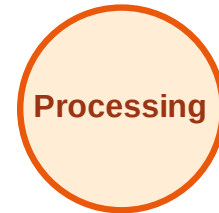
Visited: A, B, C, D, E, F, G, H, J, O



BFS Traversal

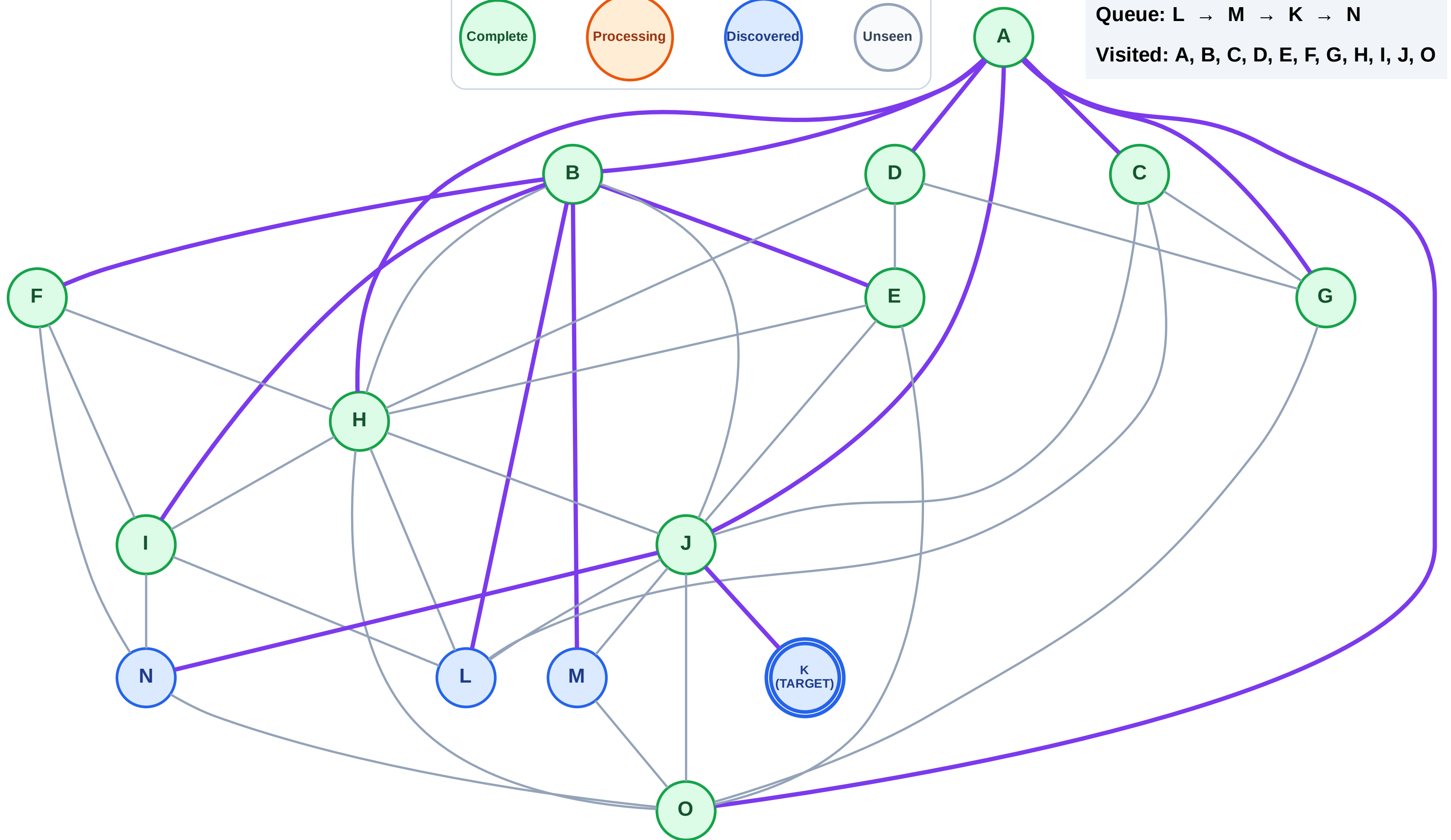
Step 23: No new neighbors from I

Legend



Queue: L → M → K → N

Visited: A, B, C, D, E, F, G, H, I, J, O



BFS Traversal

Step 24: Process node L

Legend

Complete

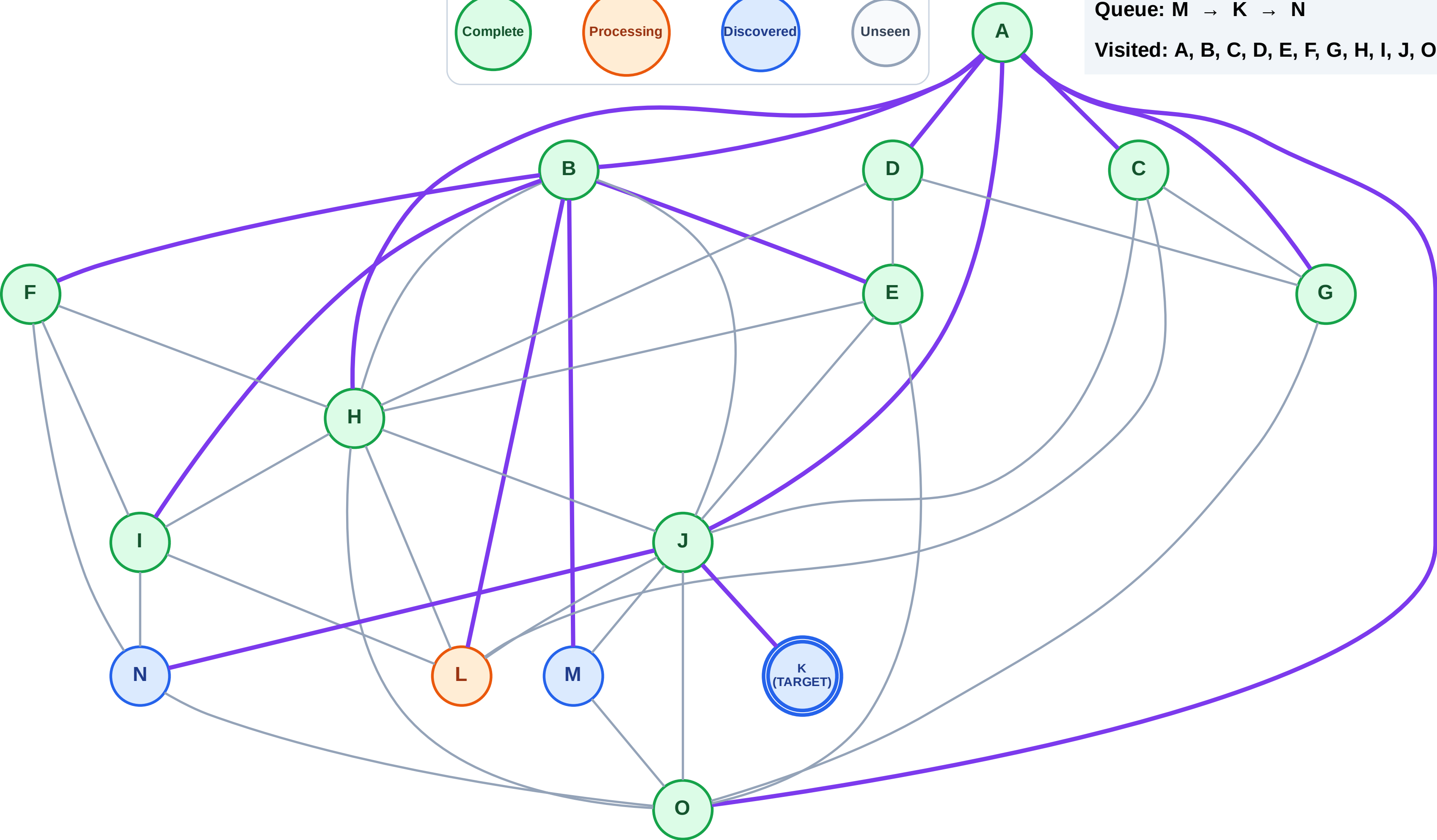
Processing

Discovered

Unseen

Queue: M → K → N

Visited: A, B, C, D, E, F, G, H, I, J, O



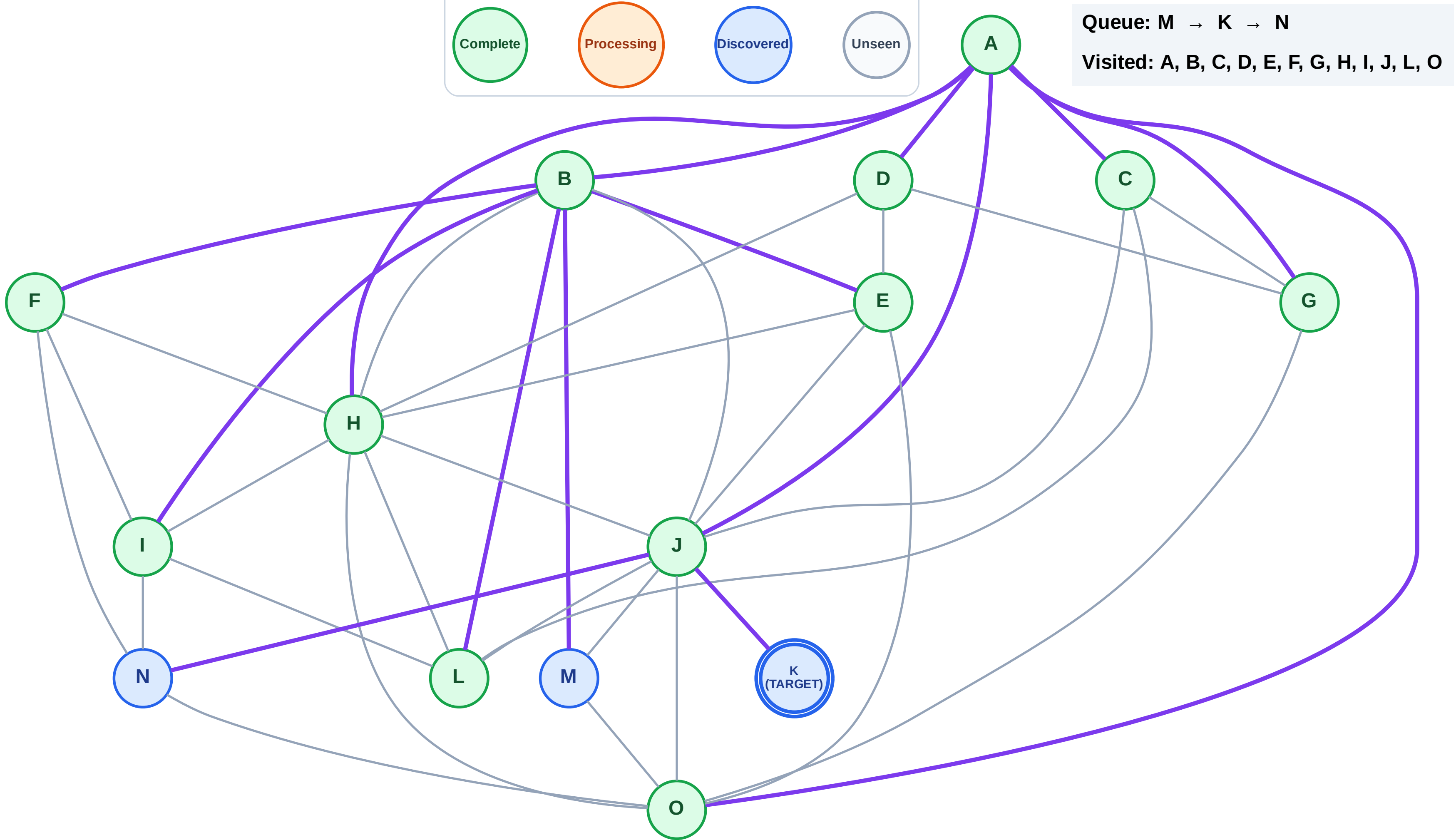
Step 25: No new neighbors from L

Legend

Processing

Unseen

Visited: A, B, C, D, E, F, G, H, I, J, L, O



BFS Traversal

Step 26: Process node M

Legend

Complete

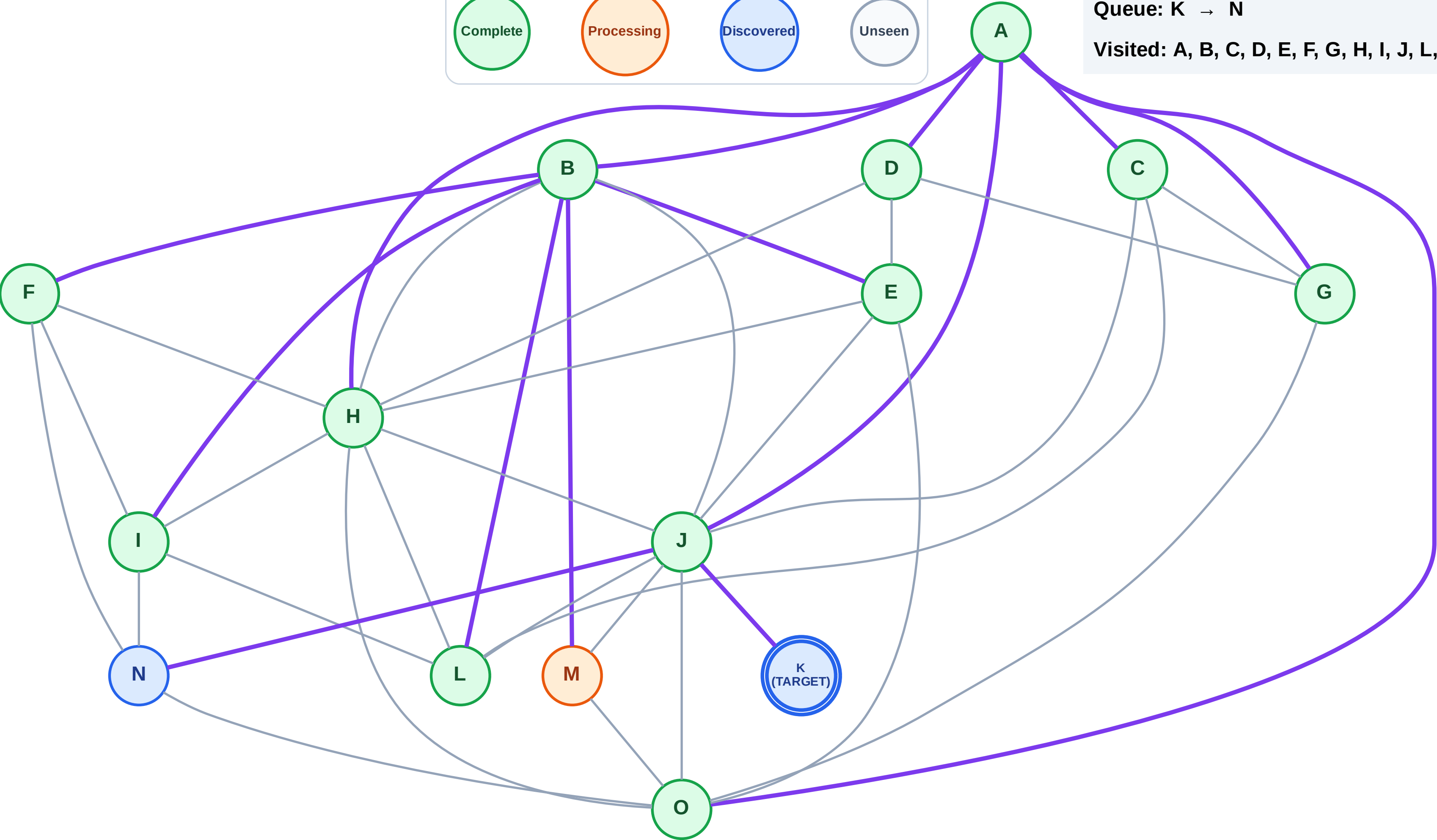
Processing

Discovered

Unseen

Queue: K → N

Visited: A, B, C, D, E, F, G, H, I, J, L, O



Step 27: No new neighbors from M

Legend

Complete

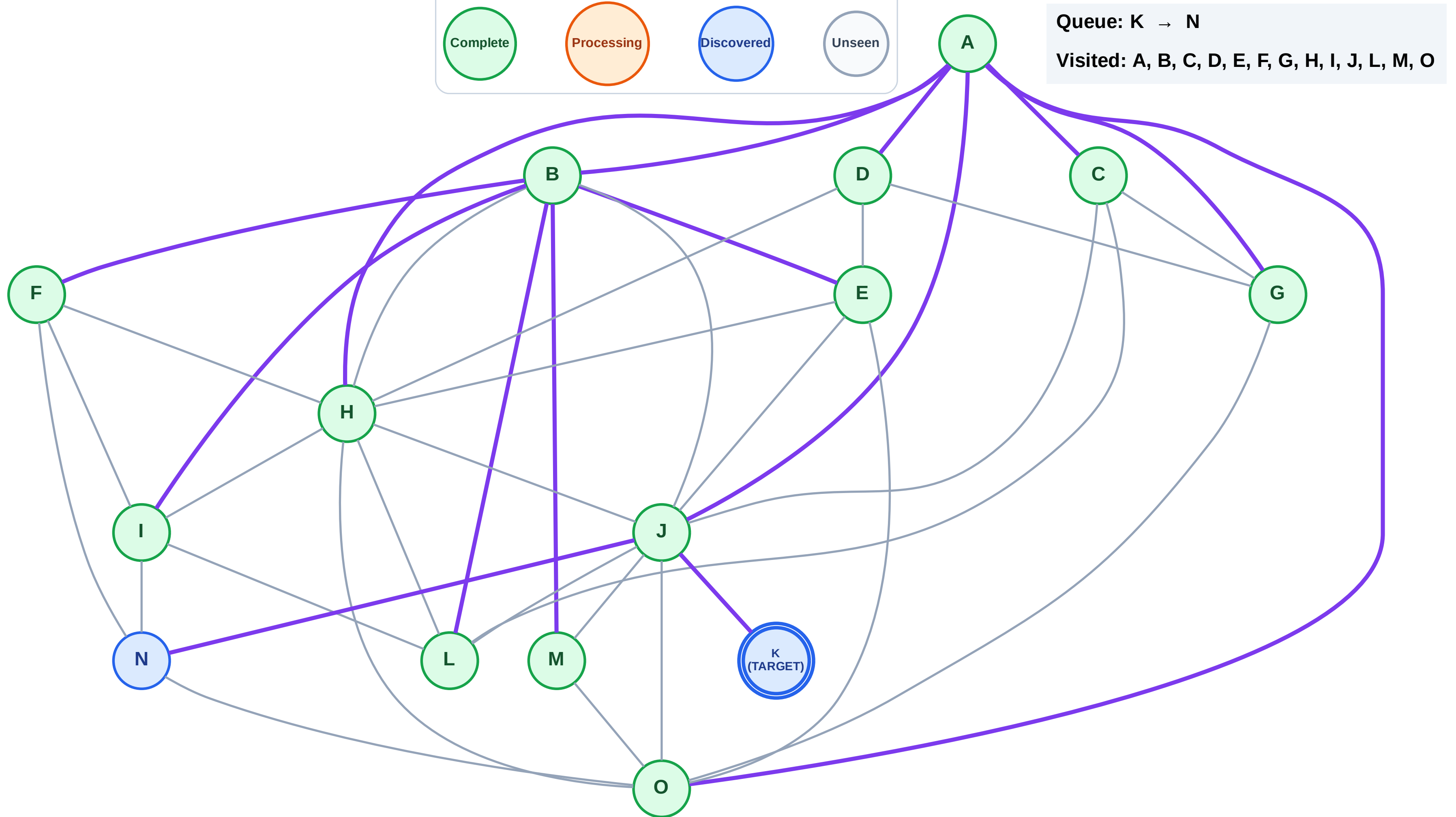
Processing

Discovered

Unseen

Queue: $K \rightarrow N$

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



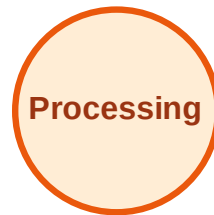
BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



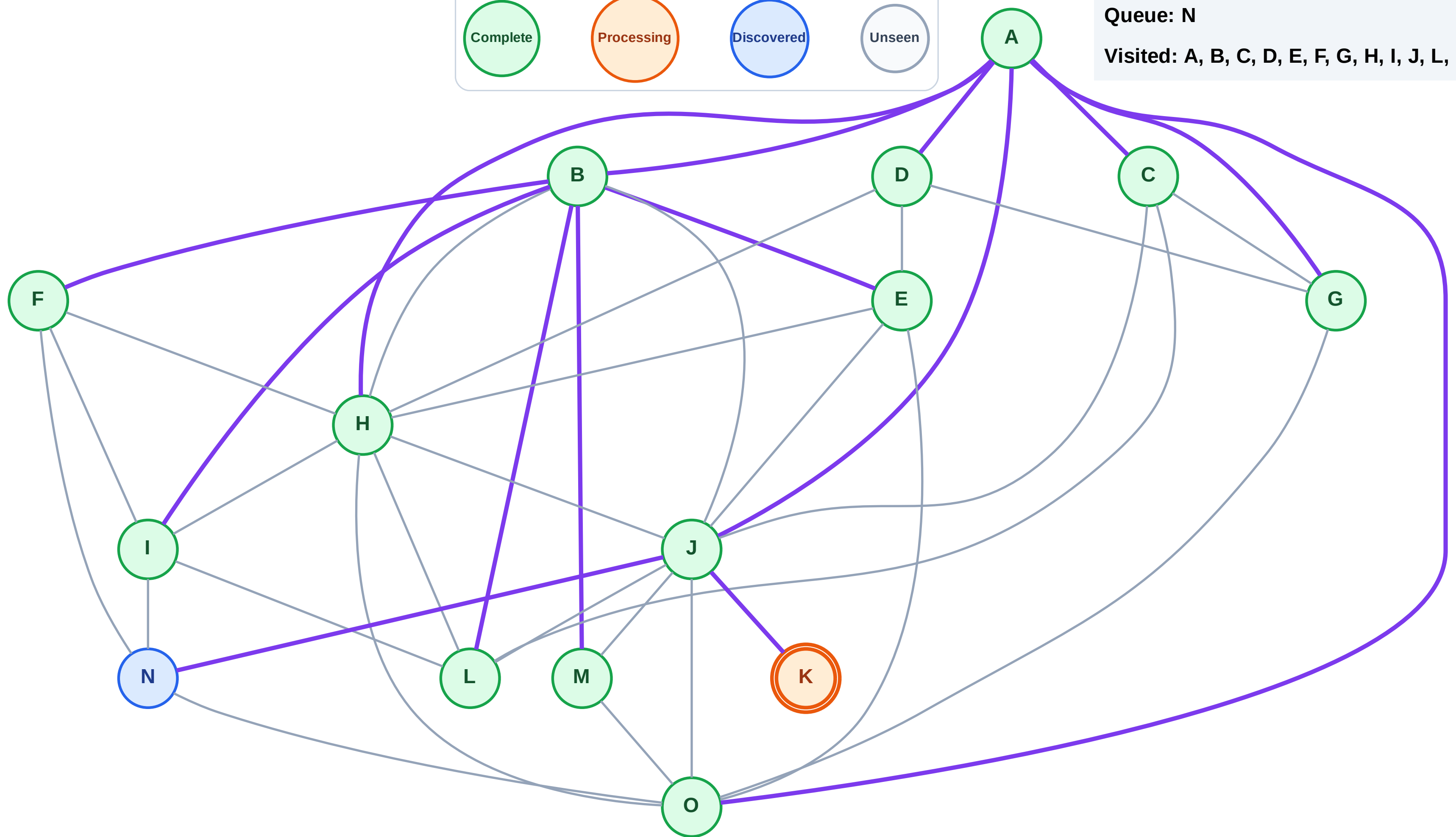
Discovered



Unseen

Queue: N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → J → K

Legend

Complete

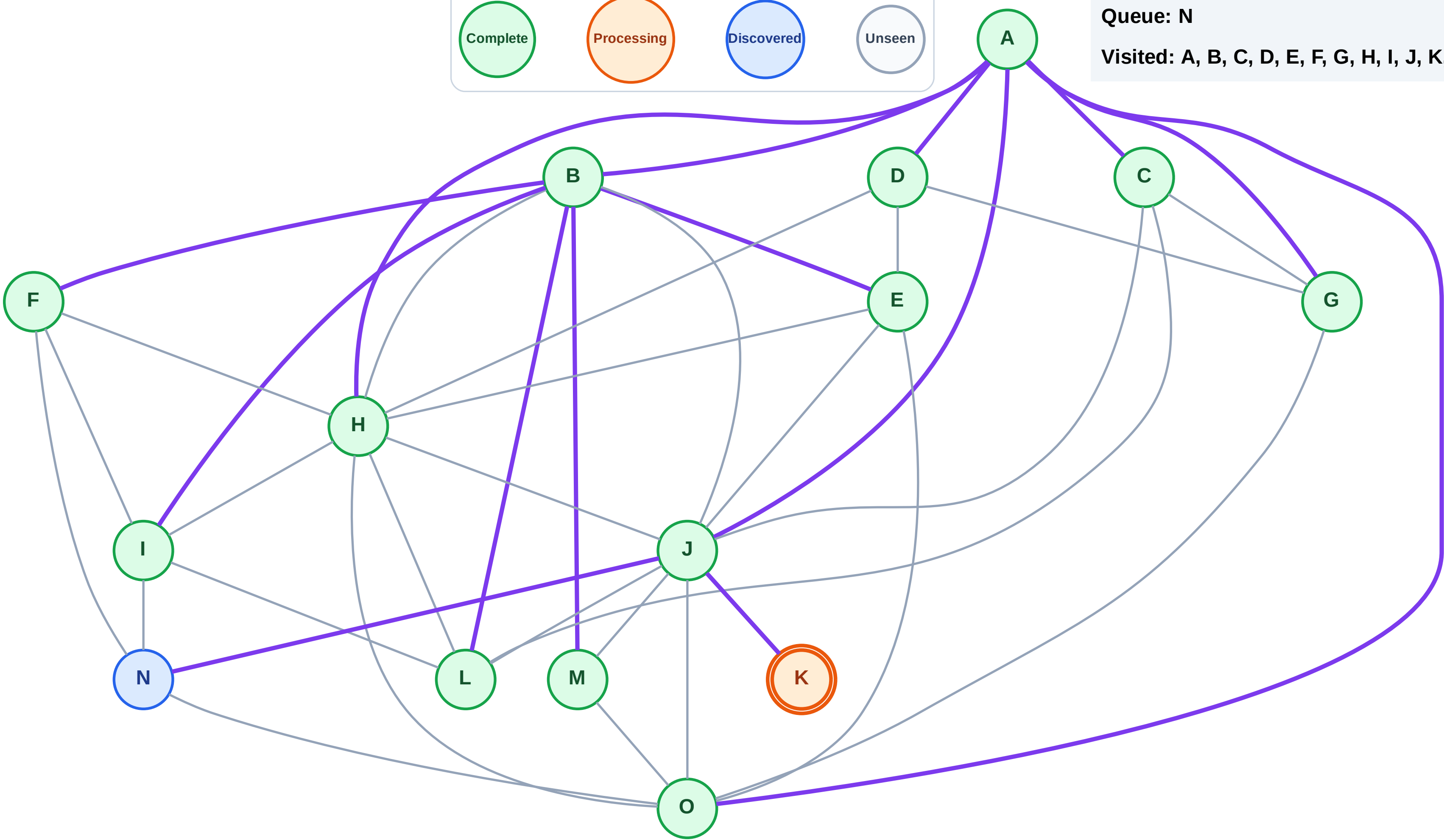
Processing

Discovered

Unseen

Queue: N

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

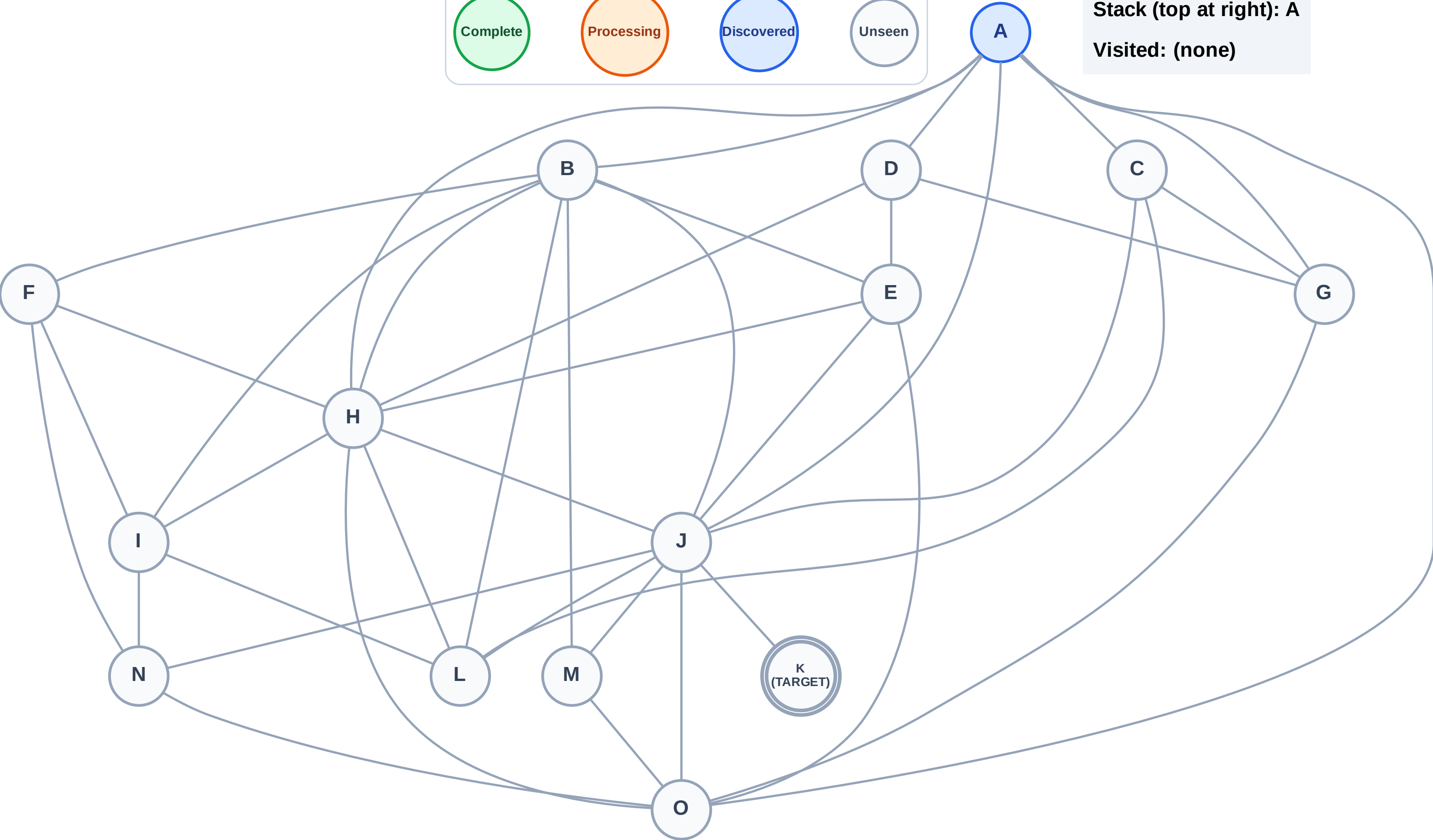
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

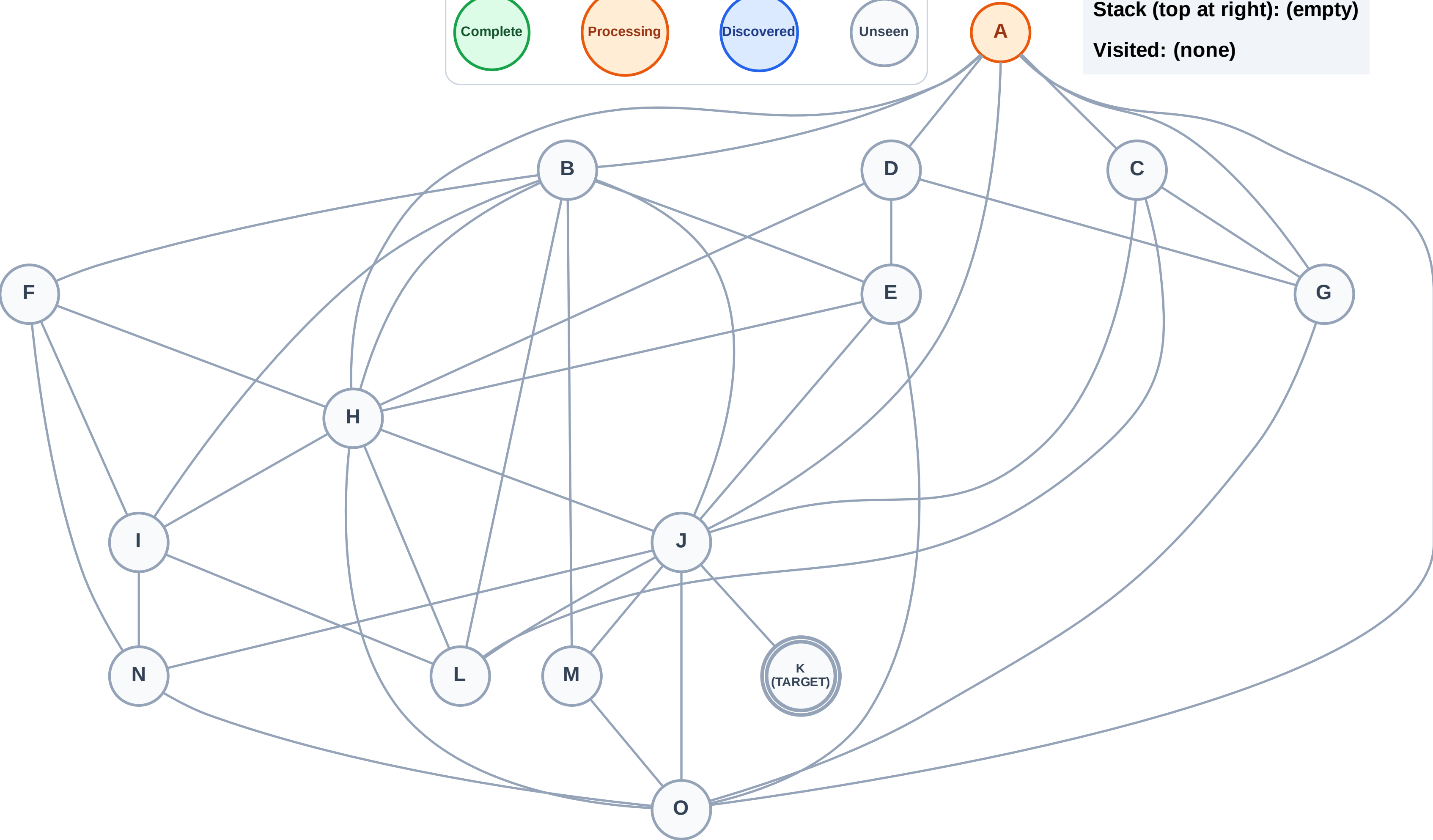
Complete

Processing

Discovered

Unseen

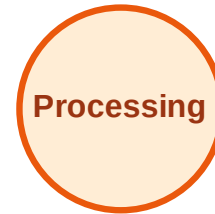
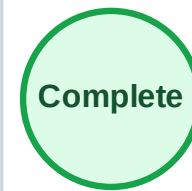
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

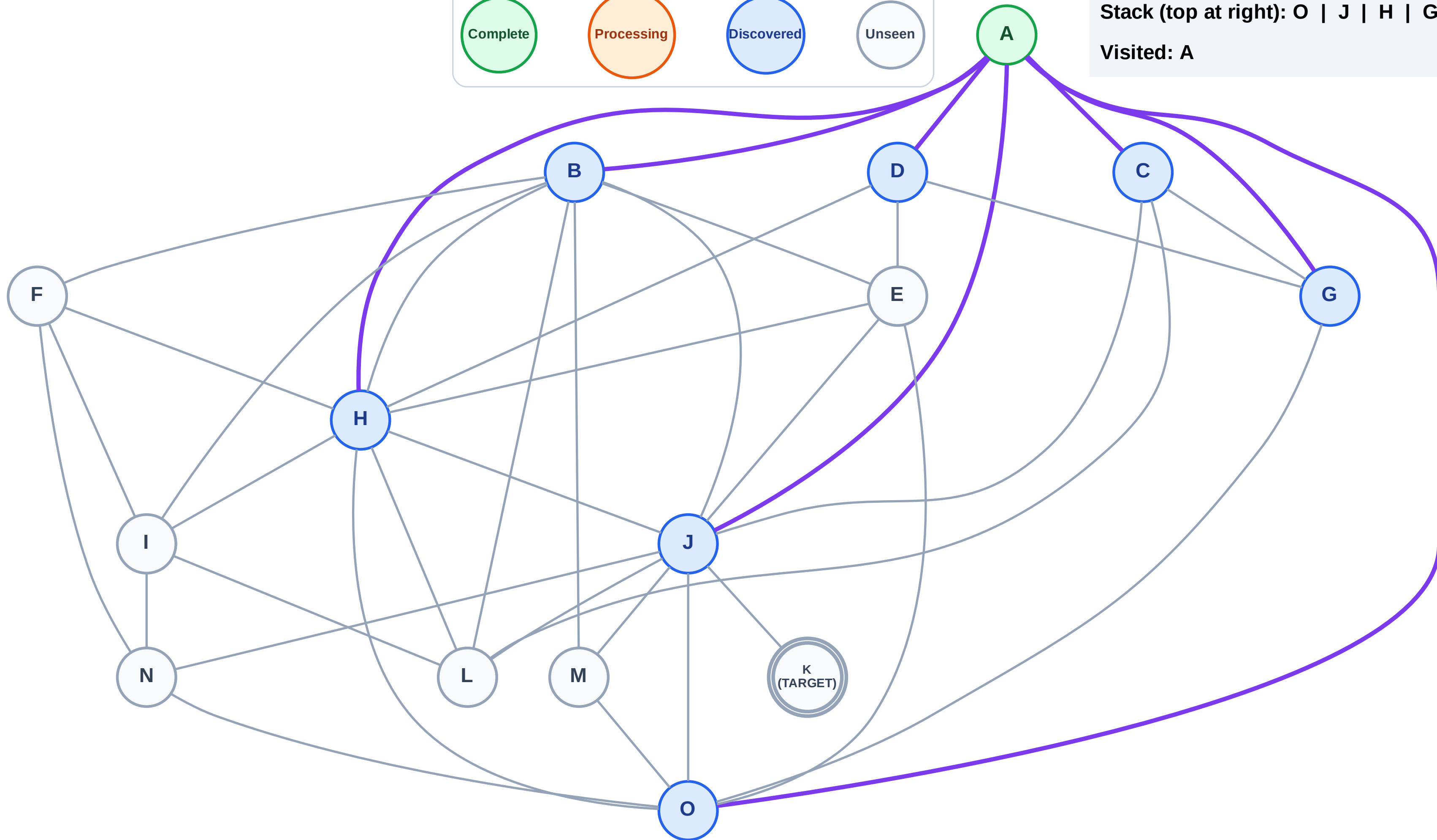
Step 3: Discover O, J, H, G, D, C, B from A

Legend



Stack (top at right): O | J | H | G | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

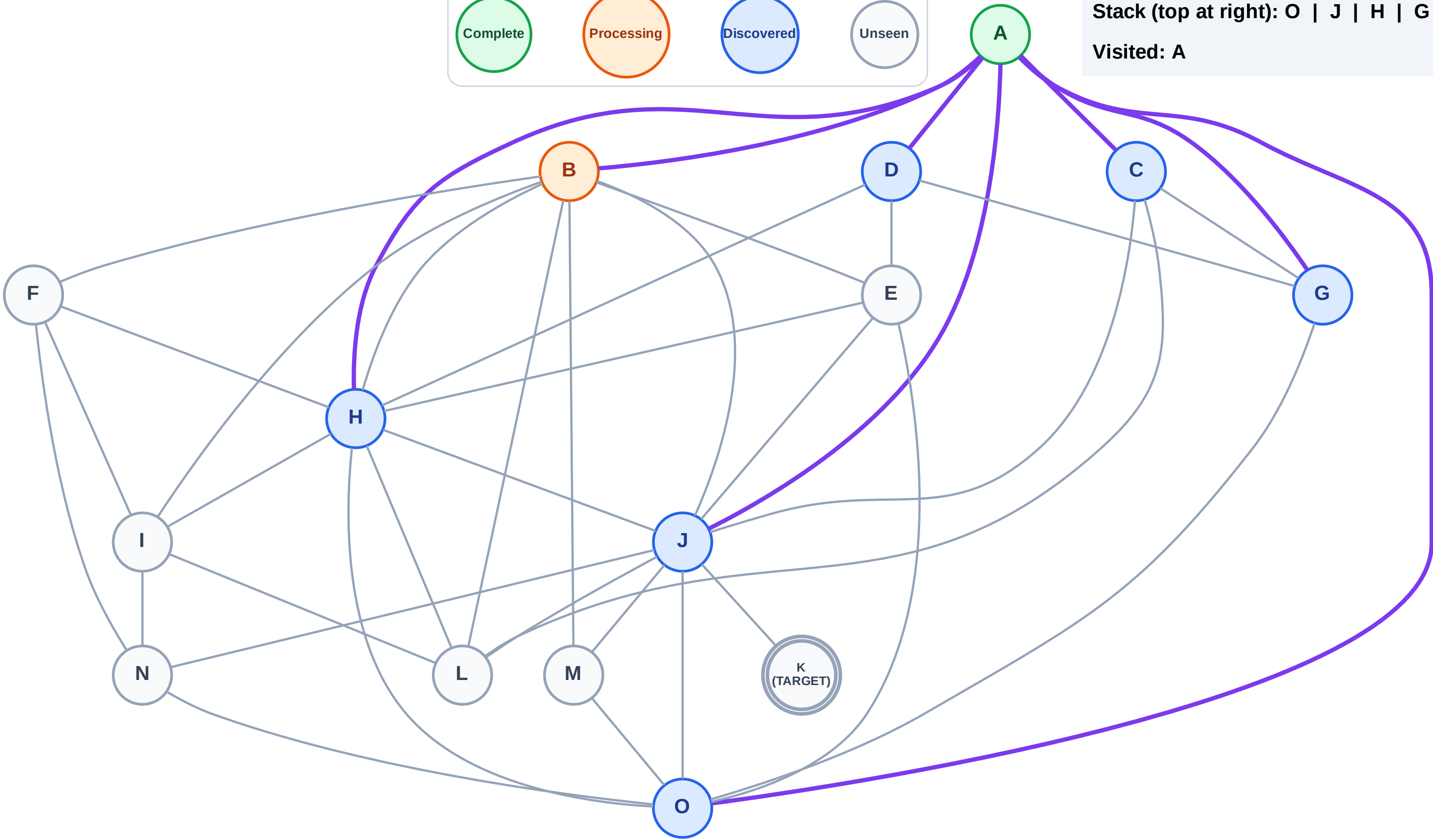
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C

Visited: A



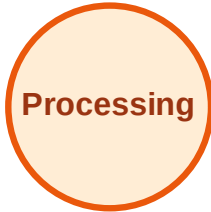
DFS Traversal

Step 5: Discover M, L, I, F, E from B

Legend



Complete



Processing



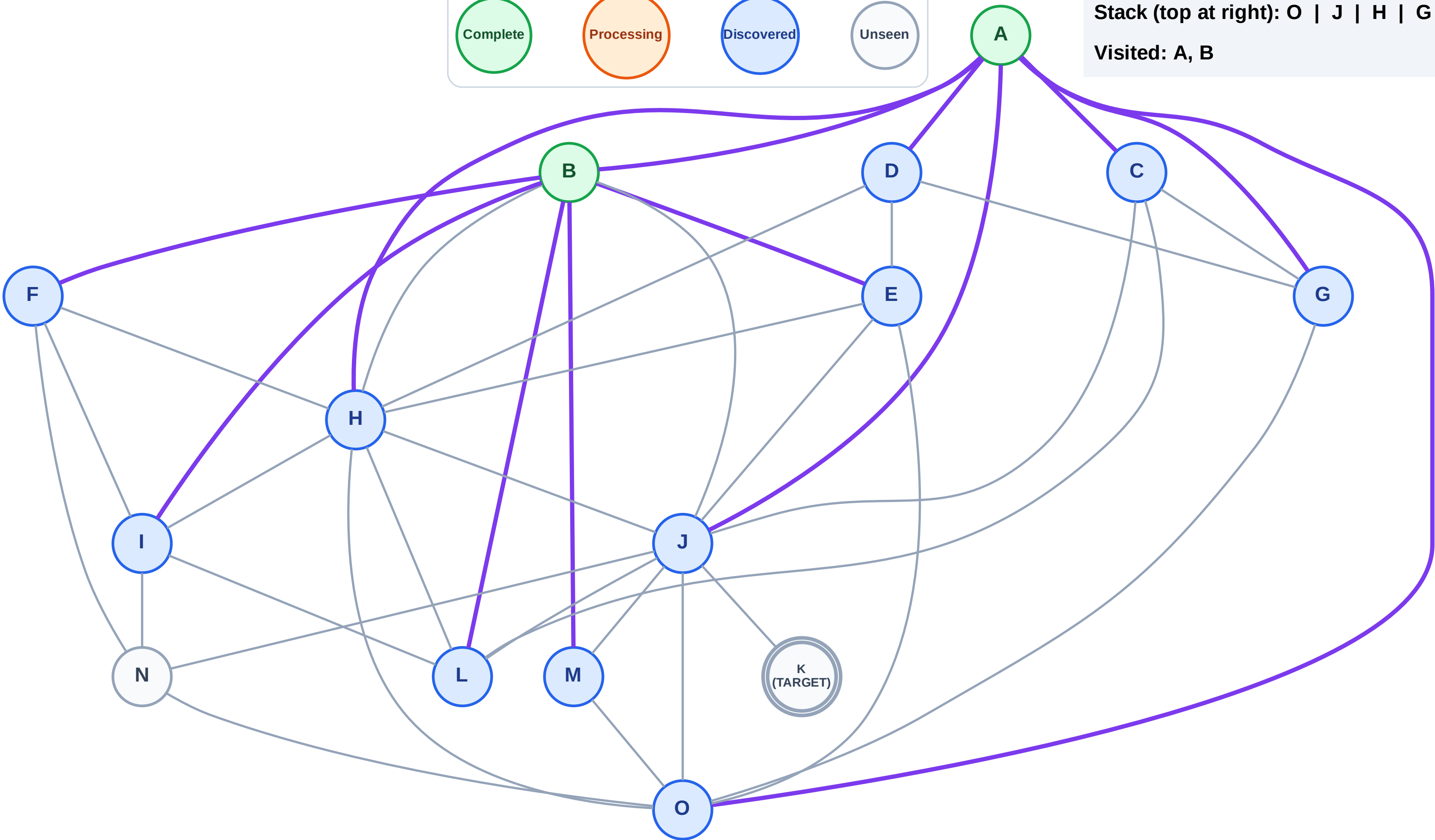
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | F | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



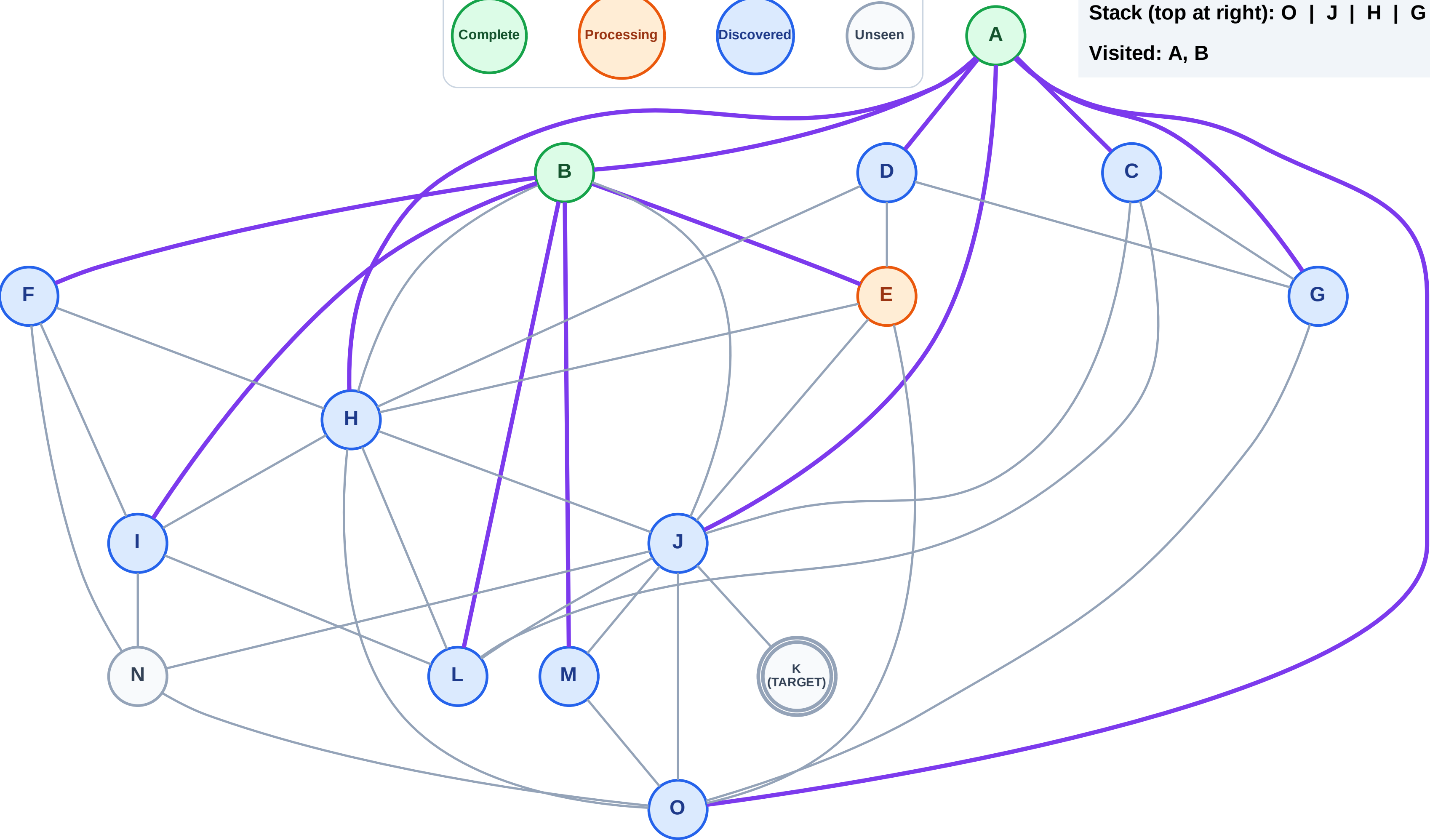
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | F

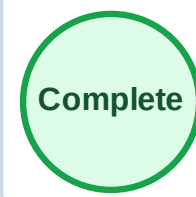
Visited: A, B



DFS Traversal

Step 7: No new neighbors from E

Legend



Complete



Processing



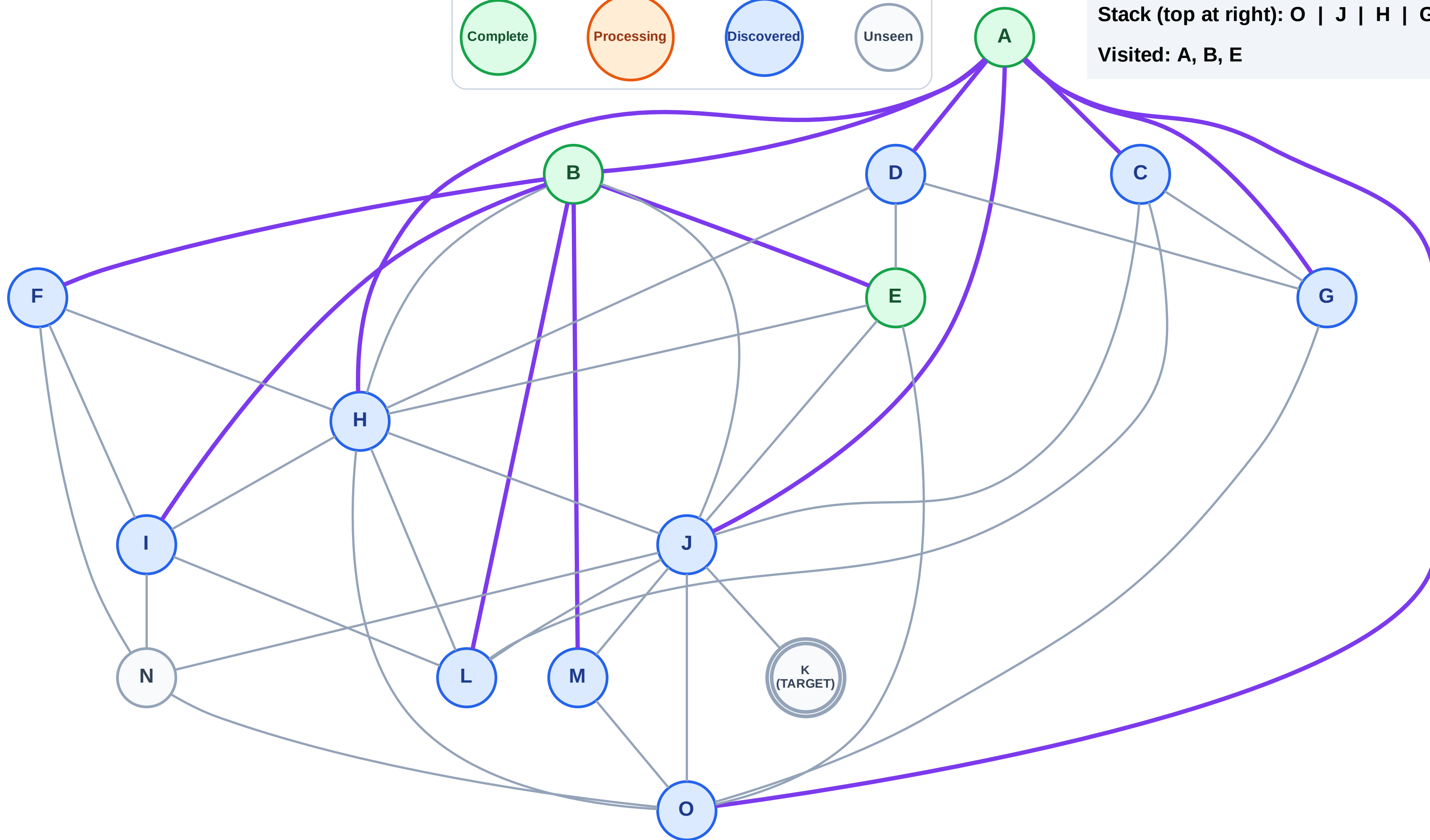
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

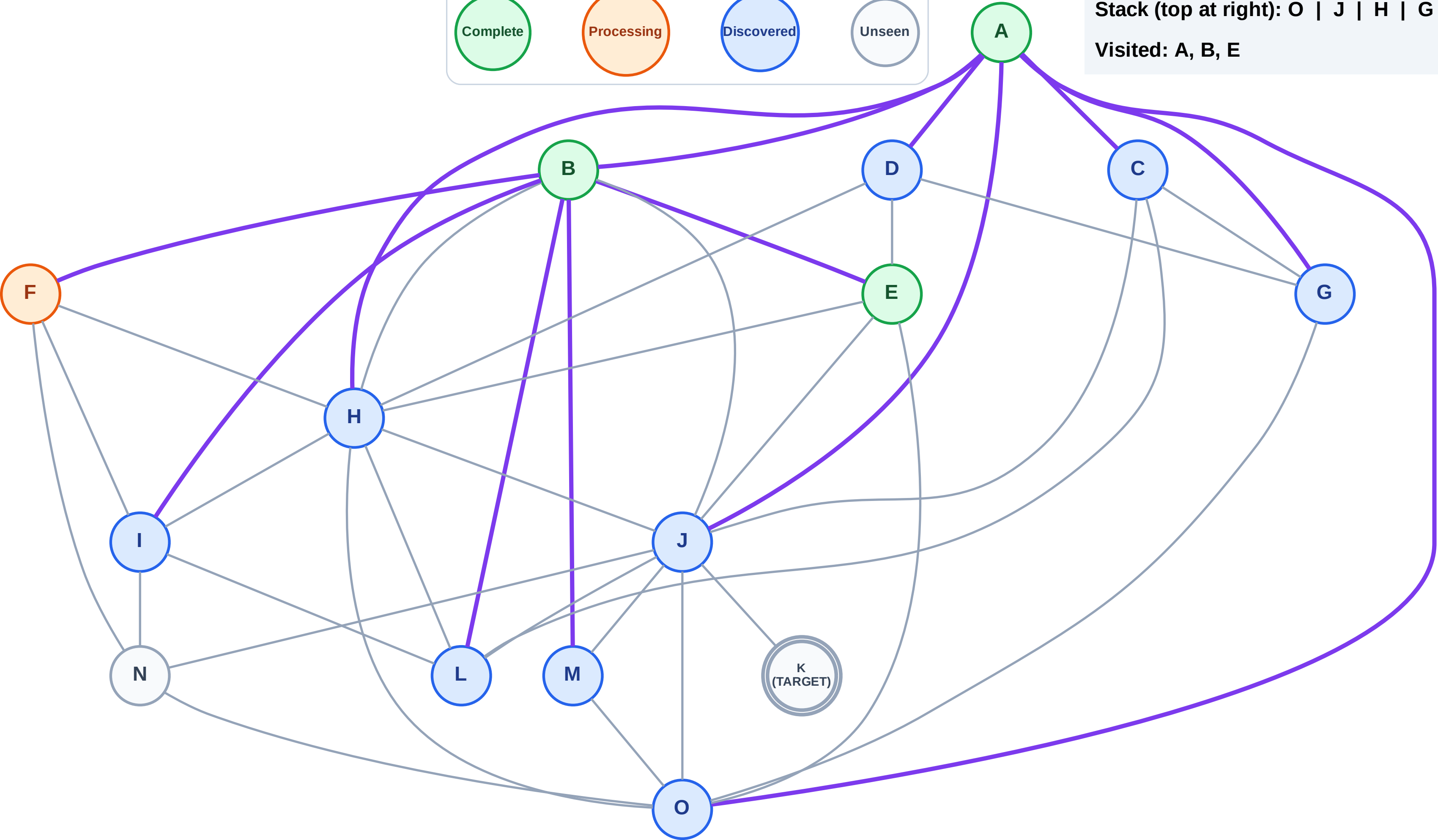
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I

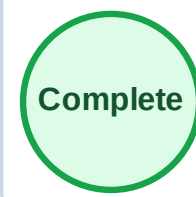
Visited: A, B, E



DFS Traversal

Step 9: Discover N from F

Legend



Complete



Processing



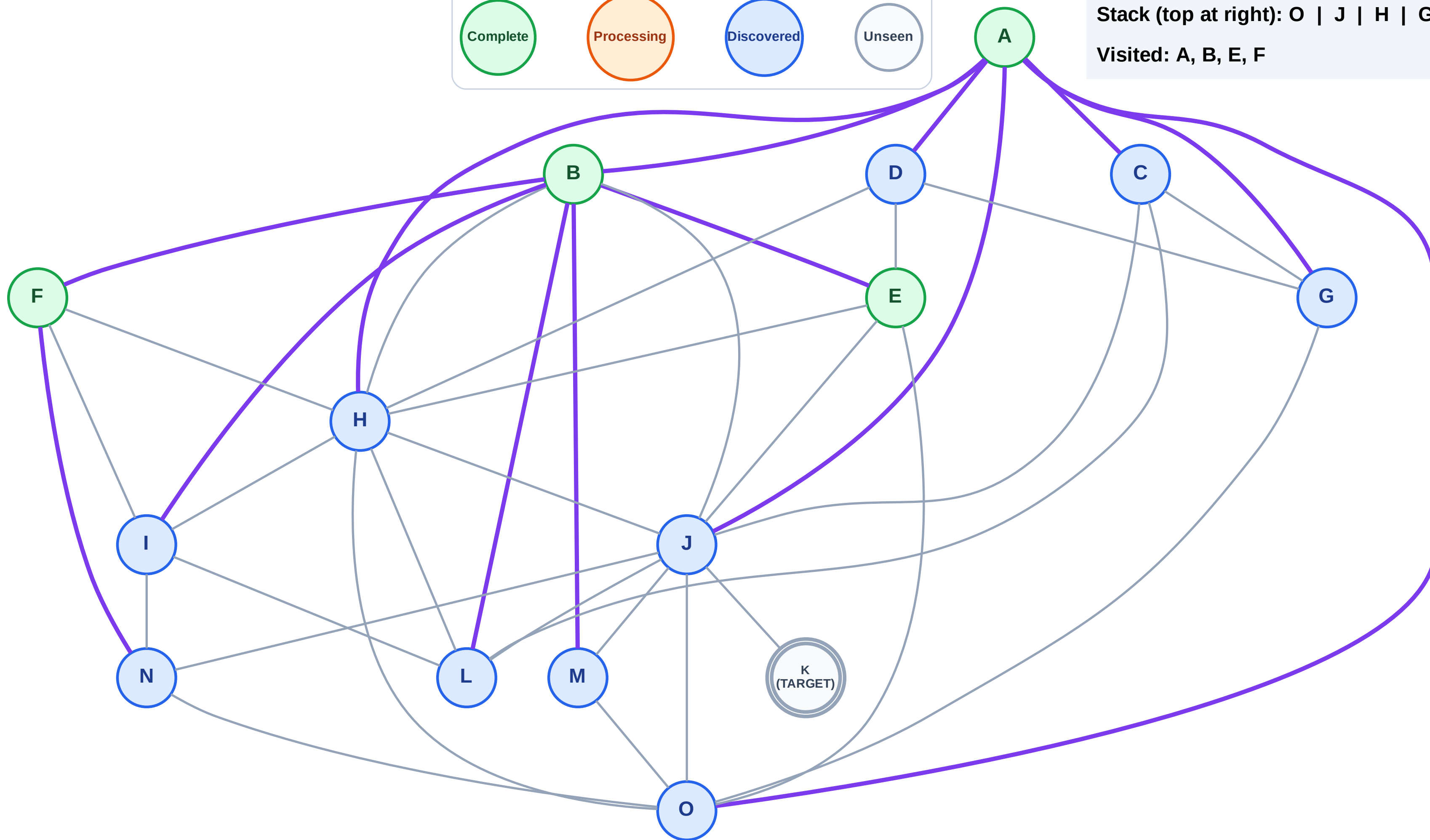
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I | N

Visited: A, B, E, F



DFS Traversal

Step 10: Process node N

Legend

Complete

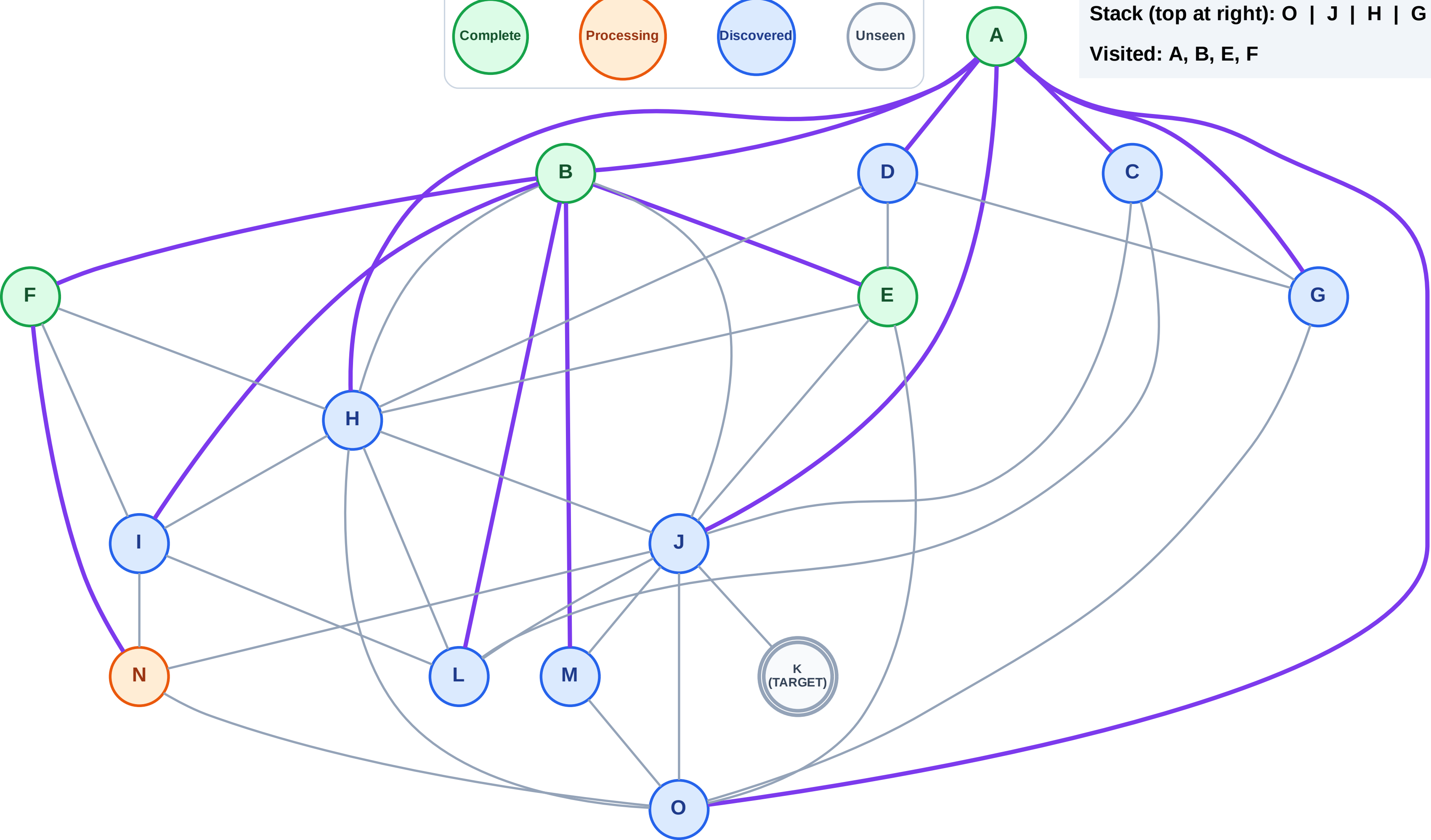
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I

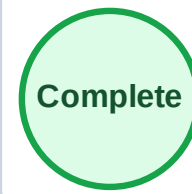
Visited: A, B, E, F



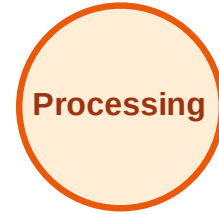
DFS Traversal

Step 11: No new neighbors from N

Legend



Complete



Processing



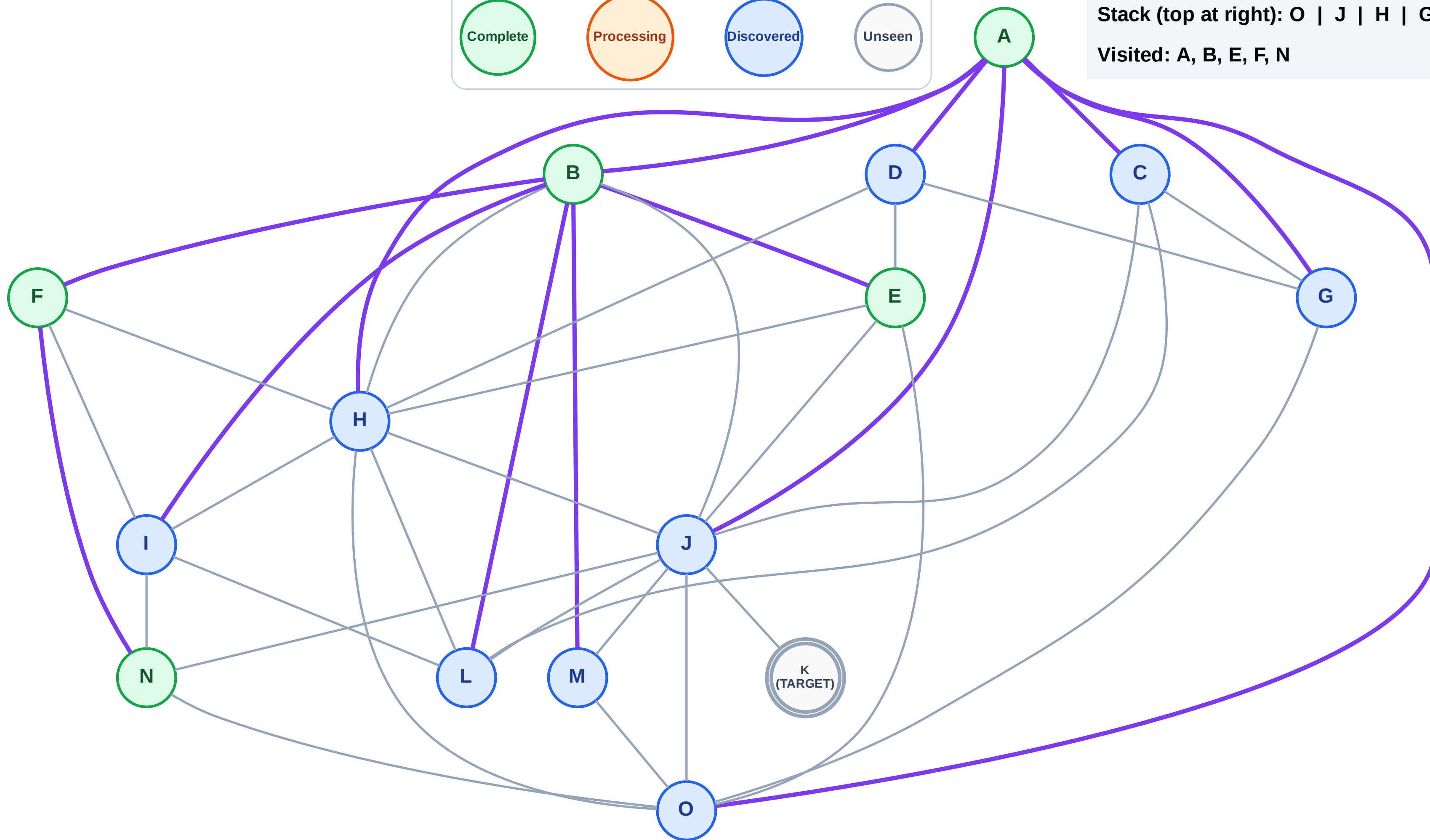
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L | I

Visited: A, B, E, F, N



DFS Traversal

Step 12: Process node I

Legend

Complete

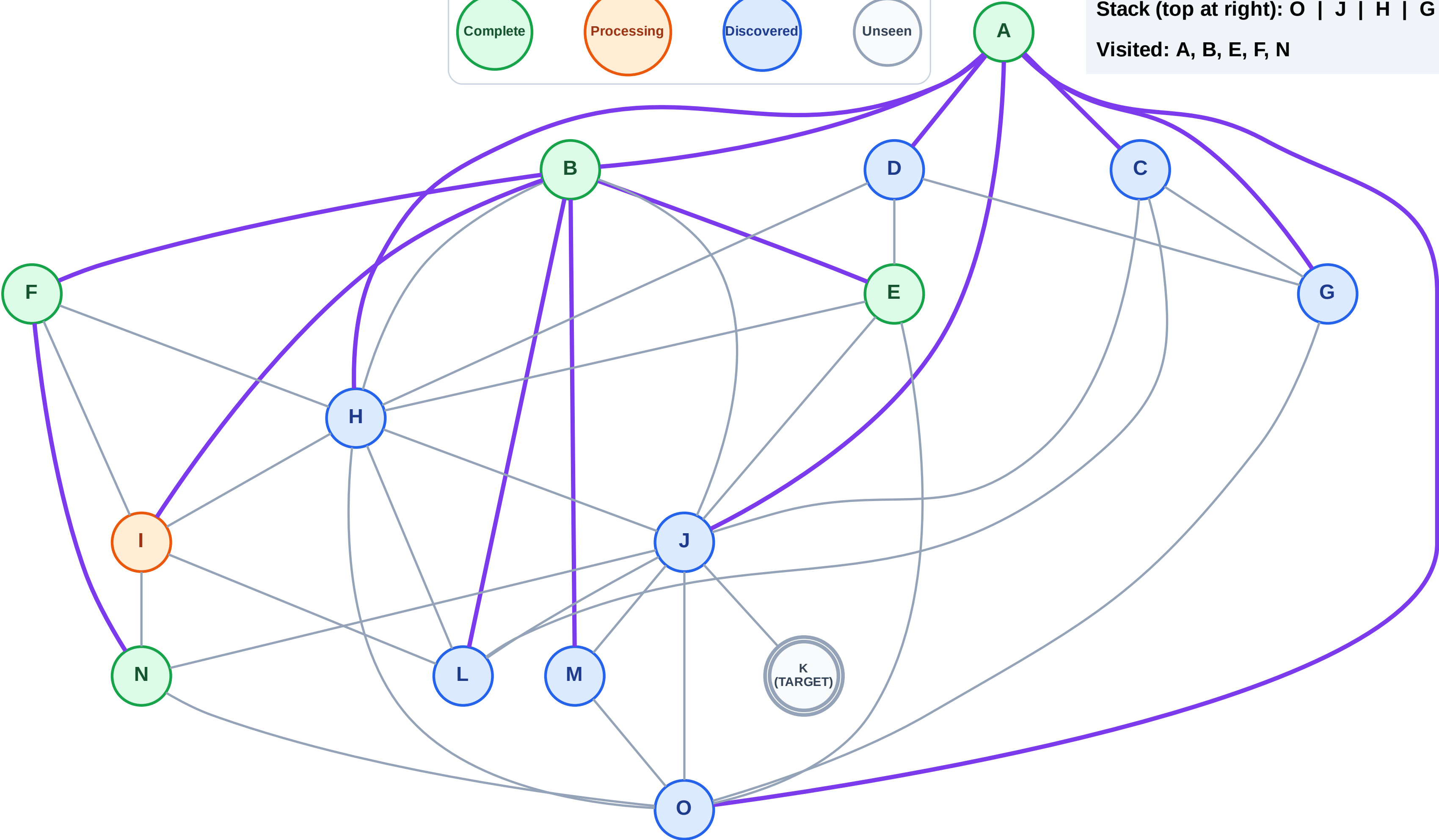
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D | C | M | L

Visited: A, B, E, F, N



DFS Traversal

Step 13: No new neighbors from I

Legend



Complete



Processing



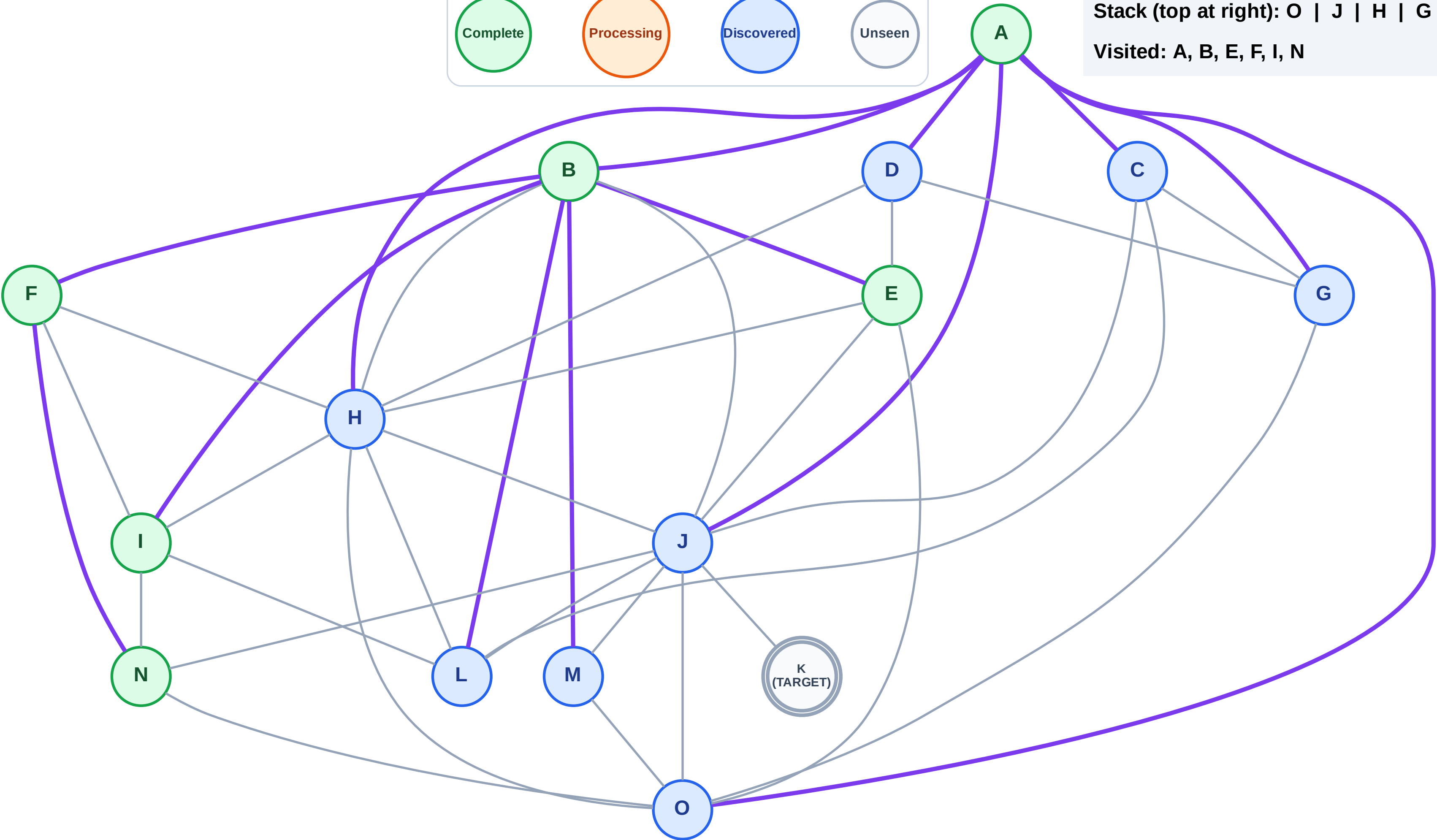
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M | L

Visited: A, B, E, F, I, N



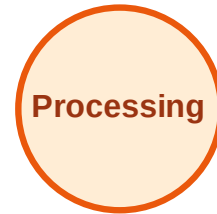
DFS Traversal

Step 14: Process node L

Legend



Complete



Processing



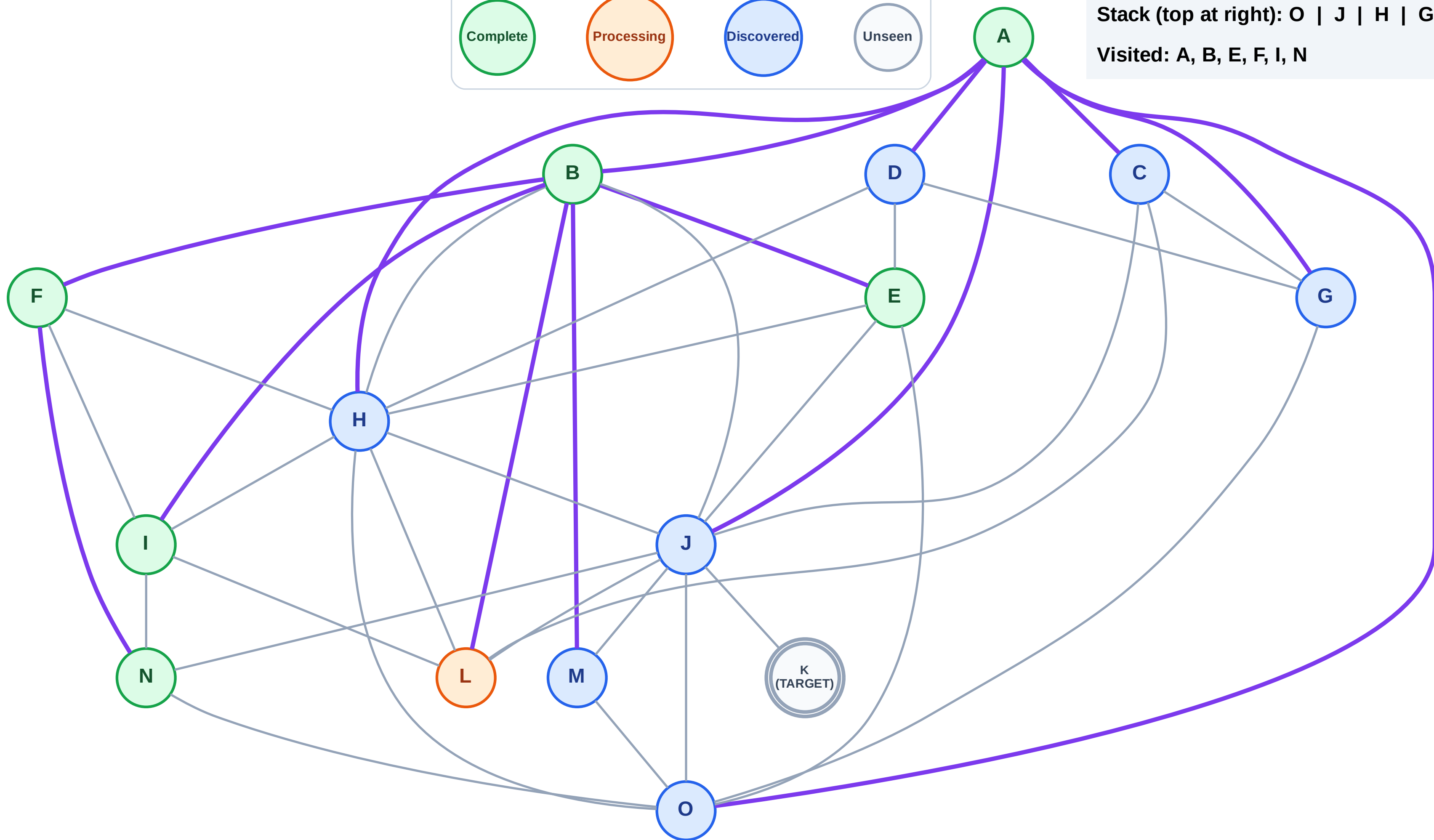
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C | M

Visited: A, B, E, F, I, N



DFS Traversal

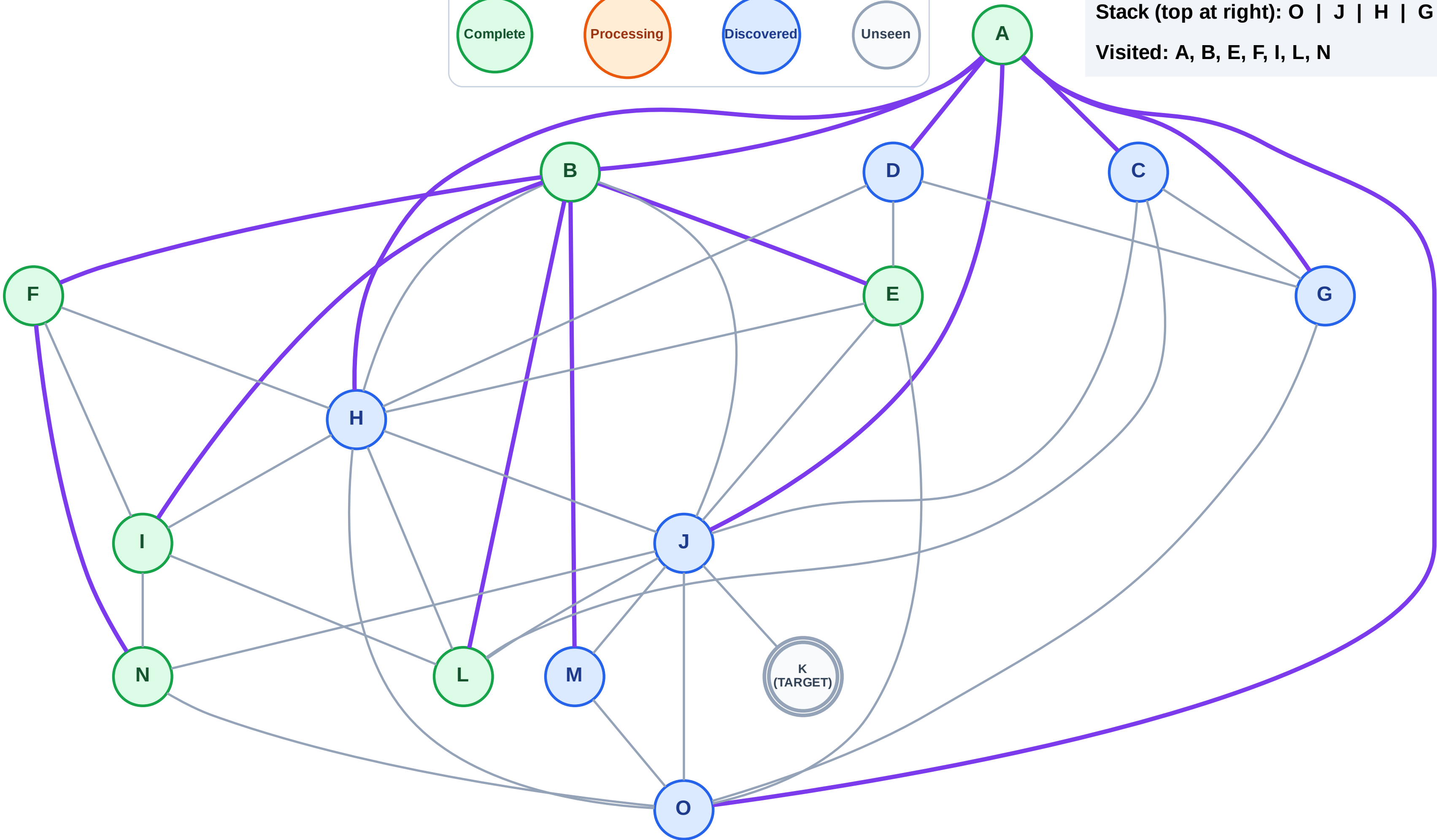
Step 15: No new neighbors from L

Legend



Stack (top at right): O | J | H | G | D | C | M

Visited: A, B, E, F, I, L, N



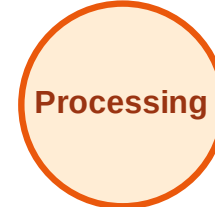
DFS Traversal

Step 16: Process node M

Legend



Complete



Processing



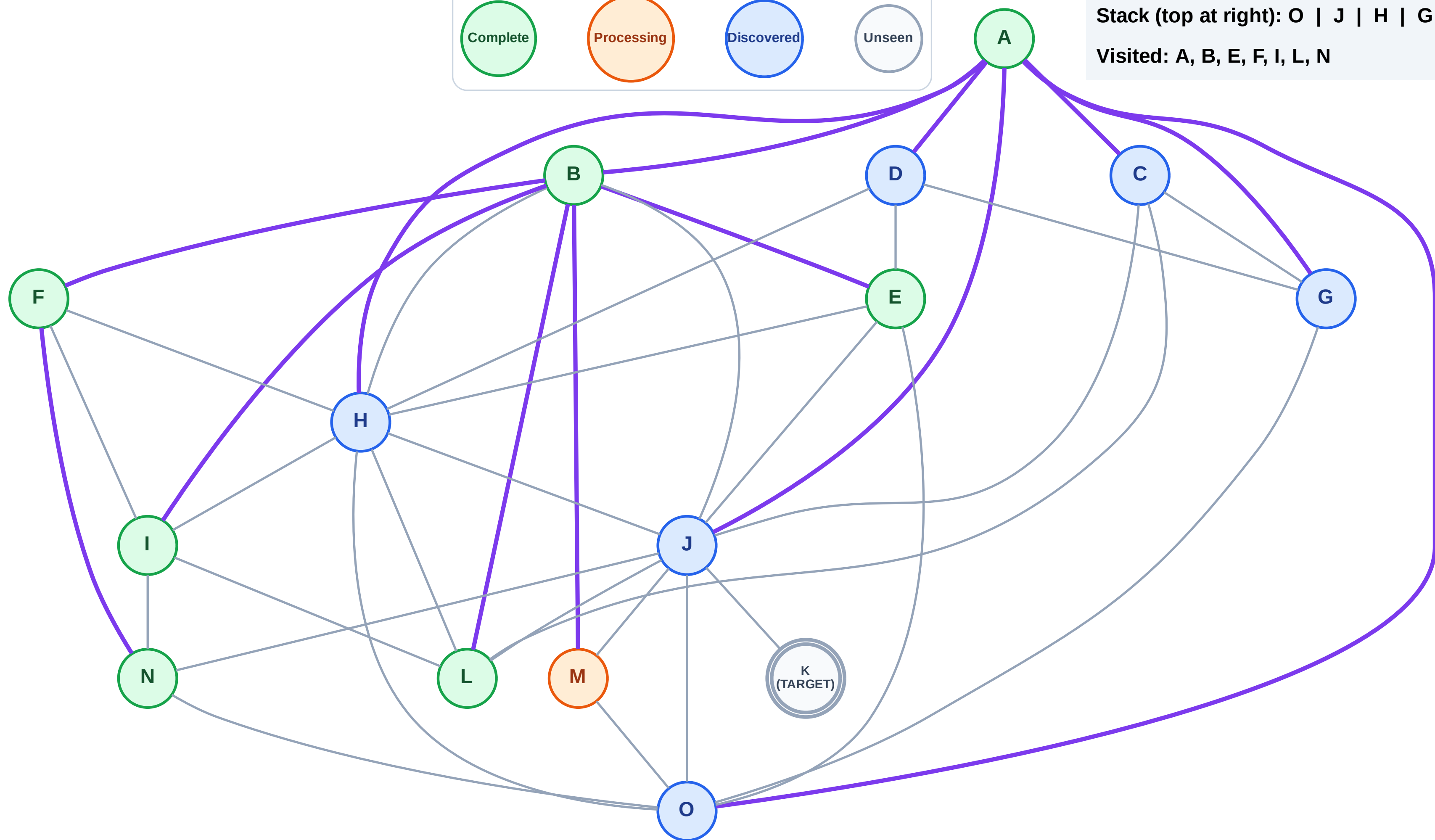
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C

Visited: A, B, E, F, I, L, N



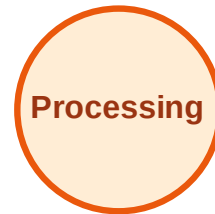
DFS Traversal

Step 17: No new neighbors from M

Legend



Complete



Processing



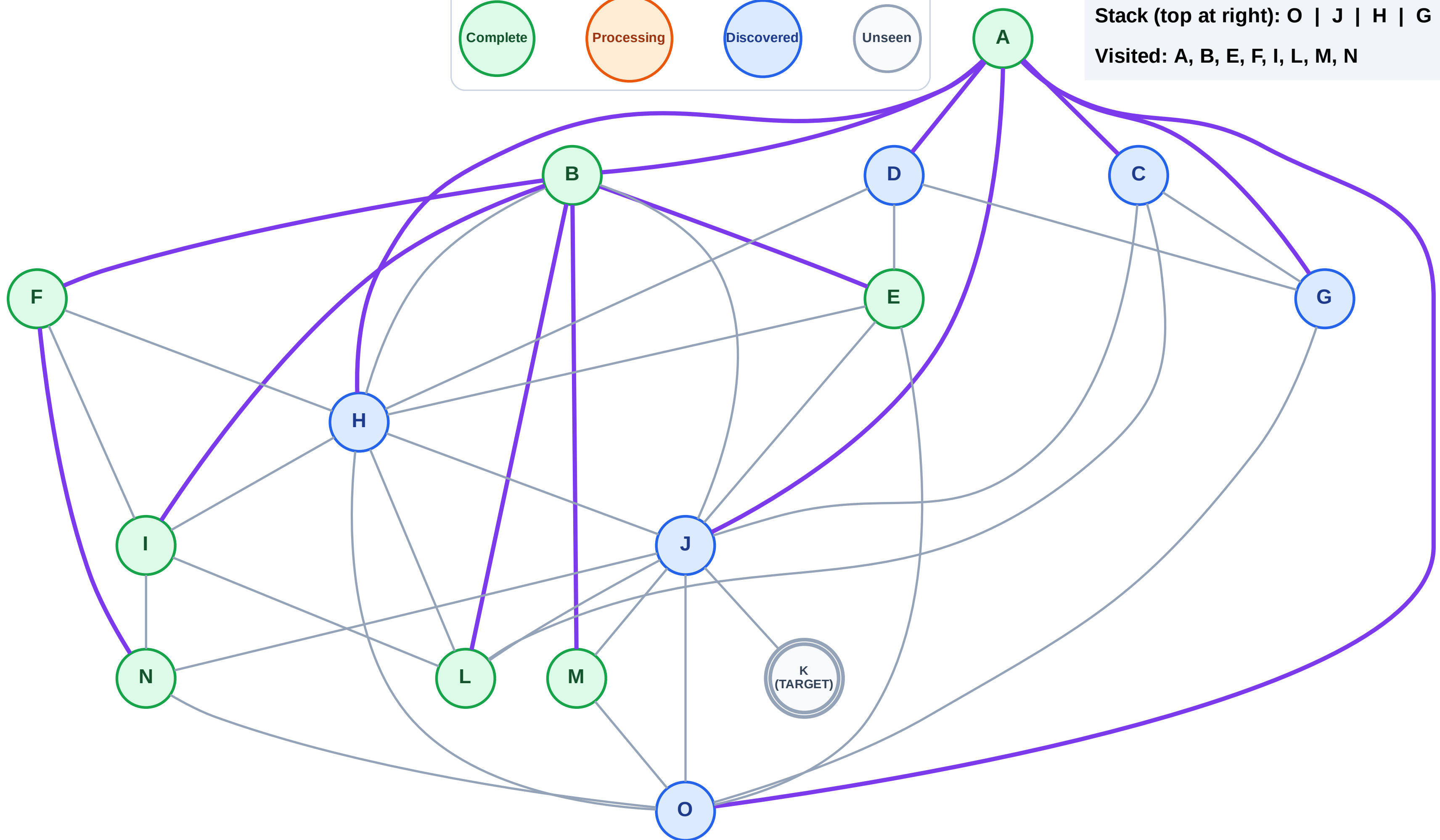
Discovered



Unseen

Stack (top at right): O | J | H | G | D | C

Visited: A, B, E, F, I, L, M, N



DFS Traversal

Step 18: Process node C

Legend

Complete

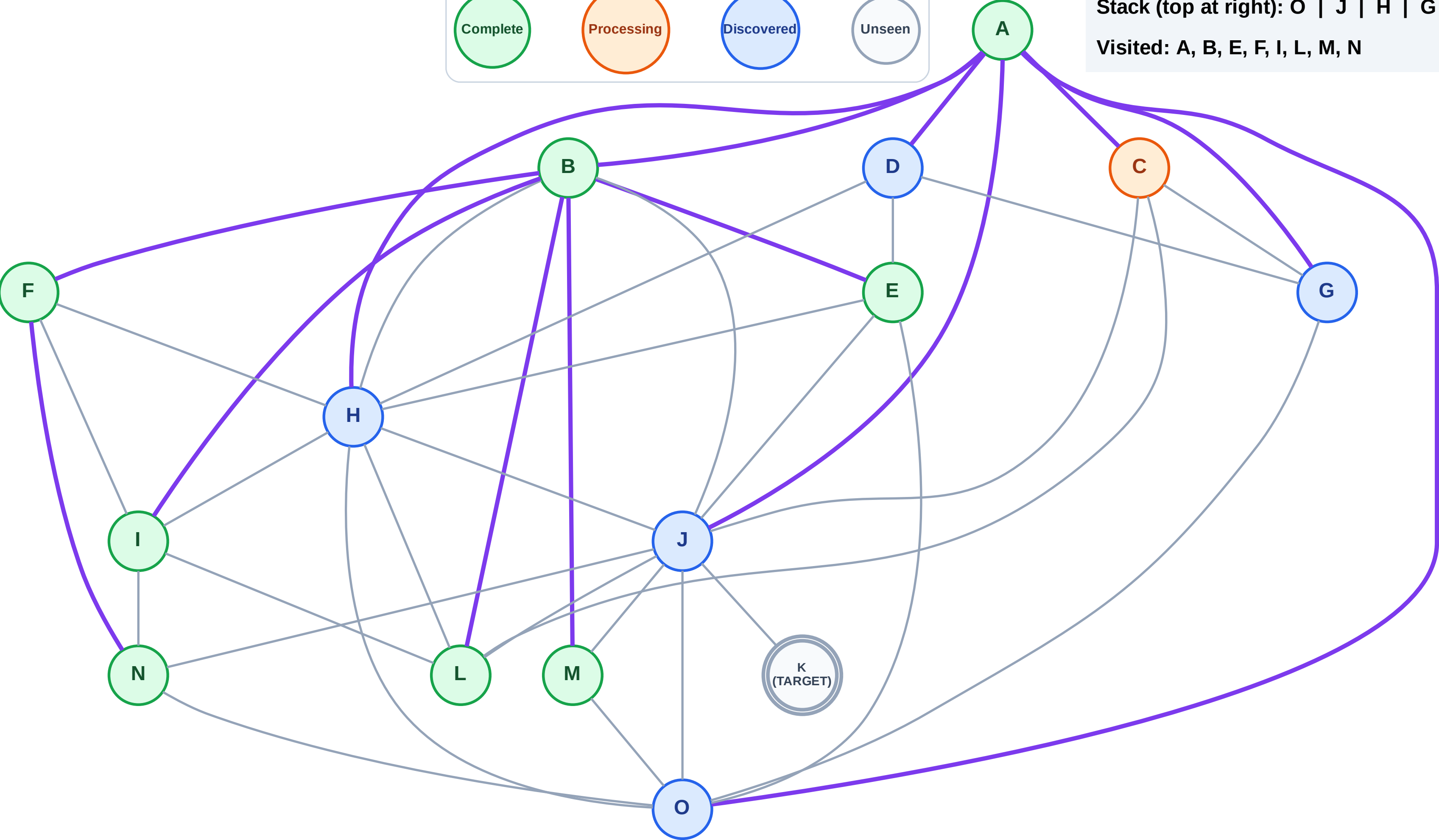
Processing

Discovered

Unseen

Stack (top at right): O | J | H | G | D

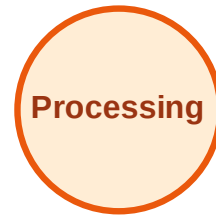
Visited: A, B, E, F, I, L, M, N



DFS Traversal

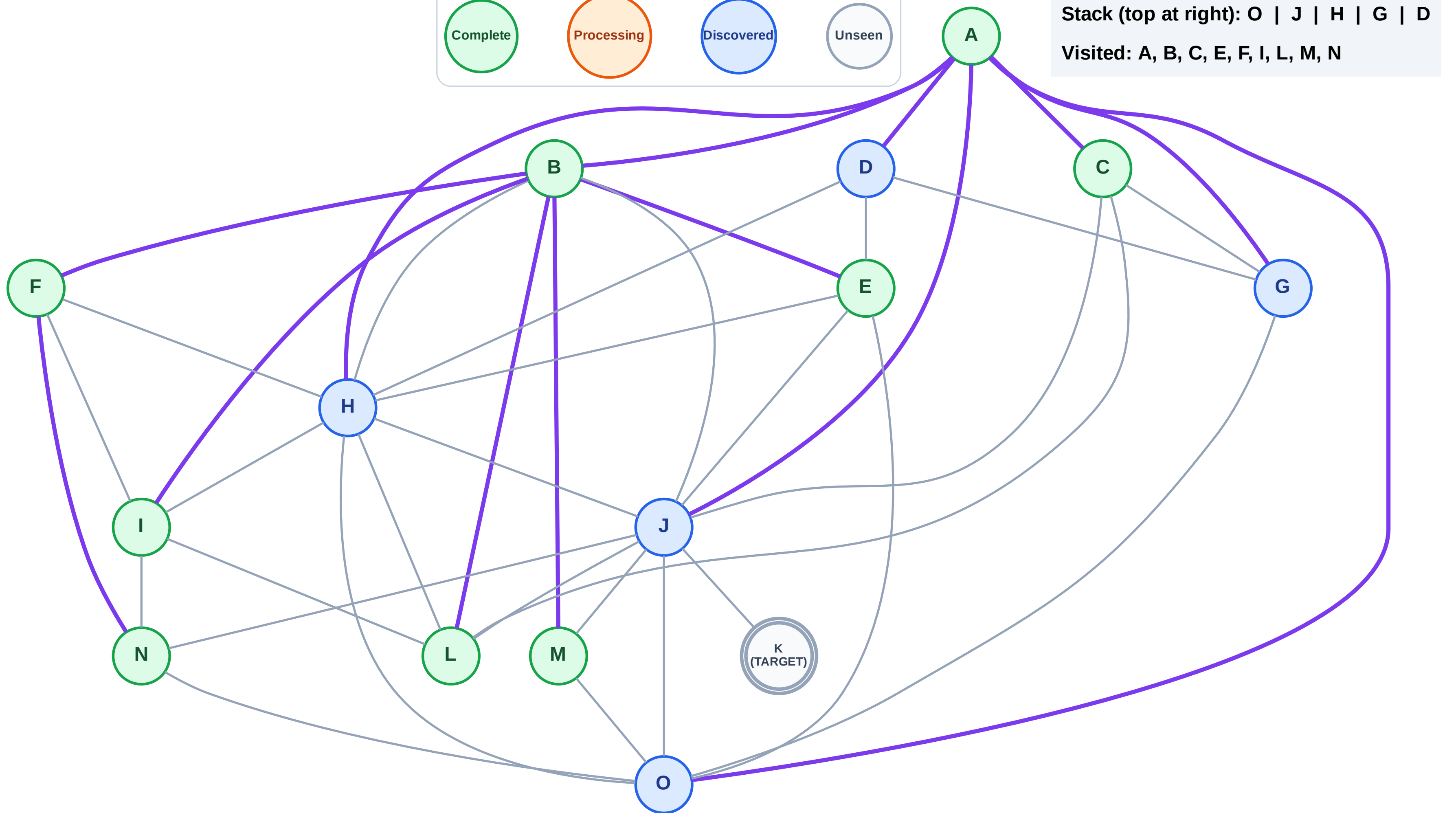
Step 19: No new neighbors from C

Legend



Stack (top at right): O | J | H | G | D

Visited: A, B, C, E, F, I, L, M, N



Step 20: Process node D

Step 20: Process node D

Legend

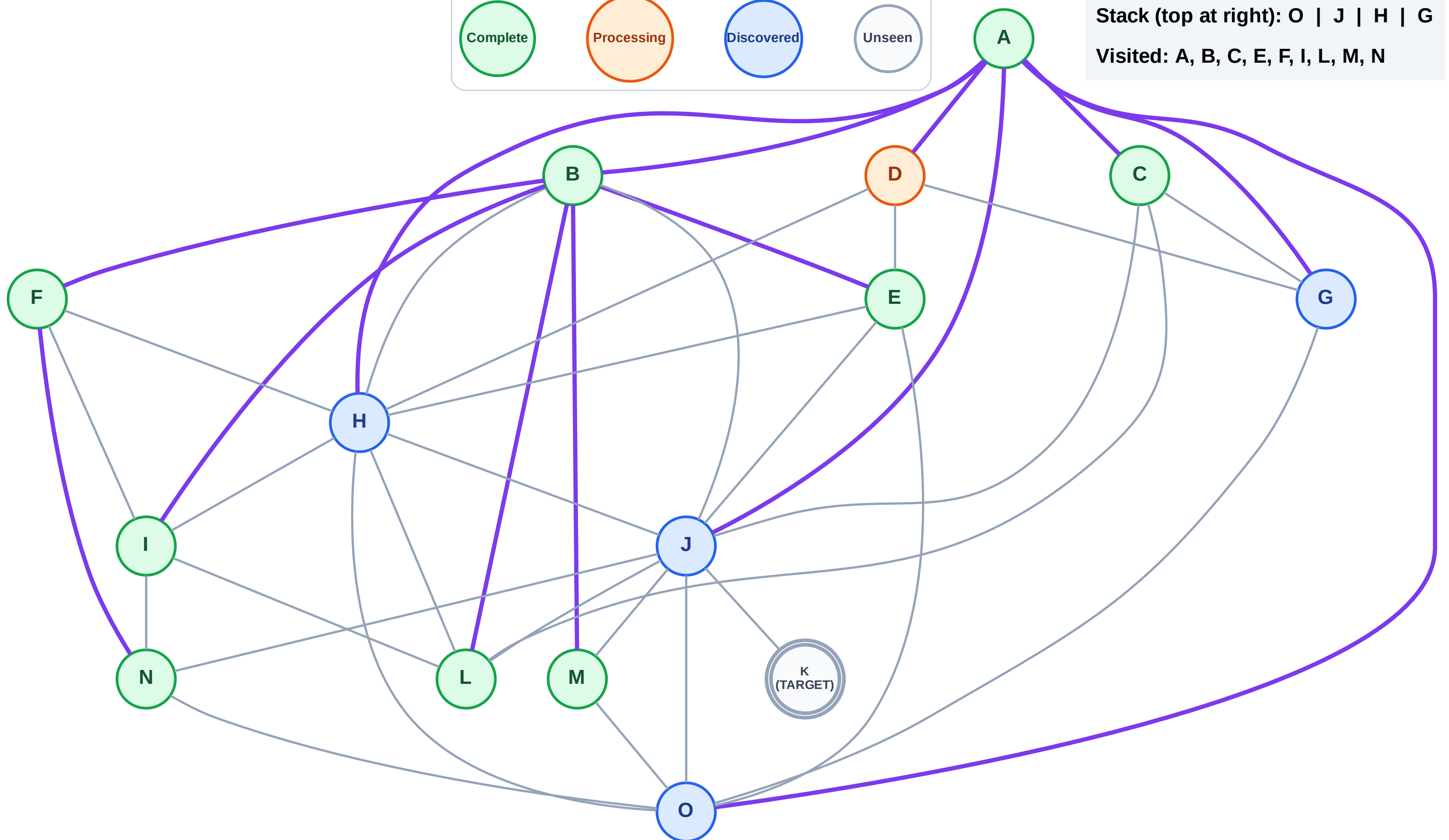


Complete Processing Discovered Unseen

Processing

Unseen

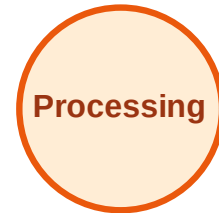
Visited: A, B, C, E, F, I, L, M, N



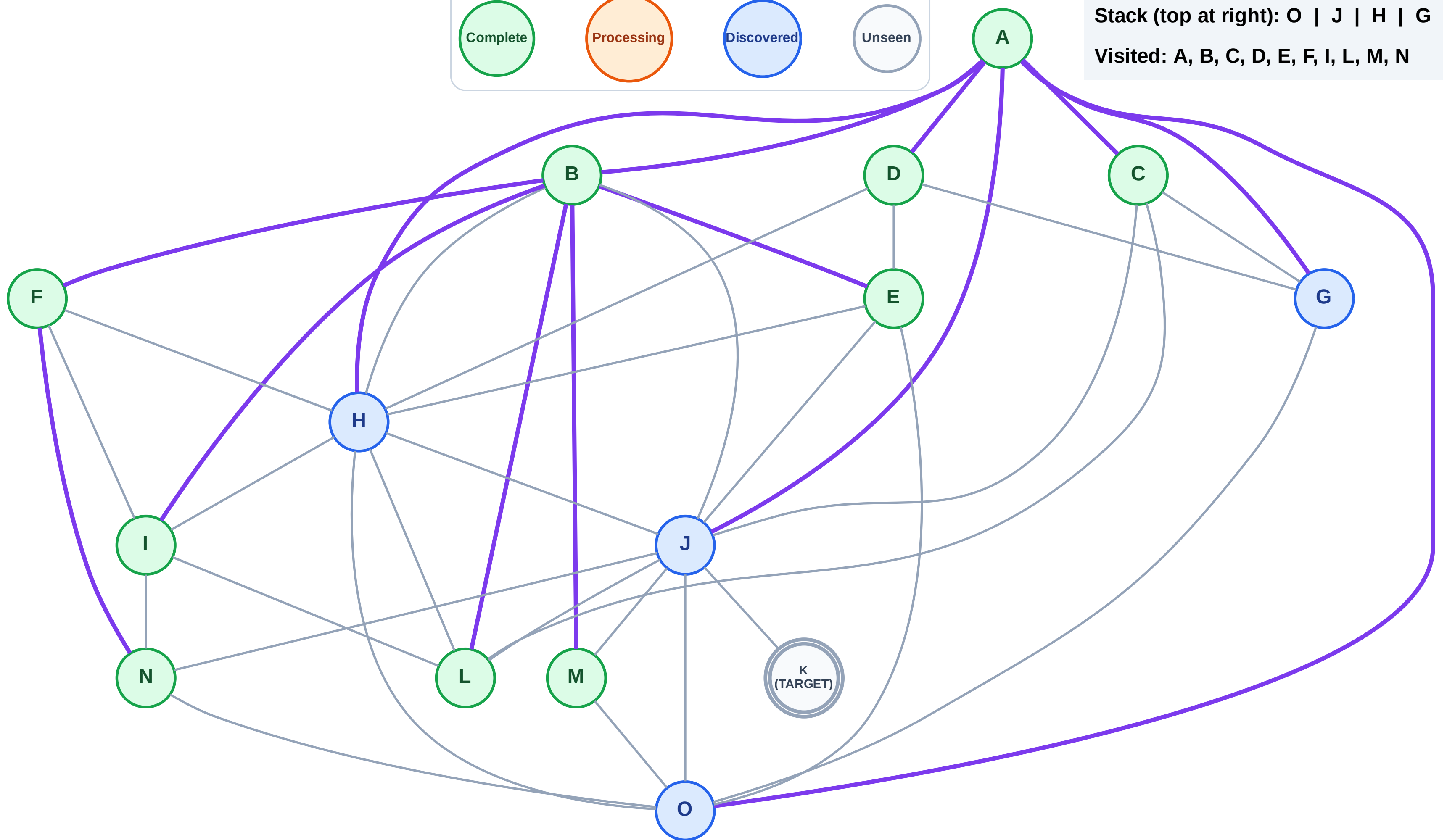
DFS Traversal

Step 21: No new neighbors from D

Legend



Stack (top at right): O | J | H | G
Visited: A, B, C, D, E, F, I, L, M, N



DFS Traversal

Step 22: Process node G

Legend

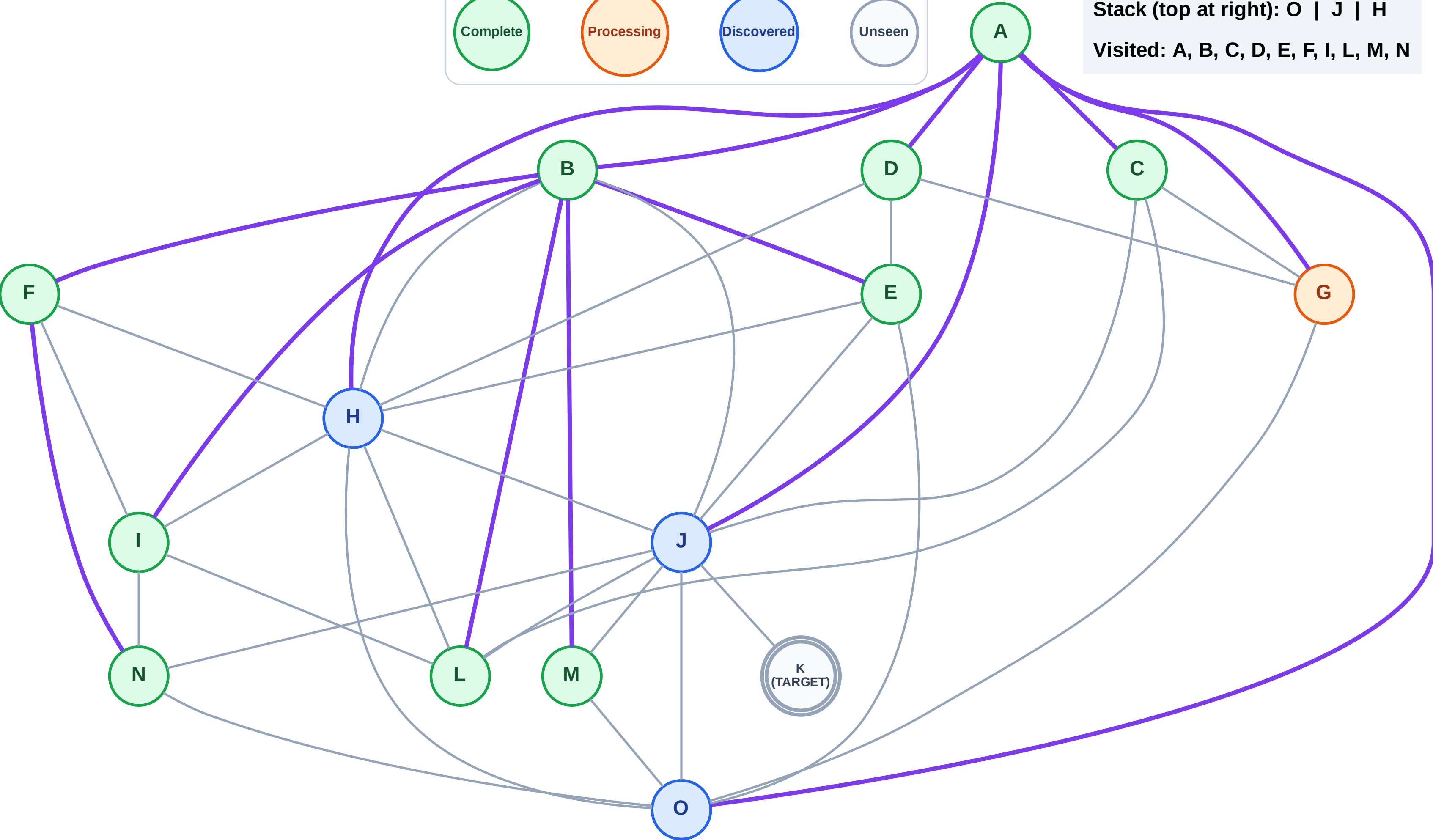
Complete

Processing

Discovered

Unseen

Stack (top at right): O | J | H
Visited: A, B, C, D, E, F, I, L, M, N



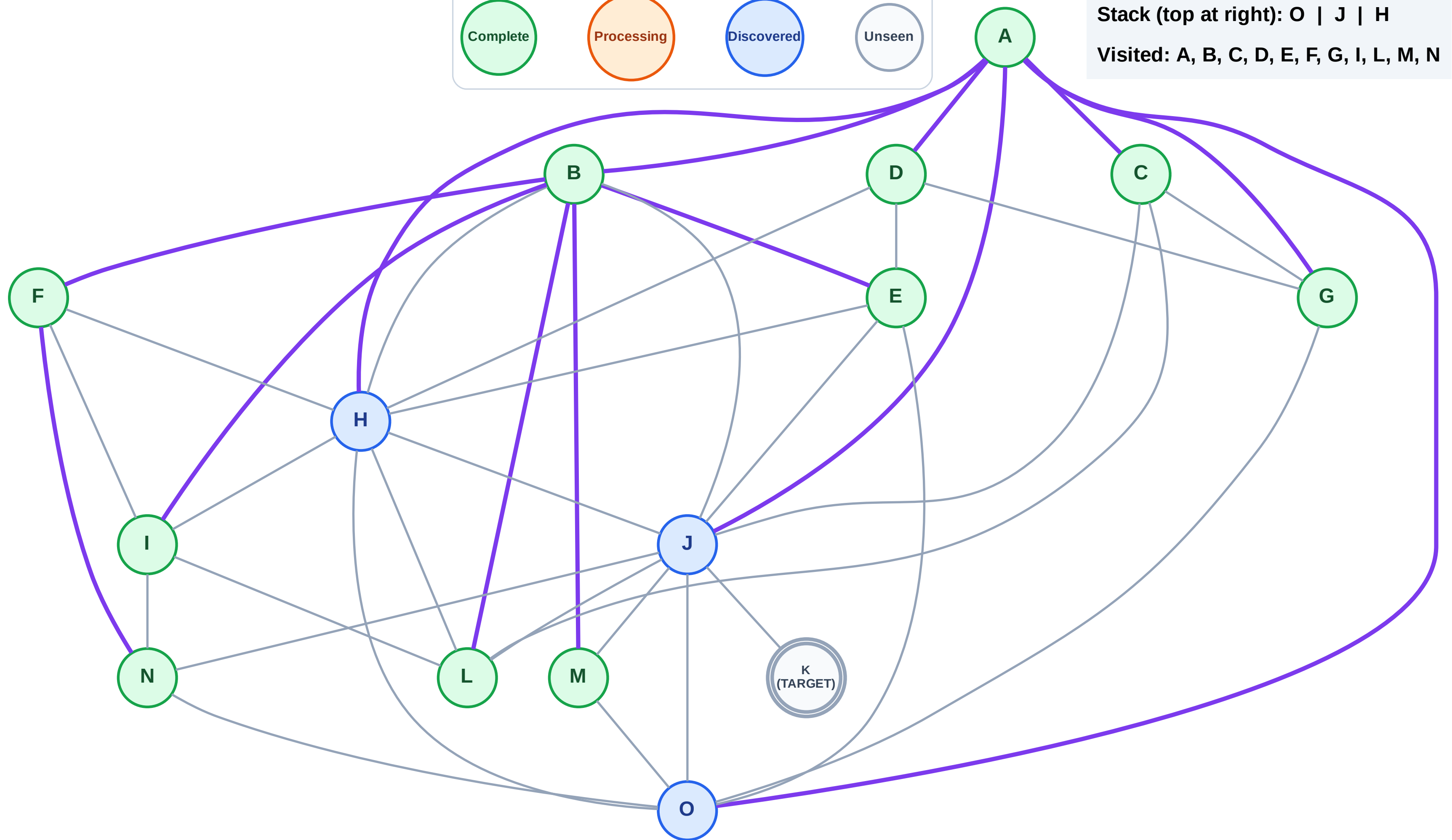
DFS Traversal

Step 23: No new neighbors from G

Legend



Stack (top at right): O | J | H
Visited: A, B, C, D, E, F, G, I, L, M, N



DFS Traversal

Step 24: Process node H

Legend

Complete

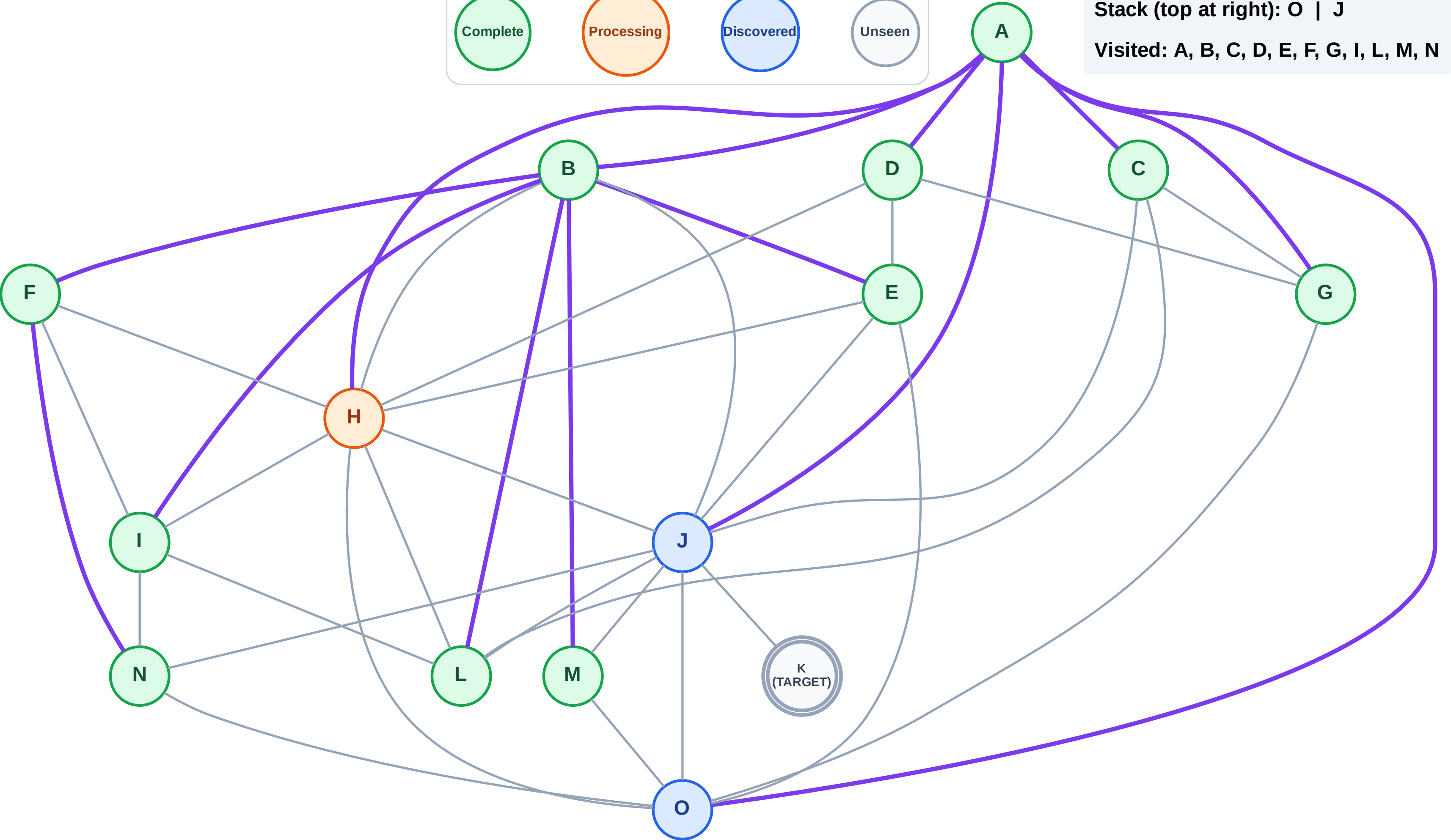
Processing

Discovered

Unseen

Stack (top at right): O | J

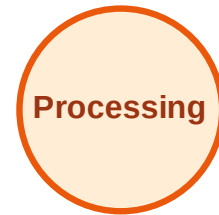
Visited: A, B, C, D, E, F, G, I, L, M, N



DFS Traversal

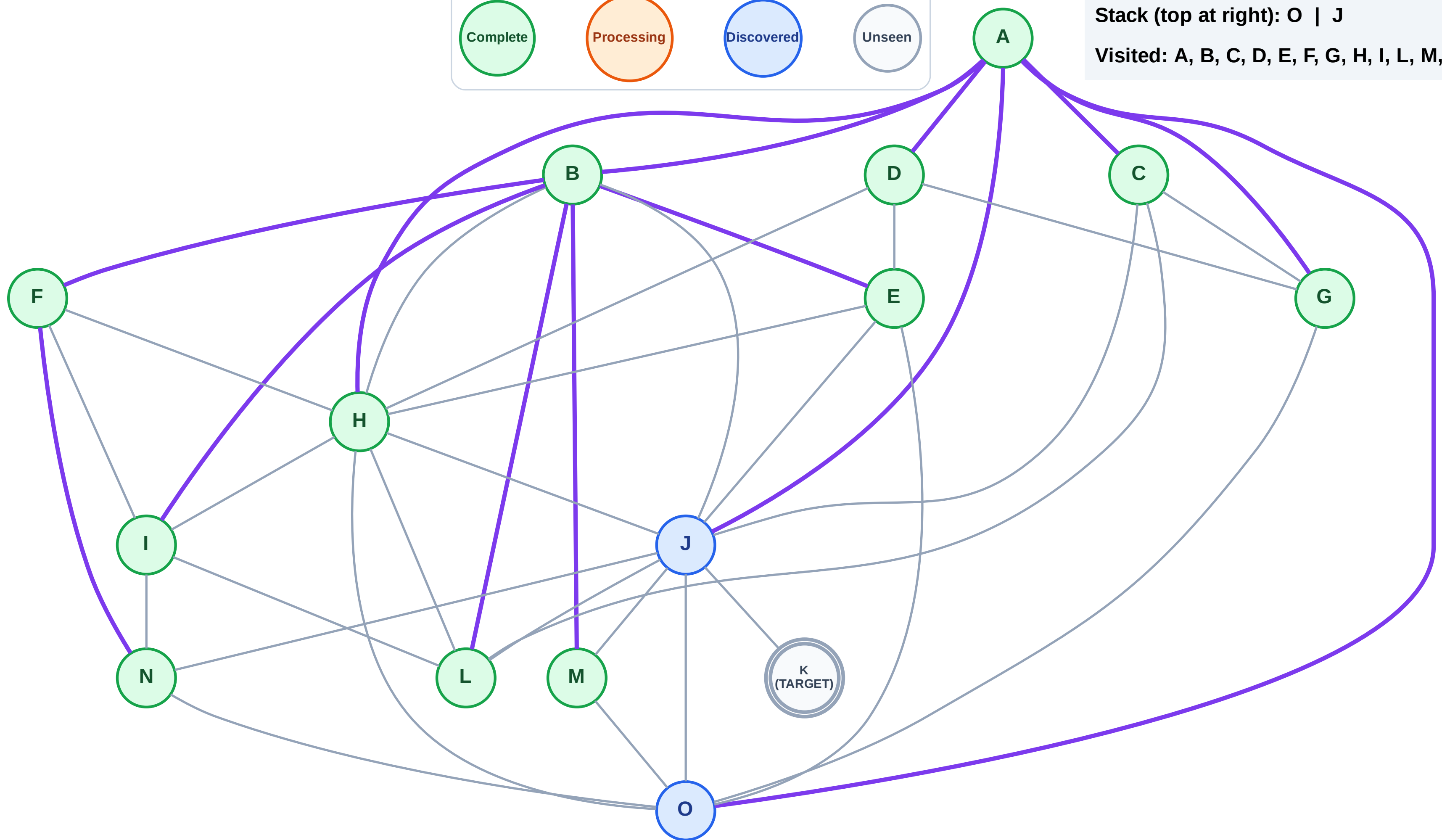
Step 25: No new neighbors from H

Legend



Stack (top at right): O | J

Visited: A, B, C, D, E, F, G, H, I, L, M, N



DFS Traversal

Step 26: Process node J

Legend

Complete

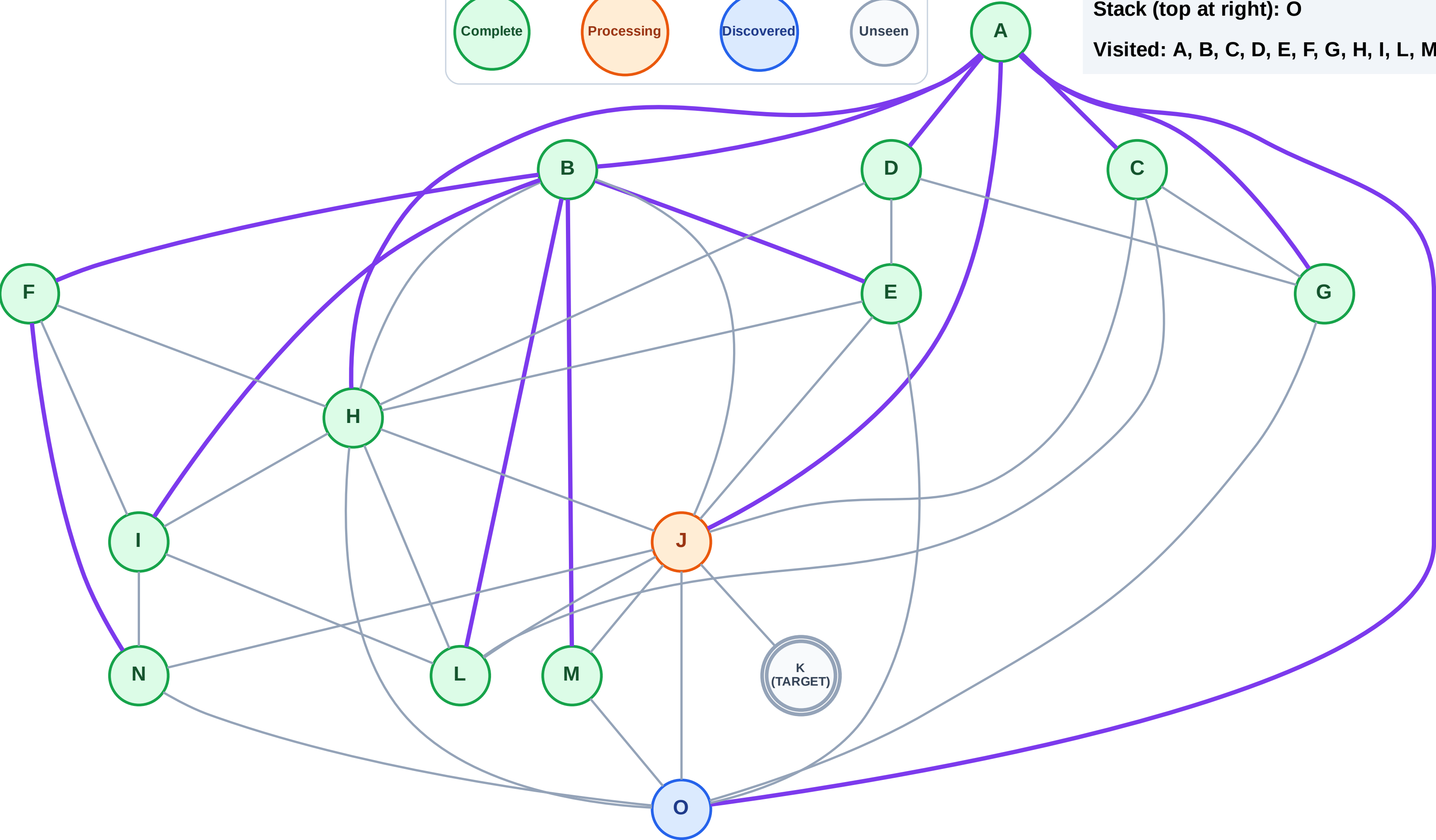
Processing

Discovered

Unseen

Stack (top at right): O

Visited: A, B, C, D, E, F, G, H, I, L, M, N



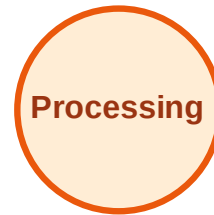
DFS Traversal

Step 27: Discover K from J

Legend



Complete



Processing



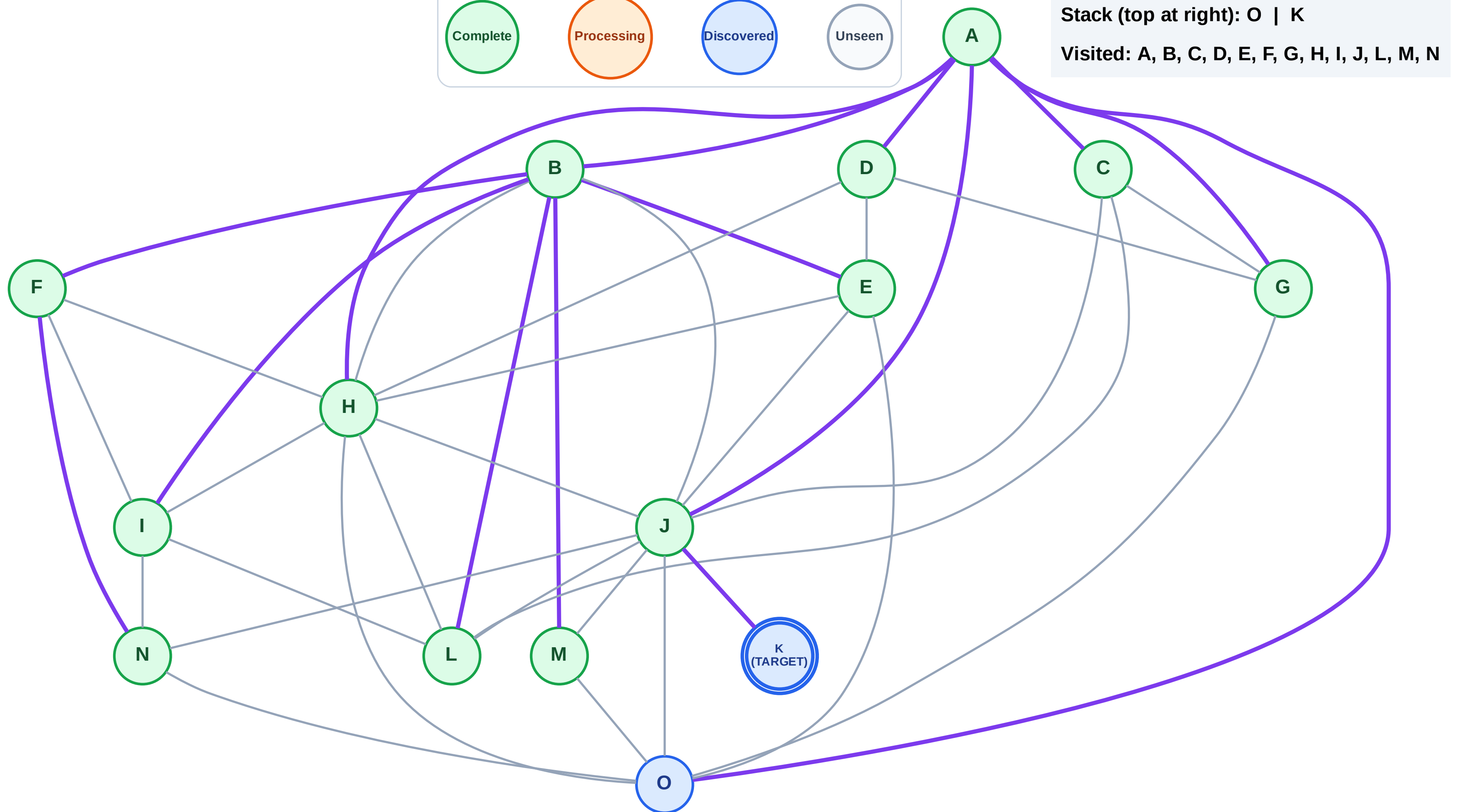
Discovered



Unseen

Stack (top at right): O | K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



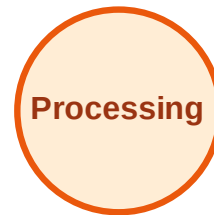
DFS Traversal

Step 28: Process node K

Legend



Complete



Processing



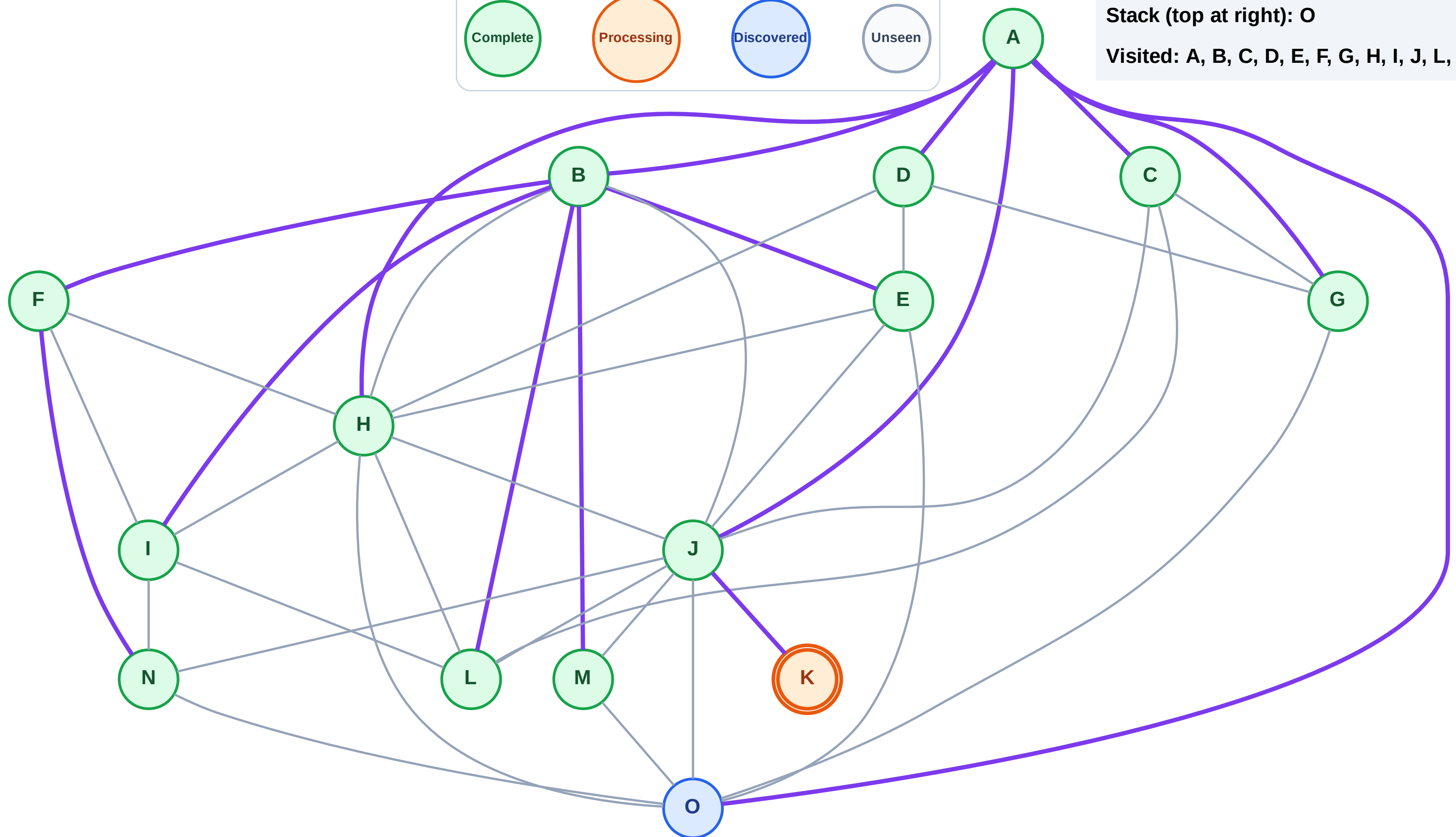
Discovered



Unseen

Stack (top at right): O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



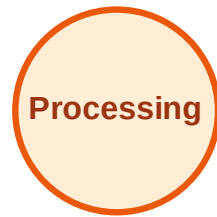
DFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → J → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): O

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N

